CENTER FOR DRUG EVALUATION AND RESEARCH

APPLICATION NUMBER:

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MEDICAL REVIEW(S)

Clinical Review
Shrimant Mishra, MD, MPH (Efficacy)
Hiwot, Hiruy, MD, PhD (Safety)
BLA 761046
Bezlotoxumab

CLINICAL REVIEW

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Division/Office	DAIP/OAP
Reviewer Names	Shrimant Mishra, MD, MPH (Efficacy) Hiwot Hiruy, MD, PhD (Safety)
Review Completion Date	09/16/2016
Established Name	Bezlotoxumab (a.k.a. MK-6072)
(Proposed) Trade Name	Zinplava
Applicant	Merck, Sharp, & Dohme Corp.
Formulation	Intravenous
Dosing Regimen	Single Infusion 10mg/kg of MK-6072
Applicant Proposed Indication(s)/Population(s)	Prevention of *Clostridium difficile* Infection (CDI) recurrence in patients 18 years or older receiving antibiotic therapy for CDI.
Recommendation on Regulatory Action	Approval
Recommended Indication/Population	Prevention of *Clostridium difficile* Infection (CDI) recurrence in patients 18 years or older receiving antibiotic therapy for CDI.

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Glossary

AC	advisory committee
AE	adverse event
BLA	biologics license application
BPCA	Best Pharmaceuticals for Children Act
BRF	Benefit Risk Framework
CBER	Center for Biologics Evaluation and Research
CDER	Center for Drug Evaluation and Research
CDRH	Center for Devices and Radiological Health
CDTL	Cross-Discipline Team Leader
CFR	Code of Federal Regulations
CMC	chemistry, manufacturing, and controls
COSTART	Coding Symbols for Thesaurus of Adverse Reaction Terms
CRF	case report form
CRO	contract research organization
CRT	clinical review template
CSR	clinical study report
CSS	Controlled Substance Staff
DMC	data monitoring committee
ECG	electrocardiogram
eCTD	electronic common technical document
ETASU	elements to assure safe use
FDA	Food and Drug Administration
FDAAA	Food and Drug Administration Amendments Act of 2007
FDASIA	Food and Drug Administration Safety and Innovation Act
GCP	good clinical practice
GRMP	good review management practice
ICH	International Conference on Harmonization
IND	Investigational New Drug
ISE	integrated summary of effectiveness
ISS	integrated summary of safety
ITT	intent to treat
MedDRA	Medical Dictionary for Regulatory Activities
mITT	modified intent to treat
NCI-CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Event
NDA	new drug application
NME	new molecular entity
OCS	Office of Computational Science
OPQ	Office of Pharmaceutical Quality

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OSE	Office of Surveillance and Epidemiology
OSI	Office of Scientific Investigation
PBRER	Periodic Benefit-Risk Evaluation Report
PD	pharmacodynamics
PI	prescribing information
PK	pharmacokinetics
PMC	postmarketing commitment
PMR	postmarketing requirement
PP	per protocol
PPI	patient package insert
PREA	Pediatric Research Equity Act
PRO	patient reported outcome
PSUR	Periodic Safety Update report
REMS	risk evaluation and mitigation strategy
SAE	serious adverse event
SAP	statistical analysis plan
SGE	special government employee
SOC	standard of care
TEAE	treatment emergent adverse event

1 Executive Summary

1.1. Product Introduction

Merck, Sharp, & Dohme, Corp. is seeking the approval of a New Molecular Entity (NME). This NME is a monoclonal antibody directed at Toxin B of *Clostridium difficile* and has been designated as MK-6072. The applicant is seeking the indication of prevention of *Clostridium difficile* Infection (CDI) recurrence in patients 18 years or older receiving antibiotic therapy for CDI.

1.2. Conclusions on the Substantial Evidence of Effectiveness

This reviewer recommends that MK-6072 be approved for the prevention of recurrence of CDI in subjects 18 years and older. The drug's effect on preventing recurrence appears variably persuasive, both statistically and clinically, in studies P001 and P002. Though, a potentially harmful effect of MK-6072 on initial clinical cure cannot be ruled out given the inconsistent findings in study P001 and P002, further analysis appears to diminish the clinical significance of several of those deemed failures of initial treatment. Still, the potential negative findings relative to clinical cure should be made clear in labelling, and clinicians should use clinical judgment in deciding the best candidates for this therapy.

1.3. Benefit-Risk Assessment

Benefit-Risk Summary and Assessment

MK-6072 is an antibody directed at Toxin B of *Clostridium difficile*. It is meant to be given during a subject's CDI episode along with standard of care (SOC) in order to prevent recurrence of CDI. In two phase 3 trials, studies P001 and P002, the drug demonstrated a significant effect on prevention of recurrence. However, it also appeared to have an uncertain effect on the initial CDI episode. In study P001, it appeared to have a possible harmful effect on the clinical cure of the initial CDI. In study P002, no harmful effect on the clinical cure of the initial CDI infection was noted. A major amendment allowed for further analysis of both the severity/frequency of recurrences as well as the clinical variables associated with clinical failures. After this analysis, this reviewer felt there was reasonable comfort that potential negative effects on initial clinical cure were minor and that the benefit of reducing the frequency/severity of recurrences outweighed possible negative effects on initial clinical cure.

Dimension	Evidence and Uncertainties	Conclusions and Reasons
Analysis of Condition	• *Clostridium difficile* infection is a serious condition resulting in significant morbidity and mortality. It has a high recurrence rate even with initial satisfactory treatment, and the treatment of recurrences is markedly heterogeneous and unstandardized.	This is a serious condition for which an approved therapy that met regulatory evidentiary and safety standards would be of significant benefit to patients and clinicians.
Current Treatment Options	• There is no approved treatment for prevention of *C. difficile recurrences*. Fecal transplant has been used for the prevention of CDI recurrences but this strategy has not been approved.	An approved therapy would bring needed clarity to the clinical approach of this condition and would provide clinicians/patients with significantly improved data upon which to assess treatment of recurrences.
Benefit	• The sponsor conducted two large phase 3 trials in a reasonably representative patient population. Though the definition of recurrence of CDI (used as the primary endpoint) was not robustly defined, there appeared to be a general trend toward reducing recurrences irrespective of the definition used or population studied. However, clinical cure appeared to be adversely affected by MK-6072 in one	The evidentiary standard has been met with regards to the prevention of CDI recurrence by MK-6072. Though its effect on initial clinical cure remains unclear, there appears to be a positive risk benefit calculus (i.e., the beneficial effects on recurrences outweigh the potential

Dimension	Evidence and Uncertainties	Conclusions and Reasons
	trial but not in the other.	negative effects on initial clinical cure).
Risk	• The safety database for bezlotoxumab included all patients that received full or partial infusion of bezlotoxumab in the two phase 3 trials. There was adequate number of subjects exposed to bezlotoxumab at the proposed dose during the drug development program. The rates of treatment emergent adverse events (TEAE), serious adverse events (SAE) and deaths were comparable between the bezlotoxumab-treated and placebo study arms. Of note, a safety signal of a higher number of TEAE, SAE and deaths in a subgroup of patients with baseline history of congestive heart failure (CHF) was noted in subjects who were treated with bezlotoxumab compared to placebo.	Overall, bezlotoxumab, given as a single IV infusion, had a favorable safety profile. However, the safety signal noted in patients with a history of CHF requires caution in the use of bezlotoxumab in these patients.
Risk Management	• The safety signal finding in patients with a history of CHF treated with bezlotoxumab may lead to the inclusion of a warning in the bezlotoxumab label regarding use in this patient subgroup.	In subjects with baseline congestive heart failure, bezlotoxumab should be reserved for use in situations where the benefit clearly outweighs the risk.

2 Therapeutic Context

2.1. Analysis of Condition

Clostridium difficile infection is a well-recognized cause of colitis associated with morbidity and mortality, particularly in hospitalized patients and patients in long-term care facilities. It is implicated in 20-30% of patients with antibiotic-associated diarrhea, and 50-75% of those with antibiotic-associated colitis. It is a gram positive, spore forming, anaerobic rod-shaped bacterium which colonizes subjects through the fecal-oral route. Toxin-producing *C. difficile* may expand and cause infection in the colon, particularly in those persons with recent disruptions to their colonic flora by antimicrobial or chemotherapeutic exposure. It mediates its disease activity through toxin production, particularly through toxins A and B (though other unknown mechanisms of action/toxin actions are likely at play as well). These toxins are thought to have harmful effects on the colonic epithelium and induce deleterious secretory, inflammatory, and apoptotic processes. The toxins appear to disrupt cytoskeletal regulation eventually resulting in cell rounding in death. Toxin A mediates fluid secretion and an inflammatory response. Toxin B's mechanism of action is less well-understood but appears to be more potent that Toxin A. CDI can be distinguished by strain, particularly by ribotyping of the spacer region 16S and 23S rDNA. Notably, particular strains such as NAP1/027 have been associated with epidemic disease.

Subjects suffering from infection may range from having mild diarrhea, to significant colitis leading to colectomy and, possibly, death. Symptoms are often associated with fever and leukocytosis. Occasionally, subjects may not have diarrhea but may instead present with a toxic megacolon, a potentially life threatening complication. Diagnosis of the disease is most often made with a combination of clinical suspicion (due to diarrhea in a hospitalized elderly patient, etc.) and a positive ELISA for Toxin A or B or both. These ELISA tests have variable sensitivity but are commonly used due to their ease of testing and cheap expense. Increasingly, PCR-based tests that assess for the toxin-producing genes are also being used.

The initial infection episode is often successfully managed by fluid replacement, discontinuation of the offending agent such as an antimicrobial (if possible), and oral/IV antimicrobial therapy with oral metronidazole, vancomycin, or fidaxomicin (and occasionally IV metronidazole or rectally delivered vancomycin or metronidazole). Second –line agents used very infrequently include nitazoxanide and rifamycin as well as cytotoxin binding agents such as cholestyramine or colestipol. However, a significant proportion (estimates are wide-ranging from a quarter to almost half of patients) of treated subjects will experience a recurrence of CDI, including elderly subjects, subjects with previous episodes of diarrhea or underlying morbidity, and subjects infected with particular strains. There are no approved therapies for the treatment of recurrences.

2.2. Analysis of Current Treatment Options

Though treatment of an initial episode of CDI is relatively straightforward, generally involving 10-14 day courses of oral vancomycin, oral fidaxomicin, or oral metronidazole (sometimes combined with intravenous metronidazole in severe cases), there is no approved or well-studied definitive treatment for prevention of recurrences. As to the treatment of recurrent episodes of CDI, a repeat of the initial regimen is commonly given and is often successful for the first recurrence. In cases of repeat recurrences, prolonged/tapering oral vancomycin regimens, pulsed oral vancomycin therapy, administration of immunoglobulins (particularly in cases of hypogammglobulinemia), probiotic administration and fecal transplant have all been advocated. While fecal transplant also shows promise for prevention of recurrences, this therapy has not been rigorously studied or received regulatory approval. Also, its production has yet to be streamlined.

3 Regulatory Background

3.1. U.S. Regulatory Actions and Marketing History

MK-6072 is an NME that has not been approved or marketed anywhere in the world.

3.2. Summary of Presubmission/Submission Regulatory Activity

The initial investigational new drug application (IND) for human monoclonal antibodies against *C. difficile* toxin A and toxin B was submitted on November 25, 2005. Early in product development, the applicant hypothesized that an anti-toxin A antibody or the combination of anti-toxin A and B antibodies would be the most effective in prevention of CDI recurrence.

The first Phase 2 trial evaluated the efficacy and safety of a single infusion of an anti-toxin A antibody versus placebo. The trial was conducted in 2005-2006 and was terminated early after enrolling 46 subjects because emerging non-clinical data suggested that the combination of antibodies against toxin A and B would be more effective for the prevention of CDI recurrence.

A Phase 2 study comparing a combination of *C. difficile* toxin A and B antibodies with placebo in patients receiving antibacterial therapy for CDI was conducted in 2006-2008. The study enrolled 101 patients in the combination arm and 99 patients in the placebo arm. The recurrence rate was lower among patients treated with the combination therapy (7% vs. 25%; $p < 0.001$). No Phase 2 studies evaluated bezlotoxumab alone.

Subsequently, the applicant proposed to conduct Phase 3 studies comparing the combination of monoclonal antibodies to *C. difficile* toxins A and B with placebo. On May 12, 2010 the

combination was granted fast track designation for the treatment of patients with *C. difficile* infection and/or for the prevention of CDI recurrence.

FDA, in keeping with the "combination rule" outlined in CFR 300.50(a), recommended studying recurrence by conducting a four arm study that compared the combination of monoclonal antibodies to *C. difficile* toxins A and B, antibody to *C. difficile* toxin A alone, antibody to *C. difficile* toxin B alone, and placebo, all given in conjunction with standard of care antimicrobial therapy for CDI.

On December 21, 2010 a Special Protocol Assessment (SPA) agreement was reached on the design of study P001, a Phase 3, randomized, double-blind, placebo-controlled, adaptive-design study of the efficacy in the prevention of CDI recurrence of a single infusion of human monoclonal antibody to *C. difficile* toxin A, human monoclonal antibody to *C. difficile* toxin B, and the combination of human monoclonal antibodies to *C. difficile* toxin A and toxin B in patients receiving antibacterial therapy for *C. difficile* infection. The primary endpoint in the protocol was CDI recurrence following clinical cure of baseline CDI episode. The proportion was calculated as the number of subjects with CDI recurrence divided by the total number of subjects in the analysis population regardless of whether or not they had a successful resolution of the initial episode of CDI.

However, upon review of Protocol P002 FDA recommended changing the primary endpoint from CDI recurrence to global cure. The reason for the recommendation was the concern that in the calculation of the CDI recurrence rate, subjects who failed treatment of the initial CDI episode would be counted as not having a recurrence. For this binary endpoint, a successful outcome would group those who were initially cured and did not have a recurrence with those who failed initial treatment. Subsequently a greater number of initial cure failures in a treatment group would result in a smaller number of subjects who could potentially develop recurrence. Therefore, an imbalance between treatment groups in the number of subjects who failed treatment of the initial CDI episode could confound the assessment of recurrence rates. This would be most concerning if the monoclonal antibody arm had a lower initial cure rate than placebo. The endpoint of global cure rate avoids this problem by defining treatment success as clinical cure of the initial CDI episode AND the absence of CDI recurrence. In August 2012 FDA conveyed their recommendations to the applicant.

The applicant disagreed with changing the primary endpoint. Global Cure remained a key secondary endpoint in both trials. FDA acknowledged that the Phase 3 trials would preserve the use of CDI recurrence rate as the primary efficacy analysis but noted that interpretation of the results of these trials would depend on the number of subjects with clinical cure of the initial infection.

In November 2015 the applicant submitted a BLA with the results of two Phase 3 pivotal trials to support approval of bezlotoxumab for the prevention of *C. difficile* infection recurrence.

Though the original PDUFA date was in July 2016, on the recommendations of an AC (conducted in June) as well as through internal discussions, a Major Amendment was requested in order to allow review of further information related to the severity of recurrences as well as the clinical characteristics of subjects who failed initial clinical cure. The PDUFA data was thus extended to October 23, 2016.]

3.3. Foreign Regulatory Actions and Marketing History

This product is not marketed anywhere in the world though it is undergoing regulatory evaluation overseas, including by the European Medecines Agency (EMA) and by the Pharmaceuticals and Medical Devices Agency (PMDA- Japan).

4 Significant Issues from Other Review Disciplines Pertinent to Clinical Conclusions on Efficacy and Safety

4.1. Office of Scientific Investigations (OSI)

Please see the review by Dr. John Lee. Four study sites and one sponsor site were inspected. At all four CI sites, overall study conduct appeared GCP-compliant, including study oversight by the sponsor and by the institutional review board (**IRB**). All audited NDA data were adequately verifiable against source records and case report forms (CRFs). The data from the four CI sites appeared reliable as reported in the BLA.
It should be noted that one investigator, Dr. Jose A. Giron, was noted to have significant flaws in trial conduct, including delayed sending of blood samples, insufficiently obtaining stool samples at baseline, and breaches involving informed consent. All 15 subjects enrolled at this site (in study P001) were not included in the FAS, and the investigator received an OAI warning letter.

4.2. Product Quality

Bezlotoxumab is a (b) (4) human IgG1 monoclonal antibody that binds to *C. difficile* toxin B. The drug substance is manufactured at (b) (4), (b) (4). Bezlotoxumab drug product is manufactured at Merck, Sharp, and Dohme (MSD) in Carlow, Ireland. The manufacturing process of bezlotoxumab drug product includes (b) (4), (b) (4). The product quality reviewer, Dr. William Hallet, found the manufacture of bezlotoxumab to be well controlled and meeting FDA's sterility standards. The manufacturing process has been validated and has produced drug product consistently. An expiry period of 24 months was recommended for bezlotoxumab drug product when stored at 2 - 8°C.

Please see the review by Dr. William Hallet for further details

4.3. Clinical Microbiology

Please see the review by Dr. Kerian Grande for further details

4.4. Nonclinical Pharmacology/Toxicology

Single and repeat intermittent-dose toxicology studies in mice with IV bezlotoxumab administered for 14 days (5 doses/3 day interval) and 21 days (2 doses/14 day interval) were conducted. These failed to demonstrate any clinical signs or evidence of toxicity, at doses up to 50 and 125 mg/kg respectively, and at approximately 2.5 and 7 times greater exposure than observed in the clinic after a single 10 mg/kg dose. All histopathological evaluations of tissues from bezlotoxumab treated animals were similar to controls. No targets of bezlotoxumab toxicity were identified in any of the animal toxicology studies. Tissue cross-reactivity studies were conducted in vitro in at least 38 mouse and human tissues with bezlotoxumab. They showed no reactivity (positive staining) of tissue samples. No genotoxicity, reproductive/developmental toxicity, or carcinogenicity studies were conducted with bezlotoxumab.

Please see the review by Dr. Terry Miller for details

4.5. Clinical Pharmacology

4.5.1. Mechanism of Action

MK-6072 binds to toxin B secreted by toxin producing *C. difficile* during acute infection. Theoretically, such binding might limit further epithelial and inflammatory damage incited by the toxin, however exactly how binding of the toxin during initial infection prevents recurrence is unclear.

4.5.2. Pharmacodynamics

As bezlotoxumab is a monoclonal antibody to an exogenous toxin, it has no intrinsic antimicrobial activity. It binds toxin B with an affinity equilibrium dissociation constant (Kd) of <1 x 10-9 M. In Vero cells 200 mcg/mL of bezlotoxumab could block binding of 50 ng/mL toxin B. Simiarly, in Caco-2 cells the addition of bezlotoxumab (in concentraions 0-100mcg/ml) resulted in a dose-dependent neutralizing effect of 100ng/ml of toxin B as measured by transepithelial resistance.

4.5.3. Pharmacokinetics

Bezlotoxumab has a half-life of roughly 19 days and volume of distribution of 7.33L. It has characteristics typical of other monoclonal antibodies including limited extravascular distribution and tissue binding, and minimal renal clearance. It is degraded into small proteins and amino

acids by protein catabolism, and is not a substrate of hepatic metabolic enzymes Bezlotoxumab has moderate PK variability (40% and 21% CV for AUC0-inf and Cmax, respectively), though the PK does not seem to be affected by renal or hepatic impairment. The Clin Pharm reviewer noted that outside of weight no otherdose adjustments are required for external and intrinsic factors such as gender, age, race, and CDI severity. He also concluded that there was no exposure- response relationships identified for either efficacy or safety that necessitated a dose adjustment. Studies in healthy subjects and subjects with CDI indicate there may be very limited immunogenicity associated with bezlotoxumab.

Please refer to the review by Dr. Yang He for further details

4.6. Devices and Companion Diagnostic Issues

Not applicable because no device or companion diagnostic is being marketed for the study drug.

4.7. Consumer Study Reviews

None performed.

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5 Sources of Clinical Data and Review Strategy

5.1. Table of Clinical Studies

Table 1: Listing of Phase 2 and 3 Clinical Trials for BLA 761046

Trial Identity	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Follow Up	No. of patients enrolled	Study Population	No. of Centers and Countries	Import Note
Controlled Studies to Support Efficacy and Safety								
P001	Phase 3 randomized, placebo controlled, double blind, adaptive design	Single IV dose of either MK-3415 (Antitoxin A), MK-6072 (Antitoxin B), MK-3415A (Antitoxin A + Antitoxin B), or placebo	Primary: CDI Recurrence Through Week 12 Secondary: Global Cure (Clinical Cure + No CDI Recurrence); Subgroup analysis of CDI	Single Dose Only Follow-Up through Week 12 (Day 85)	Randomized: 1452 Safety: 1412 FAS: 1396	Adults with confirmed CDI within 7 days of enrollment and on/about to start SOC	184 sites in the United States (75), Italy (13), Canada (11), Spain (9), Australia (8), UK (8), Germany (7), Portugal (7), Chile (6), Denmark (6), Czech Republic (6), Israel (5),	Trial Dur 2011-201 MK-3415 dropped a interim analysis c to increas mortality to placeb low effic relative t combinat therapy

APPEARS THIS WAY ON ORIGINAL

		Active Dose: 10 mg/kg Placebo: Normal Saline All arms with SOC	Recurrence Exploratory: Clinical Cure of Initial CDI Safety				Belgium (4), Colombia (4), New Zealand (4), South Africa (4), Brazil (3), Austria (3), Mexico (1)	
P002	Phase 3 randomized, placebo controlled, double blind	Single IV dose of either MK-6072 (Antitoxin B), MK-3415A (Antitoxin A + Antitoxin B), or placebo Active Dose: 10 mg/kg Placebo: Normal Saline	Primary: CDI Recurrence Through Week 12 Secondary: Global Cure (Clinical Cure + No CDI Recurrence); Subgroup analysis of CDI Recurrence Exploratory:	Single Dose Only Follow-Up through Week 12 (Day 85) Subgroup of 300 subjects followed for 9 months after Week 12	Randomized: 1203 Safety: 1168 FAS: 1163	Adults with confirmed CDI within 7 days of enrollment and on/about to start SOC	200 sites in the United States (47); Japan (35); South Korea (15); Poland (14); France (11); Turkey (10); Argentina (9); Czech Republic (8); Spain (8); Taiwan (8); Germany (7); Russia (7); Canada (6); Sweden (6);	

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		All arms with SOC	Clinical Cure of Initial CDI Safety				Finland (5); Israel (3); and Switzerland (1)	
Studies to Support Safety								
P017	Phase 2 randomized, placebo controlled, double blind	Single IV dose of either Antitoxin B + Antitoxin A, or placebo Active dose 10 mg/kg Placebo: Normal Saline	Primary: CDI Recurrence Through Week 12 Secondary: Time to Resolution of CDI; Failure of SOC	Single Dose Only Follow-Up through Week 12 (Day 85)	Randomized: 200 Subjects	Adults with confirmed CDI within 14 days of enrollment and on/about to start SOC	30 sites in the US (23) and Canada (7)	Study Pe 2006-200 Combina had reduc rates of C recurrenc No differ in failure SOC for initial epi
P018	Phase 2 randomized, placebo controlled, double blind	Single IV dose of either Antitoxin A, or placebo Active dose	Primary: Resolution of CDI; Recurrence of CDI Through Week 8	Single Dose Only Follow-Up through Week 8 (Day 56)	Randomized: 46 subjects	Adults with CDI and on/about to start SOC	16 sites in the US	Trial dura 2005-200 Study terminate early for business

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		10 mg/kg Placebo: Normal Saline All arms with SOC						reasons (wanted t evaluate combinat vs. placeb due to an study findings) No differ in efficac study not powered

5.2. Review Strategy

The Clinical Review of this application was divided between two different reviewers. Dr. Shrimant Mishra reviewed clinical efficacy while Dr. Hiwot Hiruy reviewed clinical safety.

Efficacy

The review of clinical efficacy was primarily focused on the two pivotal Phase 3 studies – studies P001 and P002. Because the sponsor ultimately chose to pursue MK-6072 for this indication, earlier Phase 2 trials were less useful in evaluating efficacy because either they did not have a MK-6072 alone arm or did not evaluate MK-6072 at all. Similarly, in evaluation of the Phase 3 trials, almost all of the focus in this review has been on the comparison of MK-6072 with placebo. The MK-3415 (human monoclonal antibodies to *C. difficile* toxin A) and MK-3415A (human monoclonal antibodies to *C. difficile* toxin A and toxin B) arms are discussed sparsely.

For the purposes of clinical efficacy, the sponsor's results were evaluated and interpreted in detail by both the statistical reviewer, Dr. Cheryl Dixon, and by Dr. Mishra. Because there was concern over the use of the primary endpoint of recurrence and how recurrence was defined (see below), numerous sensitivity analyses were performed by both the clinical and statistics reviewers using SAS, JMP, JReview, etc. in order to more fully inform the reported results. Both the statistical and clinical reviewer attempted to redefine recurrence more robustly and thus their respective major findings may differ to a degree from the sponsor as well as from each other.

6 Review of Relevant Individual Trials Used to Support Efficacy

P001: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Adaptive Design Study of the Efficacy, Safety, and Tolerability of a Single Infusion of MK-3415 (Human Monoclonal Antibody to *Clostridium difficile* toxin A), MK-6072 (Human Monoclonal Antibody to *C. difficile* toxin B), and MK-3415 A (Human Monoclonal Antibodies to *C. difficile* toxin A and toxin B) in Patients Receiving Antibiotic Therapy for C. *difficile* infection (MODIFY I)

P002: A Phase III, Randomized, Double-Blind, Placebo- Controlled Study of the Efficacy, Safety, and Tolerability of a Single Infusion of MK-6072 (Human Monoclonal Antibody to *Clostridium difficile* toxin B), and MK3415A (Human Monoclonal Antibodies to *Clostridium difficile* toxin A and B) in Patients Receiving Antibiotic Therapy for *Clostridium difficile* Infection (MODIFY II)

6.1.1. Study Design

Overview and Objective

P001: The purpose of the MODIFY I trial was to determine whether the infusion of the combined monoclonal antibodies to *C. difficile* toxins A and B (MK-3415A) in conjunction with standard of care was superior to the infusion of monoclonal antibody to toxin A only (MK-3415) plus standard of care , the infusion of monoclonal antibody to toxin B only (MK-6072) plus standard of care , and placebo plus standard of care in preventing the recurrence of *C. difficile* infection over a period of 12 weeks. The trial was also designed to potentially show the superiority of MK-3415 plus standard of care or MK-6072 plus standard of care over placebo plus standard of care in preventing recurrence.

P002: Similar to that of P001 except that there was no antitoxin A alone arm (MK-3415).

Trial Design

P001

This was a randomized, double-blind, placebo controlled, multicenter, adaptive design study evaluating the recurrence of *C. difficile* infection (CDI) in subjects receiving a single intravenous infusion of monoclonal antibodies to both toxin A and toxin B, individual monoclonal antibodies to toxin A or toxin B alone, or placebo along with standard of care antimicrobial therapy . The study was designed to evaluate recurrence over a 12 week period post infusion. The trial involved 184 sites worldwide including sites in North and South America, Europe, Africa, and Australia; there were 75 sites in the United States. 1600 subjects were planned for enrollment.

Appropriately, the study was placebo controlled/blinded as there is no currently approved drug to prevent recurrence of *C. difficile* infection (CDI). The study was also designed to meet the combination rule (in this case demonstrating the superiority of the combination antibody arm over the individual antibody arms in preventing recurrence). The follow up time of 12 weeks would be expected to capture a significant portion of recurrences though CDI recurrences have been known to occur after even longer intervals. All subjects received standard of care antimicrobial therapy, which appropriately mimicked the likely adjunct use of the study drug that would occur in a clinical setting.

A subject could be included in the trial if he/she had CDI as defined by 3 or more loose stools in the prior 24 hours and a positive stool test for toxigenic *C. difficile* in the seven days prior to infusion. This is similar to the diagnostic criteria used in the recent approval for fidaxomicin, although in those studies the toxin test had to have been positive in the previous 48 hours.

The inclusion and exclusion criteria were appropriately narrow. Elderly subjects and subjects with significant illness (such as renal or hepatic dysfunction) were not necessarily excluded however there were some exclusions based on the use of concomitant medications that might

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interfere with assessment of the primary endpoint such as immunoglobulins, fecal transplant, probiotics, and antiperistaltic agents (It is notable that these exclusions might not conform with actual clinical practice but is understandable given the need to eliminate confounding when evaluating effect of the study drug).

For this study, all doses of monoclonal antibodies were 10 mg/kg; thus a subject could get an active arm consisting of 10 mg/kg MK-6072, 10 mg/kg MK-3415, or 10 mg/kg MK-6072+ 10 mg/kg MK-3415 (MK-3415A). The 10 mg/kg dose was chosen based on pharmacodynamic/ PK targets identified in animal studies and confirmed in phase 1 studies, namely a target peak serum concentration of ≥200 μg/ml for both MK-3415 and MK-6072 (no phase 2 dose response studies were performed). Placebo was normal saline. All infusions were given as a 250 or 200ml infusion depending on investigator discretion regarding a subject's risk for fluid overload.

Standard of care antimicrobial therapy was either oral metronidazole, oral vancomycin, oral fidaxomicin, or oral vancomycin/fidaxomicin along with intravenous metronidazole. The doses chosen for SOC were typical of clinical use – 400 to 500mg every 8 hours for oral/intravenous metronidazole, 125-500mg every six hours for oral vancomycin, and 200mg twice a day for fidaxomicin. SOC therapy was to last between 10 and 14 days. SOC therapy was to have been administered prior to or within a few hours of study drug infusion; initiation of SOC alongside study drug infusion was encouraged but not mandated.

Importantly, there was no prespecified timing of study drug relative to SOC, which allowed for considerable heterogeneity regarding the timing of drug administration. Switches in SOC were allowed provided a subject had been on initial therapy for at least 3 days and certain clinical criteria were met (subject had diarrhea, ileus, or WBC > 15,000 cells/mm^3/T >38.3°C [>100.9°F]); treatment total was still to be between 10- 14 days. Concomitant medications that would interfere with efficacy assessment, such as rifaximin, nitozoxanide, antiperistaltics, immune globulin, and the probiotic *Saccharomyces boulardii* were not allowed during the study.

This trial used a centralized randomization scheme through interactive voice response system (IVRS) and randomized subjects into one of the four treatment arms. However, a prespecified interim analysis was planned at which time one or two arms may have been dropped depending on the results (the sponsor had hypothesized that the combination arm would be better than the individual arms and that those arms might be dropped after an interim analysis of efficacy). Thus, after the interim analysis randomization could have occurred into one of two or three arms. Subjects were stratified by SOC and by inpatient vs outpatient status

As noted earlier, this study had a prespecified interim analysis when at least 40% of the planned study population had been enrolled and completed their 12 week visit. The interim analysis was performed by an independent 3-7 member data monitoring committee (DMC) and had 2 objectives: 1) to evaluate if the MK-3415A arm was superior to either of the MK-6072 or MK-

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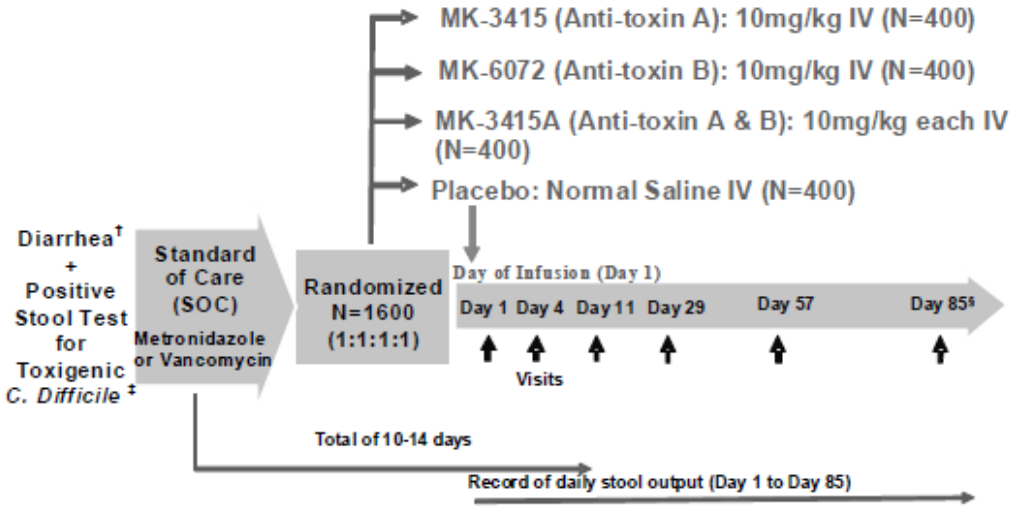
3415 arms using conservative/highly significant statistical measures of efficacy and 2) to see if any arm should be dropped due to a safety signal.

The blinding in this study was generally adequate. This is important as some the study endpoints were somewhat subjective and could be influenced by poor study blinding. For example, recurrence was based on loose stool values recorded in a subject's daily stool log; these loose stool assessments were to be based on the Bristol Stool chart. All arms used intravenous infusions. However because the monoclonal antibody arms differed slightly in appearance from placebo, all infusion bags were covered by an opaque sleeve (apparently differences could not be noted in the tubing). It should be noted that the possibility of infusion reactions were more likely in the monoclonal antibody arms and thus safety and efficacy assessments could be biased in patients who had these reactions. All subjects, study investigators, study site personnel, and the sponsor were to be blinded until the study was completed and data had been screened for completeness and accuracy.

The following sponsor figure outlines the trial design including visit dates.

Figure 1: Study P001 Trial Design

Diagram of Study Design - Base Study (Day 1 to Week 12)



† Diarrhea: 3 or more loose stools in 24 or fewer hours
‡ Positive local stool test for toxigenic *C. difficile* (stool sample must be collected within 7 days prior to infusion (Day 1))
§ Extended follow-up on subset of ~200 patients through Month 12 after the interim analysis decision is communicated

As noted above, subjects had in person visits on Days 1, 4, 11, 29, 57, and Day 85. Safety labs, antidrug antibody labs, vital signs, PK measurements, etc. were done at these visits though not all procedures were done at every visit. Subjects were instructed to keep a daily stool log

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through week 12 as well as a daily temperature log through day 14, and this log was reviewed at the in person visits as well during daily/weekly phone calls with clinic personnel. Subjects who had a new episode of diarrhea after initial clinical resolution were instructed to obtain a stool sample and contact study personnel. Study subjects who withdrew or discontinued were assessed as soon as possible for their final study visit.

P002

This was essentially an identical trial to P001 with some important caveats. It had no MK-3415 arm and had a subset of 300 patients who were followed for an extra 9 months for both safety and efficacy measures (though the primary efficacy analysis covered the initial 12 week period and did not cover the extension phase). Also, this study was conducted in different regions compared to P001 at 200 trial sites (see Demographics section).

Study Endpoints

P001:

The primary endpoint for this study is the proportion of subjects in each arm [using the full analysis set (FAS) population set defined below] that have a recurrence of CDI through week 12 following clinical cure of the initial episode. Recurrence is defined as 3 or more watery stools in 24 hours associated with a positive local or central lab test for toxigenic *C difficile*.

Secondary endpoints include recurrence in particular clinically relevant subgroups as well as the proportion of subjects in each arm with a Global Cure – i.e. cure of the initial CDI episode and no recurrence through week 12.

Exploratory endpoints include time to CDI recurrence and proportion of subjects with initial clinical cure. Initial clinical cure is defined as the proportion of subjects in each arm who have taken ≤ 14 days of SOC and have ≤ 2 loose stools in 24 hours for 2 consecutive days after completing SOC. Other exploratory endpoints included time to initial clinical cure, proportion of subjects with diarrhea recurrence (regardless of whether there is an associated positive toxin test for *C. difficile*), and the proportion of subjects who had WBC ≤ 10,000 on Days 4 and 11 and temperature < 101.0° F by Day 4 and 11]

It is important to note that from the Agency perspective, recurrence was not considered the only relevant endpoint for both the P001 and P002 trials; cure of the initial episode was considered just as important especially as many subjects were getting the study drug early on in their SOC course. The sponsor's focus on recurrence as the primary endpoint does not account for potential trial outcomes in which a drug causes a decrease in initial cures while providing protection against recurrence in those subjects who are initial cures.

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P002

The study endpoints for p002 are essentially the same as those of P001 with the same issues with interpretation.

Statistical Analysis Plan

P001

The primary efficacy population was the Full Analysis Set population (FAS). This included subjects who had study drug infusion, had any post-randomization data subsequent to infusion, and had a positive toxigenic stool test for *C. difficile*. By the time of the final protocol amendment the post-randomization data criterion had been removed and another criterion added-failure to receive protocol defined SOC therapy within a 1 day window of infusion. A supportive analysis for the per-protocol population was planned but this population was not prespecified (in terms of what constituted major protocol violations) in the initial protocol. All safety analyses were performed on the Safety Population which included all subjects according to their actual, rather than planned, treatment.

P002

The analysis populations were similar.

Protocol Amendments

P001

There were three amendments to the P001 protocol; only changes of interest are discussed.

The first protocol amendment clarified the statistical requirements for dropping an arm during the interim analysis and made modifications to control multiplicity. It allowed for some changes to SOC such as switch in SOC in cases of an adverse event or in cases where a subject is switching from IV flagyl/oral vancomycin to oral vancomycin alone.

The second amendment added fidaxomicin as a possible SOC, reduced infusion time from 2 hours to 1 hour, and allowed the FAS to include those with no post infusion data points.

The third amendment allowed an IV filter size increase to 5 microns and allowed for a longer duration of SOC (up to 16 calendar days, and even longer in particular scenarios such as under dosing/lack of efficacy of SOC pre-infusion).

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P002

Amendment changes in P002 were not radically different/consequential from those of the P001 protocol amendments (some P001 amendment changes were already incorporated into the original P002 protocol).

Data Quality and Integrity: Sponsor's Assurance

For both studies, investigators were to follow GCPs and agree to financial disclosures as part of participating in the study. The following sponsor language was taken from the Clinical Study report for study P002 and provides a picture of steps taken by the sponsor to ensure data quality.

"Clinical studies conducted worldwide by the Sponsor are subject to Quality Control (QC) and Quality Assurance (QA) oversight, including independent audits, as prescribed by company and departmental Standard Operating Procedures.

Quality Control & Quality Assurance activities are intrinsic to all clinical trial-related activities. Such activities include, but are not limited to, on-site monitoring inclusive of source data verification, medical monitoring of clinical trial data and resultant databases and quality reviews of regulatory submission documents.

The Clinical Quality Management (CQM) group implements QC Plans to assess the state of compliance (real-time measurements) and to identify any potential process gaps (risk-based selection of processes). CQM analyzes the data from these QC Plans together with the data from internal compliance audits and Health Authority inspections. The goal is to identify any emerging risks / trends of non-compliance to support continuous process improvement as well as to prevent the recurrence of any identified quality issues. An independent compliance oversight function exists within the Sponsor's MRL Compliance organization. The Global Clinical & Pharmacovigilance Compliance (GC&PVC) group is part of Merck Compliance. This group independently assesses compliance through a comprehensive, risk-based audit program to ensure compliance with applicable GCP, Good Pharmacovigilance Practices (GPvP) regulations and applicable company policies and procedures.

The Merck Code of Conduct within the protocol describes how the trial was monitored to ensure compliance with GCP. All authors reviewed this CSR for accuracy of scientific content. Independent Ethics Committees (IECs) reviewed and approved the protocol and applicable amendments. All Independent Ethics Committees (IECs), also referred to as an Ethical Review Committees (ERCs) or Institutional Review Boards (IRBs) used for this trial meet the following definition of an "IEC" consistent with the definition outlined in the Food and Drug Administration Code of Federal Regulations (CFR) Title 21, Part 312.3; All IECs used for this trial were adequately constituted in accordance with local regulations to provide assurance of human subject/patient protection.

The trial had several investigator meetings to review all protocol procedures and investigator responsibilities under Good Clinical Practices (GCP). At the meetings, the conduct of the trial was explained and instructions were provided to ensure accuracy and consistency in

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data collection. Investigators understood that by signing the Protocol Investigator Signature Page (and, if applicable, any country-specific trial agreements), they provided their commitment to comply with applicable GCP regulations and guidances and to conduct the trial in accordance with the protocol.

This trial was conducted in conformance with GCP requirements and applicable country and/or local statutes and regulations regarding ethical committee review, informed consent, and the protection of human subjects participating in biomedical research. The original, generic, Sponsor-approved, written informed consent form (ICF) was obtained prior to any study activities for each subject. For all non-United States, non-Investigational New Drug (IND) sites, informed consent at all trial sites was obtained in accordance with the principles and provisions in Section 4.8 of the International Conference of Harmonisation (ICH)-E6 Guideline for GCP. Local country and/or cultural consent practices and/or requirements were followed where applicable. For all United States-IND sites, informed consent was obtained in accordance with United States CFR, Title 21, Part 50, Protection of Human Subjects.

The generic Sponsor-approved written ICF document(s) were developed in accordance with the ICH E6 4.8.10 and include a description of the type of incentives, if any, that were provided to trial subjects. The specific amount and schedule of payments were individually determined by each investigative site and approved by the overseeing IEC in accordance with local country regulations. The details of such payments and incentives are available upon request."

It should be noted that OSI found no major deficiencies in its review of clinical sites, the sponsor, etc.

6.1.2. Study Results

Compliance with Good Clinical Practices

Please see above.

Financial Disclosure

There were two investigators/sub-investigators at (b) (4) that could not be certified by Merck as having no financial conflicts despite due diligence: Site (b) (4) investigators (b) (6) (b) (6). However, these individuals were certified as having no interests when working on the P001 trial. For the P001 and P002 study, 3 individuals were noted to have financial interests as seen in the sponsor table below:

Table 2: Table of All Clinical Investigators/Subinvestigators in Studies P001 and P002 Who Hold Financial Interests and/or Arrangements Requiring Disclosure

Protocol/Site	Investigator/ Subinvestigator	Financial Interests or Arrangements
(b) (4)	(b) (6)	Significant Payments of Other Sorts: Amount: $75,000.00 $75,000.00 in payments from Merck for research grants, consulting fees and speaking honorarium as reported by the investigator on (b) (4) Amount: less than $25,000.00 in last year as reported by investigator on (b) (4)
(b) (4)	(b) (6)	Equity Interest: Amount: $35,285.00 (b) (6) maintains 948 shares of Merck stock as part of (b) (6) 401k (b) (6) no longer a Merck employee as reported by investigator on (b) (4)
(b) (4)	(b) (6)	Significant Payments of Other Sorts: Amount: $226,000.00 $226,000.00 in payments from Merck between (b) (4) for presentations, chair of sessions and advisory boards as reported by investigator on (b) (4)

-Financial Disclosure Summary_Merck, pg. 44

Together these 3 sites only account for 45 subjects in study P001 (43 [3.1%] subjects in the FAS) so it is unlikely that the findings will be changed by their removal from the analysis. Similarly, only 29 subjects in Study P002 were involved (27 [2.3%] subjects in the FAS).

Patient Disposition

P001

Screen failures occurred in only 5% of screened subjects (74 of 1526 subjects). There were 1452 randomized subjects of which only 40 (2.8%) did not receive study medication; 50 subjects total were discontinued from study medication. All four arms had essentially equivalent numbers of randomized subjects except for the MK-3415 arm. This arm had substantially less subjects because of an interim analysis, after which the DMC recommended stopping further randomization into this arm because of safety concerns. All four arms had a study discontinuation rate around 15%. The primary reason for study discontinuation was death followed by loss to follow up and withdrawal of consent. Of those who received study medication, the study discontinuation rate in the four arms was between 12 and 14%, with the lowest rate of discontinuation in the MK-3415A arm (11.6%). Again, the primary reasons for discontinuation were death, loss to follow up, and withdrawal of consent. Relative to the other 3 arms, death accounted for a higher percentage of discontinuations in the MK-3415 arm. Only 10 subjects (.08%) in the All treated population did not receive a full dose of study medication (8 subjects had technical problems, and 2 subjects had an AE; only one such subjects was in the MK-6072 arm). Close to half of subjects who discontinued from the FAS, did so in the first four

weeks. Please note the following two sponsor tables

Table 3: Study P001: Disposition of Subjects: All Randomized Subjects

	MK-3415A		MK-3415		MK-6072		Placebo		Total	
	n	(%)	n	(%)	n	(%)	n	(%)	n	(%)
Subjects in population	403		242		403		404		1,452	
Study Disposition										
Completed	343	(85.1)	201	(83.1)	340	(84.4)	340	(84.2)	1,224	(84.3)
Discontinued	60	(14.9)	41	(16.9)	63	(15.6)	64	(15.8)	228	(15.7)
Adverse Event	0	(0.0)	1	(0.4)	1	(0.2)	0	(0.0)	2	(0.1)
Death	20	(5.0)	26	(10.7)	30	(7.4)	25	(6.2)	101	(7.0)
Lack Of Efficacy	0	(0.0)	1	(0.4)	0	(0.0)	0	(0.0)	1	(0.1)
Lost To Follow-Up	15	(3.7)	2	(0.8)	11	(2.7)	16	(4.0)	44	(3.0)
Physician Decision	2	(0.5)	2	(0.8)	4	(1.0)	3	(0.7)	11	(0.8)
Progressive Disease	1	(0.2)	0	(0.0)	0	(0.0)	2	(0.5)	3	(0.2)
Protocol Violation	6	(1.5)	0	(0.0)	2	(0.5)	1	(0.2)	9	(0.6)
Subject Withdrew Consent	14	(3.5)	7	(2.9)	15	(3.7)	15	(3.7)	51	(3.5)
Technical Problems	2	(0.5)	2	(0.8)	0	(0.0)	2	(0.5)	6	(0.4)
Study Medication Disposition										
Completed	385	(95.5)	229	(94.6)	391	(97.0)	397	(98.3)	1,402	(96.6)
Discontinued	3	(0.7)	6	(2.5)	1	(0.2)	0	(0.0)	10	(0.7)
Adverse Event	0	(0.0)	1	(0.4)	1	(0.2)	0	(0.0)	2	(0.1)
Technical Problems	3	(0.7)	5	(2.1)	0	(0.0)	0	(0.0)	8	(0.6)
Did Not Take Study Medication	15	(3.7)	7	(2.9)	11	(2.7)	7	(1.7)	40	(2.8)

Each subject is counted once for Study Disposition and once for Study Medication Disposition.
MK-3415A = actoxumab + bezlotoxumab, MK-3415 = actoxumab alone, MK-6072 = bezlotoxumab alone

-CSR P001, pg. 106; Table 10-1

Table 4: Study P001: Disposition of Subjects: All Treated Subjects

	MK-3415A		MK-3415		MK-6072		Placebo		Total	
	n	(%)	n	(%)	n	(%)	n	(%)	n	(%)
Subjects in population	388		235		392		397		1,412	
Study Disposition										
Completed	343	(88.4)	201	(85.5)	340	(86.7)	340	(85.6)	1,224	(86.7)
Discontinued	45	(11.6)	34	(14.5)	52	(13.3)	57	(14.4)	188	(13.3)
Adverse Event	0	(0.0)	1	(0.4)	0	(0.0)	0	(0.0)	1	(0.1)
Death	19	(4.9)	26	(11.1)	30	(7.7)	25	(6.3)	100	(7.1)
Lack Of Efficacy	0	(0.0)	1	(0.4)	0	(0.0)	0	(0.0)	1	(0.1)
Lost To Follow-Up	15	(3.9)	2	(0.9)	11	(2.8)	16	(4.0)	44	(3.1)
Physician Decision	2	(0.5)	1	(0.4)	1	(0.3)	2	(0.5)	6	(0.4)
Progressive Disease	1	(0.3)	0	(0.0)	0	(0.0)	2	(0.5)	3	(0.2)
Protocol Violation	1	(0.3)	0	(0.0)	0	(0.0)	1	(0.3)	2	(0.1)
Subject Withdrew Consent	7	(1.8)	3	(1.3)	10	(2.6)	11	(2.8)	31	(2.2)
Study Medication Disposition										
Completed	385	(99.2)	229	(97.4)	391	(99.7)	397	(100.0)	1,402	(99.3)
Discontinued	3	(0.8)	6	(2.6)	1	(0.3)	0	(0.0)	10	(0.7)
Adverse Event	0	(0.0)	1	(0.4)	1	(0.3)	0	(0.0)	2	(0.1)
Technical Problems	3	(0.8)	5	(2.1)	0	(0.0)	0	(0.0)	8	(0.6)

Each subject is counted once for Study Disposition and once for Study Medication Disposition.
MK-3415A = actoxumab + bezlotoxumab, MK-3415 = actoxumab alone, MK-6072 = bezlotoxumab alone

CSR P001, pg. 108; Table 10-2

P002

1270 subjects were screened and 67 were screen failures, the vast majority of which were due to a lack of a clear CDI diagnosis, lack/withdrawal of consent, or a subject having a condition that would jeopardize his/her safety and/or confound the interpretation of results.

A total of 1203 subjects were randomized into one of three treatment groups: 397 (33.0%) to the MK-3415A group, 407 (33.8%) to the MK-6072 group, and 399 (33.2%) to the placebo group. Among the randomized subjects, 37 (3.1%) did not complete the study medication, 35 of which did not receive the trial medication. The most common reasons a subject did not receive trial medications were withdrawal of consent (n=23 out of 35, 65.7%).
In the All Treated population, there was a slightly higher study completion rate in the MK-6072 arm. In all three arms, the most common reasons for study discontinuation were death, withdrawal by subject, and loss to follow up.

Overall, 197 (16.9%) of the 1163 subjects included in the FAS population were prematurely discontinued from the study. Of these 197 subjects, approximately half (103 subjects, 8.8%

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overall) were discontinued during the first 4 weeks of the follow-up period. Please note the sponsor tables below.

Table 5: Study P002: Disposition of Subjects: All Randomized Subjects

	MK-3415A n (%)	MK-6072 n (%)	Placebo n (%)	Total n (%)
Subjects in population	397	407	399	1203
Main Study Disposition				
Completed	322 (81.1)	337 (82.8)	311 (77.9)	970 (80.6)
Discontinued	75 (18.9)	70 (17.2)	88 (22.1)	233 (19.4)
Adverse Event	1 (0.3)	1 (0.2)	2 (0.5)	4 (0.3)
Death	29 (7.3)	22 (5.4)	32 (8.0)	83 (6.9)
Lost To Follow-Up	11 (2.8)	10 (2.5)	6 (1.5)	27 (2.2)
Physician Decision	4 (1.0)	4 (1.0)	4 (1.0)	12 (1.0)
Protocol Violation	2 (0.5)	2 (0.5)	2 (0.5)	6 (0.5)
Technical Problems	1 (0.3)	2 (0.5)	0 (0.0)	3 (0.2)
Withdrawal By Subject	27 (6.8)	29 (7.1)	42 (10.5)	98 (8.1)
Study Medication Disposition				
Completed Study Medication	390 (98.2)	395 (97.1)	381 (95.5)	1166 (96.9)
Discontinued Study Medication	1 (0.3)	1 (0.2)	0 (0.0)	2 (0.2)
Technical Problems	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.1)
Withdrawal By Subject	0 (0.0)	1 (0.2)	0 (0.0)	1 (0.1)
Did Not Take Study Medication	6 (1.5)	11 (2.7)	18 (4.5)	35 (2.9)
Each subject is counted once for Trial Disposition and once for Subject Study Medication Disposition MK-3415A = actoxumab + bezlotoxumab, MK-6072 = bezlotoxumab alone				

CSR P002, pg. 116; Table 10-1

Table 6: Study P002: Disposition of Subjects: All Treated Subjects

	MK-3415A n (%)	MK-6072 n (%)	Placebo n (%)	Total n (%)
Subjects in population	391	396	381	1168
Main Study Disposition				
Completed	322 (82.4)	337 (85.1)	311 (81.6)	970 (83.0)
Discontinued	69 (17.6)	59 (14.9)	70 (18.4)	198 (17.0)
Adverse Event	1 (0.3)	1 (0.3)	2 (0.5)	4 (0.3)
Death	29 (7.4)	22 (5.6)	32 (8.4)	83 (7.1)
Lost To Follow-Up	11 (2.8)	10 (2.5)	6 (1.6)	27 (2.3)
Physician Decision	3 (0.8)	2 (0.5)	1 (0.3)	6 (0.5)
Protocol Violation	2 (0.5)	0 (0.0)	1 (0.3)	3 (0.3)
Withdrawal By Subject	23 (5.9)	24 (6.1)	28 (7.3)	75 (6.4)
Study Medication Disposition				
Completed Study Medication	390 (99.7)	395 (99.7)	381 (100.0)	1166 (99.8)
Discontinued Study Medication	1 (0.3)	1 (0.3)	0 (0.0)	2 (0.2)
Technical Problems	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.1)
Withdrawal By Subject	0 (0.0)	1 (0.3)	0 (0.0)	1 (0.1)
Each subject is counted once for Trial Disposition and once for Subject Study Medication Disposition MK-3415A = actoxumab + bezlotoxumab, MK-6072 = bezlotoxumab alone				

CSR P001, pg. 118; Table 10-2

Protocol Violations/Deviations

P001

In this trial, a major protocol deviation was defined as "any protocol deviation which may significantly/adversely impact the completeness, accuracy, and/or reliability of the trial data or that may significantly/adversely affect a subject's rights, safety, or well-being." Overall, 45% of subjects had a major protocol violation; the sponsor states that roughly 20% of the treated subjects (277) had violations that could substantially affect primary efficacy endpoints. 13 of these subjects were found to not have had a major protocol violation after database unlocking but were kept out of the per protocol analysis regardless. In the sponsor table below outlining major protocol violations, it is difficult to say with certainty what may/may not have impacted efficacy assessments. However, there was generally little disparity between the MK-6072 arm and placebo arm for the individual violations except for "site personnel or sponsor partial or fully unblinded" and use of a "prohibited concomitant medication," both of which were slightly higher in the placebo arm. These two categories are likely to have opposing effects on the efficacy assessment.

Table 7: Study P001 Summary of Major Protocol Violations, All Treated Subjects

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	MK-3415A	MK-3415	MK-6072	Placebo	Total
Subjects in population	388	235	392	397	1412
Subjects with any protocol deviation	**175/388 (45.1)**	**117/235 (49.8)**	**170/392 (43.4)**	**169/397 (42.6)**	**631/1412 (44.7)**
Clinical supplies	**7/388 (1.8)**	**23/235 (9.8)**	**9/392 (2.3)**	**24/397 (6)**	**63/1412 (4.5)**
Improper reconstitution of study medication	2/388 (0.5)	0/235 (0)	0/392 (0)	0/397 (0)	2/1412 (0.1)
Improper storage of study medication	0/388 (0)	5/235 (2.1)	2/392 (0.5)	0/397 (0)	7/1412 (0.5)
Received less than 75% of intended dose	2/388 (0.5)	6/235 (2.6)	1/392 (0.3)	0/397 (0)	9/1412 (0.6)
Site personnel or Sponsor partially or fully unblinded	2/388 (0.5)	12/235 (5.1)	4/392 (1)	24/397 (6)	42/1412 (3.0)
Wrong study treatment administered	1/388 (0.3)	0/235 (0)	2/392 (0.5)	0/397 (0)	3/1412 (0.2)
Efficacy Assessment	**104/388 (26.8)**	**57/235 (24.3)**	**107/392 (27.3)**	**95/397 (23.9)**	**363/1412 (25.7)**
Missed 2 consecutive scheduled visits	6/388 (1.5)	1/235 (0.4)	3/392 (0.8)	7/397 (1.8)	17/1412 (1.2)
Missing daily temperatures, baseline central laboratory test for stool sample, or follow-up phone calls	52/388 (13.4)	30/235 (12.8)	57/392 (14.5)	47/397 (11.8)	186/1412 (13.2)
Missing stool tests associated with return of diarrhea	30/388 (7.7)	13/235 (5.5)	29/392 (7.4)	25/397 (6.3)	97/1412 (6.9)
SoC > 16 days without continued symptoms	10/388 (2.6)	4/235 (1.7)	13/392 (3.3)	7/397 (1.8)	34/1412 (2.4)
Substantial number of post-baseline loose stool counts missing	18/388 (4.6)	11/235 (4.7)	13/392 (3.3)	22/397 (5.5)	64/1412 (4.5)
Entry Criteria	**33/388 (8.5)**	**32/235 (13.6)**	**32/392 (8.2)**	**22/397 (5.5)**	**119/1412 (8.4)**
Baseline toxigenic *C. difficile* test out of window	10/388 (2.6)	5/235 (2.1)	8/392 (2)	2/397 (0.5)	25/1412 (1.8)
Diarrhea criteria not met for baseline CDI episode	0/388 (0)	0/235 (0)	0/392 (0)	1/397 (0.3)	1/1412 (0.1)
Did not have positive local laboratory test for toxigenic *C. difficile*	1/388 (0.3)	0/235 (0)	0/392 (0)	0/397 (0)	1/1412 (0.1)
Did not receive protocol defined SoC therapy within 1 day window of the infusion	1/388 (0.3)	0/235 (0)	0/392 (0)	0/397 (0)	1/1412 (0.1)
Duration of SoC therapy less than 10 days	14/388 (3.6)	16/235 (6.8)	15/392 (3.8)	9/397 (2.3)	54/1412 (3.8)
Eligibility criteria not met	2/388 (0.5)	0/235 (0)	1/392 (0.3)	0/397 (0)	3/1412 (0.2)

	MK-3415A	MK-3415	MK-6072	Placebo	Total
Subjects in population	388	235	392	397	1412
Prohibited prior medication	1/388 (0.3)	1/235 (0.4)	1/392 (0.3)	1/397 (0.3)	4/1412 (0.3)
SoC initially prescribed for > 14 days	0/388 (0)	0/235 (0)	0/392 (0)	2/397 (0.5)	2/1412 (0.1)
SoC route or combination not per protocol	5/388 (1.3)	12/235 (5.1)	9/392 (2.3)	8/397 (2)	34/1412 (2.4)
GCP NonCompliance	**3/388 (0.8)**	**3/235 (1.3)**	**6/392 (1.5)**	**2/397 (0.5)**	**14/1412 (1.0)**
GCP NonCompliance	3/388 (0.8)	3/235 (1.3)	6/392 (1.5)	2/397 (0.5)	14/1412 (1.0)
Informed Consent	**26/388 (6.7)**	**12/235 (5.1)**	**22/392 (5.6)**	**18/397 (4.5)**	**78/1412 (5.5)**
Did not give appropriate informed consent	26/388 (6.7)	12/235 (5.1)	22/392 (5.6)	18/397 (4.5)	78/1412 (5.5)
Prohibited medication	**31/388 (8)**	**23/235 (9.8)**	**27/392 (6.9)**	**43/397 (10.8)**	**124/1412 (8.8)**
Prohibited concomitant medication	31/388 (8)	23/235 (9.8)	27/392 (6.9)	43/397 (10.8)	124/1412 (8.8)
Safety Assessment	**19/388 (4.9)**	**12/235 (5.1)**	**26/392 (6.6)**	**26/397 (6.5)**	**83/1412 (5.9)**
Missing Day 1 ECGs, safety laboratory tests or vital sign assessments	11/388 (2.8)	8/235 (3.4)	15/392 (3.8)	17/397 (4.3)	51/1412 (3.6)
SAE/AE not reported in timely fashion	8/388 (2.1)	4/235 (1.7)	12/392 (3.1)	9/397 (2.3)	33/1412 (2.3)

CSR, Study P001, pg. 126, Table 10-6

40 subjects were partially unblinded. In these cases, investigators/personnel knew that subjects were getting active treatment generally (because of overdose reporting requirements) or knew that subjects were not getting MK-3415A (through forced randomization In order to manage clinical supplies); they did not know exact treatment assignment. 4 subjects were fully unblinded.

P002

Major protocol deviations were defined similarly in this trial as in P001. 37% of subjects who received study drug had such violations. As noted in the sponsor table below, the MK-6072 and

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placebo arms were fairly matched with regards to rates of major protocol violations overall and with regards to particular major protocol violations. There was only a slight increase in the rate of subjects in the placebo arm of "site personnel or sponsor partially or fully unblinded." Please see the sponsor table below.

Table 8: Study P001 Summary of Major Protocol Violations, All Treated Subjects

	MK-3415A	MK-6072	Placebo	Total
Subjects in population	391	396	381	1168
Subjects with any protocol deviation	**131/391 (33.5)**	**156/396 (39.4)**	**146/381 (38.3)**	**433/1168 (37.1)**
Clinical Supplies	**11/391 (2.8)**	**14/396 (3.5)**	**23/381 (6.0)**	**48/1168 (4.1)**
Improper reconstitution of study medication	1/391 (0.3)	1/396 (0.3)	0/381 (0.0)	2/1168 (0.2)
Improper storage of study medication	4/391 (1.0)	6/396 (1.5)	1/381 (0.3)	11/1168 (0.9)
Received less than 75% of intended dose	0/391 (0.0)	1/396 (0.3)	0/381 (0.0)	1/1168 (0.1)
Site personnel or Sponsor partially or fully unblinded	5/391 (1.3)	6/396 (1.5)	22/381 (5.8)	33/1168 (2.8)
Wrong study treatment administered	2/391 (0.5)	1/396 (0.3)	1/381 (0.3)	4/1168 (0.3)
Efficacy Assessment	**90/391 (23.0)**	**92/396 (23.2)**	**73/381 (19.2)**	**255/1168 (21.8)**
Duration of SoC therapy less than 10 days	10/391 (2.6)	13/396 (3.3)	10/381 (2.6)	33/1168 (2.8)
Missed 2 consecutive scheduled visits	6/391 (1.5)	7/396 (1.8)	6/381 (1.6)	19/1168 (1.6)
Missing daily temperatures, baseline central laboratory test for stool sample, or follow-up phone calls	30/391 (7.7)	39/396 (9.8)	34/381 (8.9)	103/1168 (8.8)
Missing stool tests associated with return of diarrhea	26/391 (6.6)	26/396 (6.6)	14/381 (3.7)	66/1168 (5.7)
Standard of care (SoC) greater than 16 days without	12/391 (3.1)	7/396 (1.8)	9/381 (2.4)	28/1168 (2.4)
Substantial number of post-baseline loose stool counts	20/391 (5.1)	16/396 (4.0)	11/381 (2.9)	47/1168 (4.0)
Entry Criteria	**19/391 (4.9)**	**12/396 (3.0)**	**16/381 (4.2)**	**47/1168 (4.0)**
Baseline toxigenic *C. difficile* test out of window	6/391 (1.5)	3/396 (0.8)	2/381 (0.5)	11/1168 (0.9)
Diarrhea criteria not met for baseline CDI episode	1/391 (0.3)	1/396 (0.3)	0/381 (0.0)	2/1168 (0.2)
Did not have positive local laboratory test for toxigenic C. difficile	0/391 (0.0)	1/396 (0.3)	1/381 (0.3)	2/1168 (0.2)
Did not receive protocol defined SoC therapy within 1 day window of the study therapy infusion	1/391 (0.3)	0/396 (0.0)	2/381 (0.5)	3/1168 (0.3)
Eligibility criteria not met	2/391 (0.5)	0/396 (0.0)	0/381 (0.0)	2/1168 (0.2)
Prohibited prior medication	1/391 (0.3)	1/396 (0.3)	1/381 (0.3)	3/1168 (0.3)
SoC therapy route or combination not per protocol	6/391 (1.5)	6/396 (1.5)	12/381 (3.1)	24/1168 (2.1)
Unapproved stool test used to confirm eligibility	2/391 (0.5)	0/396 (0.0)	0/381 (0.0)	2/1168 (0.2)
	MK-3415A	**MK-6072**	**Placebo**	**Total**
Subjects in population	391	396	381	1168
Informed Consent	**17/391 (4.3)**	**28/396 (7.1)**	**28/381 (7.3)**	**73/1168 (6.3)**
Did not give appropriate informed consent	17/391 (4.3)	28/396 (7.1)	28/381 (7.3)	73/1168 (6.3)
Prohibited Medication	**17/391 (4.3)**	**31/396 (7.8)**	**27/381 (7.1)**	**75/1168 (6.4)**
Prohibited concomitant medication	17/391 (4.3)	31/396 (7.8)	27/381 (7.1)	75/1168 (6.4)
Safety Assessment	**18/391 (4.6)**	**17/396 (4.3)**	**22/381 (5.8)**	**57/1168 (4.9)**
Missing Day 1 ECGs, safety laboratory tests or vital sign assessments	10/391 (2.6)	12/396 (3.0)	12/381 (3.1)	34/1168 (2.9)
SAE/AE not reported in timely fashion	8/391 (2.0)	7/396 (1.8)	10/381 (2.6)	25/1168 (2.1)

CSR, Study P002, pg. 133, Table 10-6

One subject was fully unblinded (but only at Day 85 and at request of subject) and 32 partially unblinded. Partially unblinded cases occurred in cases of drug overdose (in which case a site knew the subject got an active treatment but not which specific active treatment; 8 subjects) or in cases when drug quantity wasn't available and forced randomization occurred (in which case site knew subject didn't get MK-3415A).

Demographic Characteristics

P001: Because the MK-3415 arm was stopped after the interim analysis, the discussion of

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demographics in this study will focus on the placebo, MK-6072, and MK-3415A arms of this study (primarily the former two arms as sponsor is seeking approval for MK-6072). Discussions revolve around the FAS population as this was the population used for the primary efficacy analysis.

All three arms had a slight preponderance of females with no real difference in proportions among the arms. All three arms had around half of their total population composed of subjects ≥ 65 years old. Slightly more than a quarter of each arm's population was composed of subjects ≥ 75 years old. In all three arms, the populations were overwhelmingly white, with roughly 90% of each arm coming from this subgroup. Around half of the population in each arm were enrolled in sites in North America while another third were enrolled from sites in Europe. A much smaller percentage were enrolled from sites in Latin America and Asia. Around 45% of each arm's population came from U.S. sites.

As expected all three arms had their study populations roughly split evenly between metronidazole and oral vancomycin for their SOC antimicrobial therapy. Just over a quarter of each arm's study population had a C. diff. episode in the prior 6 months. In terms of severity of baseline disease, similar numbers in each arm had a Zar score ≥2 and infection by a hypervirulent strain (defined as ribotypes 7, 78, or 244), roughly 16% and 12% in each arm, respectively. Similarly, each arm had around 28% of its subjects infected with an epidemic ribotype (defined as ribotypes 27, 14, 1, 2, 106, and 20).

In terms of baseline health, just over a fifth of each study population was defined as immunocompromomised (which was defined as a subject having an active hematological malignancy, using an antineoplastic or immunomodulating agent, using corticosteroids, having received a prior solid organ transplant, being asplenic, being neutropenic/pancytopenic, or having AIDS/immunodeficient condition. Similarly, just over a quarter of each arm's study population had a Modified Horn's Index defined as Major or Extreme. This definition meant that a qualified physician had estimated a subject's overall condition at the time of study entry as "major complications or multiple conditions requiring treatment" or "catastrophic illness likely leading to death." Interestingly, on the day of study drug infusion, over 70% of subjects in each arm had ≤ two loose stools that day. A quarter of subjects in each arm had been on SOC for > 4 days by the time they received their study drug infusion.

Overall, in P001, the arms were evenly matched (there was only a slightly higher proportion of subjects in the placebo arm that had disease/symptom onset > 4 days prior to the study drug infusion). Moreover, the demographic was moderately reflective of a population that would be prone to CDI recurrences (though there is no clear consensus on what such a population is) as there were significant numbers of elderly subjects and modest numbers of subjects with a recent history of CDI, severe/hypervirulent CDI at baseline, and with significant underlying illness. However, it did not appear that subjects were particularly symptomatic at baseline, and many of them had already been on SOC for a significant amount of time at the time of study drug

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infusion.

See Table 9 for further details.

P002

In this study, the proportion of females was higher in each arm. The placebo arm was 60% female while the other two arms had female proportions of 54%. In the placebo and MK-6072 arms, half of the subjects were ≥ 65 years old and close to 30% were ≥75 years old. However, in the MK-3415A arm these proportions increased to roughly 60% and 40%, respectively. Roughly 80% of subjects were White in each arm and roughly 16% of subjects in each arm were Asian. African American enrollment in each arm hovered around 4%. Roughly 40% of subjects in each arm were enrolled in North American sites, and another 40% were enrolled in European sites. 15% of subjects in each arm came from the Asia-Pacific. Overall, two thirds of subjects came from Ex-US sites.

Slightly fewer than 50% of subjects in each arm were stratified to metronidazole and similar proportions were noted for oral vancomycin. More hypervirulent strains were noted in the placebo arm relative to the other two arms (19% in the placebo arm vs. 13 and 12% in the other two arms). Around 30% of subjects in each arm had had an episode of *C. diff*. in the past 6 months; a quarter of subjects in each arm had had an episode in the past 3 months. Around 15% of subjects in each arm had a Zar score≥ 2 (slightly higher in the combination arm).

Slightly fewer subjects in the placebo arm were noted to be immunocompromised and around 30% of subjects in each arm were found to have a Horn's Index of Major or Extreme. Over three fourths of subjects in each arm (82% in the case of MK-6072) had ≤ 2 loose stools on the day of receiving study drug. Roughly 30% of subjects in each arm had received > 4 days of SOC by the time they received study drug.

Overall, in P002 the arms were generally evenly matched though there were some slight differences between the study arms as noted above. As with P001, the study population was moderately reflective of subjects who would be expected to have CDI recurrences. There were significant numbers of elderly subjects and modest numbers of subjects with a recent history of CDI, severe/hypervirulent CDI at baseline, and with significant underlying illness. However, it did not appear that subjects were particularly symptomatic at baseline, and many of them had already been on SOC for a significant amount of time at the time of study drug infusion. See Table 10 for further details.

Comparisons between study P001 and P002

There were some differences of note between the FAS study populations of P001 and P002. Subjects in P002 were slightly older relative to P001, had a higher proportion of Asian subjects,

a higher proportion of Ex-US subjects, and had less subjects > 70 kg. In P002, slightly more subjects in each arm had received > 4 days of SOC at the time of study drug infusion.

In P002, fewer placebo subjects were immunocompromised relative to P001, but a higher proportion of subjects had hypervirulent/epidemic strains. Also in P002, more subjects in the MK-6072 had ≤ 2 loose stools on the day of study drug infusion,

It should be noted that for some of the measured parameters, such as presence of hypervirulent/epidemic strains and recent health care utilization, there were significant portions of missing data making these baseline estimates somewhat unreliable. In other cases, such as when assessing immunocompromised states at baseline, it is unclear how much unknown data there was, again making baseline estimates somewhat unreliable.

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Table 9: Study P001: Demographics by Study Arm (FAS)

Demographic Parameters	Placebo Group (N=395) n (%)	Treatment Group		
		MK-6072 (N=386) n (%)	MK-3415A (N=383) n (%)	MK-3415 (N=232)
Sex				
Male	172 (43.5%)	157 (40.7%)	172 (44.9%)	102 (44%)
Female	223 (56.5%)	229 (59.3%)	211 (55.1%)	130 (56%)
Age				
Mean years (SD)	**62**	**60**	**62**	**64**
Min, max (years)	**19,97**	**19,100**	**18,95**	**21,99**
Age Group				
≥ 65 years	**199 (50.4%)**	**185 (47.9%)**	**200 (52.2%)**	**122 (52.6%)**
≥ 75 years	**119 (30.1%)**	**99 (25.6%)**	**111 (29%)**	**68 (29.3%)**
Race				
White	**366 (92.7%)**	**338 (87.6%)**	**350 (91.4%)**	**210 (90.5%)**
Black or African American	**18 (4.6%)**	**28 (7.2%)**	**17 (4.4%)**	**16 (6.9%)**
Asian	**1 (0.2%)**	**4 (1%)**	**4 (1%)**	**3 (1.3%)**
American Indian or Alaska Native	**3 (0.8%)**	**3 (0.8%)**	**2 (0.5%)**	**0**
Multiple	**7 (1.8%)**	**13 (3.4%)**	**10 (2.6%)**	**3 (1.3%)**
Ethnicity				
Hispanic or Latino	**56 (14.2%)**	**46 (11.9%)**	**52 (13.6%)**	**23 (9.9%)**
Not Hispanic or Latino				
Region				
North America	**208 (52.7%)**	**196 (50.8%)**	**204 (53.3%)**	**132 (56.9%)**
Latin America	**30 (7.6%)**	**26 (6.7%)**	**29 (7.6%)**	**9 (3.9%)**

Europe	**132 (33.4%)**	**139 (36%)**	**131 (34.2%)**	**80 (34.5%)**
Asia-Pacific	**23 (5.8%)**	**20 (5.2%)**	**17 (4.4%)**	**10 (4.3%)**
Africa	**2 (0.5%)**	**5 (1.3%)**	**2 (0.5%)**	**1 (0.4%)**
U.S.	**186 (47.1%)**	**171 (44.3%)**	**176 (46%)**	**120 (51.7%)**
Ex U.S.	**209 (52.9%)**	**215 (55.7%)**	**207 (54%)**	**112 (48.3%)**
Miscellaneous				
Hx. of CDI in Prior 6 months	**109 (27.6%)**	**103 (26.7%)**	**96 (25.1%)**	**69 (29.7%)**
Hx. of CDI in Prior 3 months	**94 (23.8%)**	**87 (22.5%)**	**84 (21.9%)**	**56 (24.1%)**
Zar Score ≥2	**60 (15.2%)**	**67 (17.4%)**	**62 (16.2%)**	**31 (13.4%)**
Recent Health Care Utilization*	**280 (70.9%)**	**273 (70.7%)**	**272 (71%)**	**169 (72.8%)**
Compromised Immunity	**92 (23.3%)**	**87 (22.5%)**	**78 (20.4%)**	**55 (23.7%)**
Stool Count (loose) <= 2 on Day 1	**292 (73.9 %)**	**281 (72.8%)**	**269 (70.2%)**	**175 (75.4%)**
Vancomycin	**189(47.8%)**	**182 (47.2%)**	**182 (47.5%)**	**113 (48.7%)**
Metronidazole	**192 (48.6%)**	**190 (49.2%)**	**189 (49.4%)**	**112 (48.3%)**
Hypervirulent Strains	**44 (11.1%)**	**51(13.2%)**	**44 (11.5%)**	**30 (12.9%)**
Epidemic Strain	**106 (26.8%)**	**108 (28%)**	**106 (27.7%)**	**57 (24.6%)**
>70 kg	**204 (51.6%)**	**209 (54.2%)**	**183(47.8%)**	**109 (47%)**
<=70kg	**191 (48.4%)**	**177 (45.8%)**	**200 (52.2%)**	**123 (53%)**
>100 kg	**37 (9.4%)**	**42 (10.9%)**	**33 (8.6%)**	**33 (14.2%)**
<=100kg	**358 (90.6%)**	**344 (89.1%)**	**350 (91.4%)**	**199 (85.8%)**
Horn's Index: Low or Moderate	**279 (70.6%)**	**283(73.3%)**	**280 (73.1%)**	**160 (69%)**
Horn's Index:	**116 (29.4%)**	**103 (26.7%)**	**103 (26.9%)**	**72**

Major or Extreme				**(31%)**
Pre- Infusion duration of SOC: <=4days	**292 (73.9%)**	**287 (74.4%)**	**285 (74.4%)**	**177 (76.3%)**
Disease Onset: <=4 days prior	**131 (33.2%)**	**150 (38.9%)**	**145 (37.9%)**	**100 (43.1%)**
Disease Onset: 7-8 Days prior	**77 (19.5%)**	**58 (15%)**	**38 (9.9%)**	**42 (18.1%)**
Disease Onset 11+ days prior	**64 (16.2%)**	**60 (15.5%)**	**59 (15.4%)**	**29 (12.5%)**

* For recent health care utilization either in healthcare facility at study onset or discharged in past 3 months

Table 10: Study P002: Demographic Data (FAS)

Demographic Parameters	Placebo Group (N=378) n (%)	**Treatment Group** (N=)	
		MK-6072 (N=395) n (%)	MK-3415A (N=390) n (%)
Sex			
Male	**152 (40.2%)**	**182 (46.1%)**	**178 (45.6%)**
Female	**226 (59.8%)**	**213 (53.9%)**	**212 (54.4%)**
Age			
Mean years (SD)	**64**	**62**	**65**
Median (years)			
Min, max (years)	**18,98**	**18,93**	**19,93**
Age Group			
≥ 65 years	**206 (54.5%)**	**205 (51.9%)**	**241 (61.8%)**
≥ 75 years	**116 (30.7%)**	**114 (28.9%)**	**149 (38.2%)**
Race			
White	**309 (81.8%)**	**311 (78.7%)**	**305 (78.2%)**
Black or African American	**10 (2.6%)**	**17 (4.3%)**	**18 (4.6%)**
Asian	**57 (15.1%)**	**63 (16%)**	**65 (16.7%)**
American Indian or	**1 (0.3%)**	**2 (0.5%)**	**0 (0%)**

Alaska Native			
Multiple	**1 (0.3%)**	**2 (0.5%)**	**2 (0.5%)**
Ethnicity			
Hispanic or Latino	**45 (11.9%)**	**33 (8.4%)**	**33 (8.5%)**
Region (optional)			
North America	**158 (41.8%)**	**158 (40%)**	**158 (40.5%)**
Latin America	**5 (1.3%)**	**4 (1%)**	**8 (2%)**
Europe	**161 (42.6%)**	**174 (44%)**	**161 (41.3%)**
Asia-Pacific	**54 (14.3%)**	**59 (14.9%)**	**63 (16.2%)**
US	**131 (34.7%)**	**136 (34.4%)**	**133 (34.1%)**
Ex-US	**247 (65.3%)**	**259 (65.6%)**	**257 (65.9%)**
Miscellaneous			
Hx. of CDI in Prior 6 months	**110 (29.1%)**	**113 (28.6%)**	**104 (26.7%)**
Hx. of CDI in Prior 3 months	**96 (25.4%)**	**98 (24.8%)**	**94 (24.1%)**
Zar Score ≥2	**65 (17.2%)**	**55 (13.9%)**	**80 (20.5%)**
Recent Health Care Utilization*	**285 (75.4%)**	**305 (77.2%)**	**289 (74.1%)**
Compromised Immunity	**53 (14%)**	**82 (21%)**	**75 (19%)**
Stool Count (loose) <= 2 on Day 1	**281 (74.3 %)**	**324 (82%)**	**289 (74.1%)**
Vancomycin	**184 (48.7%)**	**190 (48.1%)**	**187(48%)**
Metronidazole	**182 (48.2%)**	**189 (47.8%)**	**191 (49%)**
Hypervirulent Strains	**71 (18.8%)**	**51 (12.9%)**	**46 (11.8%)**
Epidemic Strain	**127 (33.6%)**	**102 (25.8%)**	**116 (29.7%)**
>70 kg	**168 (44.4%)**	**178 (45.1%)**	**165 (42.3%)**
<=70kg	**210 (55.6%)**	**217 (54.9%)**	**225 (57.7%)**
>100 kg	**29 (7.7%)**	**30 (7.6%)**	**23 (5.9%)**
<=100kg	**349 (92.3%)**	**365 (92.4%)**	**367 (94.1%)**

Horn's Index: Low or Moderate	**261 (69%)**	**282 (71.4%)**	**280 (71.8%)**
Horn's Index: Major or Extreme	**117 (31%)**	**112 (28.4%)**	**110 (28.2%)**
Prior SOC: <=4days	**269 (71.2%)**	**270 (68.4%)**	**269 (69%)**
Disease Onset: <=4 days prior	**139 (36.8%)**	**145 (36.7%)**	**144 (36.9%)**
Disease Onset: 7-8 Days prior	**61 (16.1%)**	**55 (13.9%)**	**55 (14.1%)**
Disease Onset: 11+ days prior	**51 (13.5%)**	**63 (16%)**	**55 (14.1%)**

*-For recent health care utilization either in healthcare facility at study onset or discharged in past 3 months

Treatment Compliance, Concomitant Medications, and Rescue Medication Use

Since there was only one infusion study drug in all arms, assessment of compliance has little relevance for both studies P001 and P002. There was very little usage of loperamide postinfusion and use of non-c diff. antimicrobials appeared to be roughly matched between the arms (data not shown because these are very crude estimates).

Efficacy Results – Primary Endpoint

As has been discussed earlier, the primary endpoint chosen by the sponsor was the proportion of subjects who had CDI recurrence over the 12 week study period. From the Division perspective, this was not considered the ideal endpoint because it did not take into account effects of study drug on initial cure (potential effect on initial cure is accounted for by Global Cure). Thus, in this section, after discussing recurrence, initial clinical cure and Global Cure will also be discussed as both components are considered primary in understanding the study drug's impact on CDI. Again, the focus will be on comparisons between placebo and MK-6072 as this is the study drug for which the sponsor is seeking approval.

The following table highlights the incidence of CDI recurrence over the 12 week study period in study P001 using both the sponsor definition of recurrence and the reviewer's expanded definition of recurrence (which, along with the subjects counted in the sponsor definition, counted as failures those who were not tested for *C. diff*. toxin upon diarrhea recurrence and those who received fecal transplant, immunoglobulin, or active antimicrobial treatment against *C diff*. after finishing standard of care.

Table 11: Study P001: CDI Recurrence (FAS)

	Placebo N= 395 n (%)	MK-6072 N=386 n (%)	MK-3415A N=383 n (%)	MK-3415 N=232 n (%)	Diff. between Placebo and MK-6072
Recurrence (Sponsor)	**109 (27.6%)**	**67 (17.4%)**	**61 (15.9%)**	**60 (25.9%)**	**10.2%**
Recurrence (Expanded)*	**189 (47.8%)**	**132 (34.2%)**	**137 (35.8%)**	**120 (51.7%)**	**13.6%**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

For the above analysis, MK-6072 and MK-3415A both appeared to reduce the incidence of recurrences relative to placebo. The use of the expanded recurrence definition led to an even more favorable difference in rates of recurrence between placebo and MK-6072/MK-3415A. The rates of recurrence in the placebo arm are in line with reported rates of recurrence in the literature, though it should be noted that reported rates are quite wide ranging.

A subgroup analysis was performed using the expanded definition of recurrence (as this is a more conservative measure). When looking at the various subgroups, the rate of recurrence in the placebo arm was higher than that of the MK-6072 arm in all examined subpopulations except in those subjects who had had disease onset for ≤ 4 days at the time of study drug infusion.

Subgroups were also analyzed to see if there was a more or less favorable difference than the FAS difference in recurrence between placebo and MK-6072 (13.6%). The subgroup arms had to have at least 50 subjects and have more or less than a 3% difference from the FAS difference in the recurrence rate between placebo and MK-6072 to be considered more or less favorable. More favorable differences were seen in subjects ≥ 65 years old, ≥75 years old, subjects enrolled in European sites, subjects receiving vancomycin as standard of care, and subjects with disease onset 7-8 days prior to study drug infusion. Less favorable differences were seen in those < 65 years old, <75 years old, subjects enrolled at North American sites, subjects with a Zar score ≥ 2 at baseline, subjects with Horn's index of Major or Extreme, subjects with disease onset ≤ 4 days prior to study drug infusion, and subjects who had received ≤ 3 days of SOC at the time of study drug infusion.

Table 12: Study P001: Recurrence Subgroup Analysis using Subgroup as Denominator (Expanded Recurrence Definition)*

Subgroups	Placebo Group	Treatment Group: Subgroup Recurrence (n)/Subgroup size in FAS (N) (%)		
		MK-6072	MK-3415A	MK-3415
Sex				
Male	**82/172 (47.7%)**	**53/157 (33.8%)**	**54/172 (31.4%)**	**54/102 (52.9%)**
Female	**107/223 (48%)**	**79/229 (34.5%)**	**83/211 (39.3%)**	**66/130 (50.8%)**
Age Group				
< 65 years	**85/196 (43.4%)**	**75/201 (37.3%)**	**68/183 (37.2%)**	**52/110 (47.3%)**
≥ 65 years	**104/199 (52.3%)**	**57/185 (30.8%)**	**69/200 (34.5%)**	**68/122 (55.7%)**
< 75 years	**129/276 (46.7%)**	**103/287 (35.9%)**	**105/272 (38.6%)**	**78/164 (47.6%)**
≥ 75 years	**60/119 (50.4%)**	**29/99 (29.3%)**	**32/111 (28.8%)**	**42/68 (61.8%)**
Race				
White	**175/366 (47.8%)**	**118/338 (34.9%)**	**124/350 (35.4%)**	**106/210 (50.5%)**
Black or African American	**10/18 (55.6%)**	**11/28 (39.3%)**	**8/17 (47.1%)**	**11/16 (68.8%)**
Asian	**0/1 (0%)**	**0/4 (0%)**	**0/4 (0%)**	**1/3 (33.3%)**
American Indian or Alaska Native	**1/3 (33.3%)**	**1/3 (33.3%)**	**1/2 50%()**	**0/0 (0%)**
Multiple	**3/7 (42.9%)**	**2/13 (15.4%)**	**4/10 (40%)**	**2/3 (66.7%)**
Ethnicity				
Hispanic or Latino	**23/56 (41.1%)**	**11/46 (23.9%)**	**16/52 (30.8%)**	**12/23 (52.2%)**
Region (optional)				

US	**95/186 (51.1%)**	**66/171 (38.6%)**	**75/176 (42.6%)**	**61/120 (50.8%)**
Ex-US	**94/209 (45%)**	**66/215 (30.7%)**	**62/207 (30%)**	**59/112 (52.7%)**
North America	**105/208 (50.5%)**	**79/196 (40.3%)**	**84/204 (41.2%)**	**70/132 (53%)**
Latin America	**14/30 (46.7%)**	**6/26 (23.1%)**	**9/29 (31%)**	**7/9 (77.8%)**
Europe	**61/132 (46.2%)**	**40/139 (28.8%)**	**41/131 (31.3%)**	**37/80 (46.2%)**
Asia-Pacific	**9/23 (39.1%)**	**7/20 (35%)**	**3/17 (17.6%)**	**5/10 (50%)**
Africa	**0/2 (0%)**	**0/5 (0%)**	**0/2 (0%)**	**1/1 (100%)**
Miscellaneous				
Hx. of CDI in Prior 6 months	**67/109 (61.5%)**	**48/103 (46.7%)**	**45/96 (46.9%)**	**40/69 (58%)**
Hx. of CDI in Prior 3 months	**56/94 (59.6%)**	**41/87 (47.1%)**	**40/84 (47.6%)**	**31/56 (55.4%)**
Zar Score ≥2	**27/60 (45%)**	**24/67 (35.8%)**	**19/62 (30.6%)**	**19/31 (61.3%)**
Compromised Immunity	**50/92 (54.4%)**	**36/87 (41.4%)**	**28/78 (35.9%)**	**25/55 (45.4%)**
Vancomycin	**107/189 (56.6%)**	**66/182 (36.3%)**	**72/182 (39.6%)**	**64/113 (56.6%)**
Metronidazole	**78/192 (40.6%)**	**56/190 (29.5%)**	**57/189 (30.2%)**	**53/112 (47.3%)**
Hypervirulent Strains	**25/44 (56.8%)**	**22/51 (43.1%)**	**16/44 (36.4%)**	**19/30 (63.3%)**
Epidemic Strain	**55/106 (51.9%)**	**40/108 (37%)**	**41/106 (38.7%)**	**31/57 (54.4%)**
>70 kg	**99/204 (48.5%)**	**72/209 (34.4%)**	**70/183 (38.2%)**	**56/109 (51.4%)**
<=70kg	**90/191 (47.1%)**	**60/177 (33.9%)**	**67/200 (33.5%)**	**64/123 (52%)**
>100 kg	**20/37 (54%)**	**14/42 (33.3%)**	**12/33 (36.4%)**	**17/33 (51.5%)**
<=100kg	**169/358 (47.2%)**	**118/344 (34.3%)**	**125/350 (35.7%)**	**103/199 (51.8%)**
Horn's Index: Low or Moderate	**133/279 (47.7%)**	**92/283 (32.5%)**	**97/280 (34.6%)**	**80/160 (50%)**

Horn's Index: Major or Extreme	**56/116 (48.3%)**	**40/103 (38.8%)**	**40/103 (38.8%)**	**40/72 (55.6%)**
Disease Onset: <=4 days prior	**54/131= (41.2%)**	**66/150= (44%)**	**46/145= (31.7%)**	**48/100 (48%)**
Disease Onset: 7-8 Days prior	**38/77= (49.4%)**	**19/58= (32.8%)**	**19/38= (50%)**	**23/42 (54.8%)**
Disease Onset: 11+ days prior	**26/64 (40.6%)**	**15/60 (25%)**	**27/59 (45.8%)**	**16/29 (55.2%)**
<=3 days of SOC	**108/241 (44.8%)**	**86/229 (37.6%)**	**89/238 (37.4%)**	**78/146 (53.4%)**
5-6 days of SOC	**46/81 (56.8%)**	**27/83 (32.5%)**	**22/78 (28.2%)**	**22/46 (47.8%)**
>6 Days of SOC	**9/22 (40.9%)**	**6/16 (37.5%)**	**10/20 (50%)**	**5/9 (55.6%)**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

Per Protocol analyses of CDI recurrence in study P001 is presented below.

Table 13: Study P001: CDI Recurrence (Per Protocol)

	Placebo N= 306 (n)(%)	MK-6072 N=332 (n)(%)	MK-3415A N=324 (n)(%)	MK-3415 N=173 (n)(%)	Diff. between Placebo and MK-6072
Recurrence (Sponsor)	**96 (31.4%)**	**63 (19%)**	**54 (16.7%)**	**51 (29.5%)**	**12.4%**
Recurrence (Expanded)*	**135 (44.1%)**	**98 (29.5%)**	**101 (31.2%)**	**85 (49.1%)**	**14.6%**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

In the below table is the proportion of recurrence noted in the treatment arms in the P002 study, again using the sponsor and expanded recurrence definition.

Table 14: Study P002: CDI Recurrence (FAS)

	Placebo N= 378 (n)(%)	MK-6072 N= 395 (n)(%)	MK-3415A N= 390 (n)(%)	Diff. between Placebo and MK-6072
Recurrence (Sponsor)	**97 (25.7%)**	**62 (15.7%)**	**58 (14.9%)**	**10.0%**
Recurrence (Expanded)*	**150 (39.7%)**	**136 (34.4%)**	**131 (33.6%)**	**5.3%**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

Using the sponsor's definition for recurrence there was an absolute reduction of 10% in the rate of recurrence between placebo and MK-6072. However, using the expanded definition of recurrence, the absolute difference was reduced by almost half to 5.3%. As in P001, the placebo recurrence rates in P002 were in line with rates reported in the literature, though placebo recurrence rates in P002 were less than those in P001, particularly using the expanded definition of recurrence.

Looking at the subgroup analysis, MK-6072 only had higher rates of recurrence in two subgroups: males and those who had disease onset 7-8 days prior to study drug infusion. Subgroups were evaluated using the same criteria as in the P001 study to analyze "more" or "less" favorable differences in recurrence relative to the difference seen in the FAS (subgroup sample of at least 50; 3% change from FAS difference using the expanded recurrence definition). More favorable subgroups were females, subjects > 65 years old, subjects with a history of CDI in the past 6 months, subjects with a history of CDI in the past 3 months, subjects who were defined as immunocompromised, subjects with hypervirulent strains, subjects who had disease onset 11 days prior to study drug infusion, and subjects who had received 5-6 of SOC at the time of study drug infusion. Less favorable subgroups were males, subjects < 65 years old, Asian subjects, subjects enrolled in Asian sites, subjects with a Zar score ≥ 2, subjects with epidemic strains, subjects with a Horn's Index of Major or Extreme, and subjects who had disease onset 7-8 days prior to study drug infusion. See Table 15.

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Table 15: Study P002: Recurrence Subgroup Analysis using Subgroup as Denominator (Expanded Recurrence Definition)*

Subgroups	Placebo Group	Treatment Group: Subgroup Recurrence (n)/Subgroup size in FAS (N) (%)	
		MK-6072	MK-3415A
Sex			
Male	**50/152 (32.9%)**	**63/182 (34.6%)**	**63/178 (35.4%)**
Female	**100/226 (44.2%)**	**73/213 (34.3%)**	**68/212 (32.1%)**
Age Group			
< 65 years	**61/172 (35.5%)**	**65/190 (34.2%)**	**46/149 (30.9%)**
≥ 65 years	**89/206 (43.2%)**	**71/205 (34.6%)**	**85/241 (35.3%)**
< 75 years	**101/262 (38.6%)**	**96/281 (34.2%)**	**73/241 (30.3%)**
≥ 75 years	**49/116 (42.2%)**	**40/114 (35.1%)**	**58/149 (38.9%)**
Race			
White	**122/309 (39.5%)**	**106/311 (34.1%)**	**95/305 (31.2%)**
Black or African American	**4/10 (40%)**	**3/17 (17.6%)**	**9/18 (50%)**
Asian	**24/57 (42.1%)**	**26/63 (41.3%)**	**26/65 (40)**
American Indian or Alaska Native	**0/1 (0%)**	**1/2 (50%)**	**0/0 (0%)**
Multiple	**0/1 (0%)**	**0/2 (0%)**	**1/2 (50%)**
Ethnicity			
Hispanic or Latino	**14/45 (31.1%)**	**8/33 (24.2%)**	**9/33 (27.3%)**
Region			
US	**53/131 (40.5%)**	**51/136 (37.5%)**	**43/133 (32.3%)**
Ex-US	**97/247 (39.3%)**	**85/259 (32.8%)**	**88/257 (34.2%)**
North America	**69/158**	**61/158 (38.6%)**	**52/158 (32.9%)**

	(43.7%)		
Latin America	**3/5 (60%)**	**1/4 (25%)**	**3/8 (37.5%)**
Europe	**56/161 (34.8%)**	**50/174 (28.7%)**	**51/161 (31.7%)**
Asia-Pacific	**22/54 (40.7%)**	**24/59 (40.7%)**	**25/63 (39.7%)**
Miscellaneous			
Hx. of CDI in Prior 6 months	**60/110 (54.6%)**	**47/113 (41.6%)**	**39/104 (37.5%)**
Hx. of CDI in Prior 3 months	**51/96 (53.1%)**	**41/98 (41.8%)**	**35/94 (37.2%)**
Zar Score ≥2	**25/65 (38.5%)**	**20/55 (36.4%)**	**28/80 (35%)**
Compromised Immunity	**25/53 (47.2%)**	**31/82 (37.8%)**	**29/75 (38.7%)**
Vancomycin	**77/184 (41.8%)**	**69/190 (36.3%)**	**65/187 (34.8%)**
Metronidazole	**66/182 (36.3%)**	**62/189 (32.8%)**	**61/191 (31.9%)**
Hypervirulent Strains	**32/71 (45.1%)**	**18/51 (35.3%)**	**15/46 (32.6%)**
Epidemic Strain	**48/127 (37.8%)**	**38/102 (37.2%)**	**44/116 (37.9%)**
> 70 kg	**70/168 (41.7%)**	**66/178 (37.1%)**	**50/165 (30.3%)**
<=70kg	**80/210 (38.1%)**	**70/217 (32.3%)**	**81/225 (36%)**
>100 kg	**13/29 (44.8%)**	**12/30 (40%)**	**7/23 (30.4%)**
<=100kg	**137/349 (39.3%)**	**124/365 (34%)**	**124/367 (33.8%)**
Horn's Index: Low or Moderate	**107/261 (41%)**	**96/282 (34%)**	**91/280 (32.5%)**
Horn's Index: Major or Extreme	**43/117 (36.8%)**	**40/112 (35.7%)**	**40/110 (36.4%)**
Disease Onset: <=4 days prior	**49/139= (35.2%)**	**40/145= (27.6%)**	**42/144= (29.2%)**
Disease Onset: 7-8 days prior	**24/61= (39.3%)**	**29/55= (52.7%)**	**22/55= (40%)**

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Disease Onset: 11+ days prior	**25/51=(49%)**	**24/63 = (38.1%)**	**22/55=(40%)**
<=3 days of SOC	**73/208 (35.1%)**	**66/212 (31.1%)**	**70/217 (32.3%)**
5-6 days of SOC	**45/91 (49.4%)**	**33/93 (35.5%)**	**26/94 (27.7%)**
>6 Days of SOC	**7/18 (38.9%)**	**13/32 (40.6%)**	**15/27 (55.6%)**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

Table 16: Study P002: CDI Recurrence (Per Protocol)

	Placebo N= 303 (n)(%)	MK-6072 N= 331 (n)(%)	MK-3415A N= 337 (n)(%)	Diff. between Placebo and MK-6072
Recurrence (Sponsor)	**89 (29.4%)**	**55 (16.6%)**	**54 (16.0%)**	**12.8%**
Recurrence (Expanded)*	**117 (38.6%)**	**97 (29.3%)**	**105 (31.2%)**	**9.3%**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

The following table highlights the rates of clinical cure of the initial CDI episode in each arm of study P001 over the 12 week study period using the sponsor's definition of clinical cure.

Table 17: Study P001: Clinical Cure (FAS)

	Placebo N= 395 (n)(%)	MK-6072 N= 386 (n)(%)	MK-3415A N= 383 (n)(%)	MK-3415 N= 232 (n) (%)	Diff. between Placebo and MK-6072 (95% CI)
Clinical Cure (Sponsor)	**327 (82.8%)**	**299 (77.5%)**	**286 (74.7%)**	**169 (72.8%)**	**5.3% (-10.9, 0.3)**

As can be seen, all three active treatment arms (including MK-6072) performed worse than the placebo arm with regards to clinical cure. The difference between placebo and the MK-6072 arm was 5.3%. The placebo rate of clinical cure was generally in line with what was seen in the pivotal approval studies for fidaxomicin (though it should be noted that there is considerable heterogeneity between those and these trials).

In the subgroup analysis, MK-6072 performed more poorly than placebo in all subgroups

(subgroup sample size of at least 50 subjects) examined except for those with baseline Horn's assessment of Major or Extreme. Using the same criteria as earlier to find subgroups that performed more or less favorably compared with the difference seen in the FAS populations (5.3%), MK-6072 performed more favorably in subjects with a baseline Zar score of ≥ 2, and subjects ≤ 70kg. MK-6072 subgroups that performed less favorably were subjects who had had a CDI episode in the prior 3 months, those infected with an epidemic strain, those given vancomycin as SOC, those > 70 kg, and those assessed with a Horn's Index of Low or Moderate at baseline.

Table 18: Study P001 Subgroup Analysis of Clinical Cure Using Subgroup as Denominator (FAS)

Subgroups	Placebo Group	**Treatment Group Subgroup Clinical Cure (n)/Subgroup size in FAS (N) (%)**		
		MK-6072	MK-3415A	MK-3415
Sex				
Male	**140/172 (81.4%)**	**119/157 (75.8%)**	**135/172 (78.5%)**	**70/102 (68.6%)**
Female	**187/223 (83.9%)**	**180/229 (78.6%)**	**151/211 (71.6%)**	**99/130 (76.2%)**
Age Group				
< 65 years	**166/196 (84.7%)**	**157/201 (78.1%)**	**134/183 (73.2%)**	**83/110 (75.4%)**
≥ 65 years	**161/199 (80.9%)**	**142/185 (76.8%)**	**152/200 (76%)**	**86/122 (70.5%)**
< 75 years	**229/276 (83.0%)**	**224/287 (78%)**	**199/272 (73.2%)**	**120/164 (73.2%)**
≥ 75 years	**98/119 (82.4%)**	**75/99 (75.8%)**	**87/111 (78.4%)**	**49/68 (72.1%)**
Race				
White	**306/366 (83.6%)**	**261/338 (77.2%)**	**263/350 (75.1%)**	**153/210 (72.9%)**
Black or African American	**13/18 (72.2%)**	**24/28 (85.7%)**	**11/17 (64.7%)**	**12/16 (75%)**

Asian	**0/1 (0%)**	**2/4 (50%)**	**3/4 (75%)**	**2/3 (66.7%)**
American Indian or Alaska Native	**3/3 (100%)**	**1/3 (33.3%)**	**1/2 (50%)**	**0/0 (0%)**
Multiple	**5/7 (71.4%)**	**11/13 (84.6%)**	**8/10 (80%)**	**2/3 (66.7%)**
Ethnicity				
Hispanic or Latino	**48/56 (85.7%)**	**35/46 (76.1%)**	**43/52 (82.7%)**	**17/23 (73.9%)**
Region				
US	**153/186 (82.3%)**	**130/171 (76.0%)**	**133/176 (75.6%)**	**87/120 (72.5%)**
Ex-US	**174/209 (83.2%)**	**169/215 (78.6%)**	**153/207 (73.9%)**	**82/112 (73.2%)**
North America	**168/208 (80.8%)**	**149/196 (76.0%)**	**154/204 (75.5%)**	**95/132 (72.0%)**
Latin America	**24/30 (80%)**	**20/26 (76.9%)**	**21/29 (72.4%)**	**6/9 (66.7%)**
Europe	**114/132 (86.4%)**	**111/139 (79.9%)**	**96/131 (73.3%)**	**61/80 (76.2%)**
Asia-Pacific	**20/23 (87%)**	**15/20 (75%)**	**14/17 (82.4%)**	**6/10 (60%)**
Africa	**1/2 (50%)**	**4/5 (80%)**	**1/2 (50%)**	**1/1 (100%)**
Miscellaneous				
Hx. of CDI in Prior 6 months	**91/109 (83.5%)**	**79/103 (76.7%)**	**72/96 (75%)**	**52/69 (75.4%)**
Hx. of CDI in Prior 3 months	**81/94 (86.2%)**	**67/87 (77.0%)**	**66/84 (78.6%)**	**44/56 (78.6%)**
Zar Score ≥2	**44/60 (73.3%)**	**48/67 (71.6%)**	**39/62 (62.9%)**	**20/31 (64.5%)**
Compromised Immunity	**78/92 (84.8%)**	**70/87 (80.5%)**	**56/78 (71.8%)**	**37/55 (67.3%)**
Vancomycin	**160/189 (84.7%)**	**137/182 (75.3%)**	**130/182 (71.4%)**	**88/113 (77.9%)**
Metronidazole	**157/192 (81.8%)**	**150/190 (79%)**	**148/189 (78.3%)**	**76/112 (67.9%)**
Hypervirulent Strains	**38/44 (86.4%)**	**39/51 (76.5%)**	**37/44 (84.1%)**	**20/30 (66.7%)**
Epidemic Strain	**90/106 (84.9%)**	**85/108 (78.7%)**	**87/106 (82.1%)**	**37/57 (64.9%)**
>70 kg	**170/204**	**155/209**	**138/183**	**78/109**

	(83.3%)	**(74.2%)**	**(75.4%)**	**(71.6%)**
<=70kg	**157/191 (82.2%)**	**144/177 (81.4%)**	**148/200 (74%)**	**91/123 (74%)**
>100 kg	**32/37 (86.5%)**	**33/42 (78.6%)**	**21/33 (63.6%)**	**25/33 (75.8%)**
<=100kg	**295/358 (82.4%)**	**266/344 (77.3%)**	**265/350 (75.7%)**	**144/199 (72.4%)**
Horn's Index: Low or Moderate	**238/279 (85.3%)**	**216/283 (76.3%)**	**213/280 (76.1%)**	**121/160 (75.6%)**
Horn's Index: Major or Extreme	**89/116 (76.7%)**	**83/103 (80.6%)**	**73/103 (70.9%)**	**48/72 (66.7%)**
<=3 days of SOC	**200/241 (83%)**	**184/229 (80.4%)**	**178/238 (74.8%)**	**112/146 (76.7%)**
5-6 days of SOC	**66/81 (81.5%)**	**64/83 (77.1%)**	**57/78 (73.1%)**	**34/46 (73.9%)**
>6 Days of SOC	**14/22 (63.6%)**	**11/16 (68.8%)**	**15/20 (75%)**	**5/9 (55.6%)**

Table 19: Study P001 Clinical Cure (Per Protocol)

	Placebo N= 306 (n)(%)	MK-6072 N= 332 (n)(%)	MK-3415A N= 324 (n)(%)	MK-3415 N= 173 (n) (%)	Diff. between Placebo and MK-6072 (95% CI)
Clinical Cure (Sponsor)	**263 (85.9%)**	**267 (80.4%)**	**248 (76.5%)**	**130 (75.1%)**	**5.5% (-11.3, 0.4)**

The following table highlights the rates of clinical cure of the initial CDI episode in each arm of study P002 over the 12 week study period using the sponsor's definition of clinical cure.

Table 20: Study P002: Clinical Cure (FAS)

	Placebo N= 378 (n)(%)	MK-6072 N= 395 (n)(%)	MK-3415A N= 390 (n)(%)	Diff. between Placebo and MK-6072 (95% CI)
Clinical Cure	**294 (77.8%)**	**326 (82.5%)**	**282 (72.3%)**	**-4.7% (-0.9, 10.4)**

Looking at clinical cure in study P002, the MK-6072 arm performed better than the placebo arm, with a difference in the rate of clinical cure between Placebo and MK-6072 of - 4.7%. The placebo cure rate was lower than that seen in the P001 study and the pivotal fidaxomicin trials.

The only subgroups (with sample size of at least 50) where the rate of clinical cure was higher in the placebo arm was in Asian subjects, subjects enrolled in sites in the Asia/Pacific region, subjects with a history of CDI in the prior 3 months, subjects with a history of CDI in the prior 6 months, and subjects with a baseline Zar score ≥2. Using the same criteria as earlier to find subgroups that performed more or less favorably compared with the difference seen in the FAS populations (-4.7%%), MK-6072 performed "less' favorably in these same subgroups. MK-6072 performed more favorably in those with a compromised immunity at baseline and those who had received > 6 days of SOC at the time of study drug infusion.

Table 21: Study P002 Subgroup Analysis of Clinical Cure Using Subgroup as Denominator (FAS)

Subgroup Analysis	Placebo Group (N=378) n (%)	**Treatment Group** (N=)	
		MK-6072 (N=395) n (%)	MK-3415A (N=390) n (%)
Sex			
Male	**114/152 (75%)**	**144/182 (79.1%)**	**136/178 (76.4%)**
Female	**180/226 (79.6%)**	**182/213 (85.4%)**	**146/212 (68.9%)**
Age Group			
< 65 years	**133/172 (77.3%)**	**157/190 (82.6%)**	**102/149 (68.5%)**
≥ 65 years	**161/206 (78.2%)**	**169/205 (82.4%)**	**180/241 (74.7%)**
< 75 years	**206/262 (78.6%)**	**234/281 (83.3%)**	**166/241 (68.9%)**
≥ 75 years	**88/116 (75.9%)**	**92/114 (80.7%)**	**116/149 (77.8%)**
Race			
White	**240/309 (77.7%)**	**259/311 (83.3%)**	**215/305 (70.5%)**

Black or African American	**6/10 (60%)**	**14/17 (82.4%)**	**12/18 (66.7%)**
Asian	**47/57 (82.5%)**	**51/63 (81%)**	**54/65 (83.1%)**
American Indian or Alaska Native	**0/1 (0%)**	**1/2 (50%)**	**0/0 (0%)**
Multiple	**1/1 (100%)**	**1/2 (50%)**	**1/2 (50%)**
Ethnicity			
Hispanic or Latino	**30/45 (66.7%)**	**25/33 (75.8%)**	**23/33 (69.7%)**
Region			
US	**99/131 (75.6%)**	**110/136 (80.9%)**	**86/133 (64.7%)**
Ex-US	**195/247 (79%)**	**216/259 (83.4%)**	**196/257 (76.3%)**
North America	**121/158 (76.6%)**	**128/158 (81%)**	**102/158 (64.6%)**
Latin America	**4/5 (80%)**	**3/4 (75%)**	**6/8 (75%)**
Europe	**124/161 (77.0%)**	**147/174 (84.5%)**	**121/161 (75.2%)**
Asia-Pacific	**45/54 (83.3%)**	**48/59 (81.4%)**	**53/63 (84.1%)**
Miscellaneous			
Hx. of CDI in Prior 6 months	**91/110 (82.7%)**	**92/113 (81.4%)**	**69/104 (66.4%)**
Hx. of CDI in Prior 3 months	**79/96 (82.3%)**	**78/98 (79.6%)**	**64/94 (68.1%)**
Zar Score ≥2	**45/65 (69.2%)**	**34/55 (61.8%)**	**50/80 (62.5%)**
Compromised Immunity	**36/53 (67.9%)**	**67/82 (81.7%)**	**43/75 (57.3%)**
Vancomycin	**137/184 (74.5%)**	**155/190 (81.6%)**	**133/187 (71.1%)**
Metronidazole	**147/182 (80.8%)**	**157/189 (83.1%)**	**144/191 (75.4%)**
Hypervirulent Strains	**52/71 (73.2%)**	**39/51 (76.5%)**	**30/46 (65.2%)**
Epidemic Strain	**101/127 (79.5%)**	**83/102 (81.4%)**	**88/116 (75.9%)**
>70 kg	**129/168 (76.8%)**	**140/178 (78.6%)**	**108/165 (65.4%)**

<=70kg	**165/210 (78.6%)**	**186/217 (85.7%)**	**174/225 (77.3%)**
>100 kg	**20/29 (69%)**	**26/30 (86.7%)**	**17/23 (73.9%)**
<=100kg	**274/349 (78.5%)**	**300/365 (82.2%)**	**265/367 (72.2%)**
Horn's Index: Low or Moderate	**218/261 (83.5%)**	**242/282 (85.8%)**	**207/280 (73.9%)**
Horn's Index: Major or Extreme	**76/117 (65%)**	**83/112 (74.1%)**	**75/110 (68.2%)**
<=3 days of SOC	**164/208 (78.8%)**	**172/212 (81.1%)**	**162/217 (74.6%)**
5-6 days of SOC	**72/91 (79.1%)**	**78/93 (83.9%)**	**65/94 (69.2%)**
>6 Days of SOC	**13/18 (72.2%)**	**27/32 (84.4%)**	**19/27 (70.4%)**

Table 22: Study P002: Clinical Cure (Per Protocol)

	Placebo N= 303 (n)(%)	MK-6072 N= 331 (n)(%)	MK-3415A N= 337 (n)(%)	Diff. between Placebo and MK-6072 (95% CI)
Clinical Cure	**252 (83.2%)**	**285 (86.1%)**	**253 (75.1%)**	**-2.9% (-2.8, 8.5)**

Global Cure

Given the above findings for recurrence and initial cure, it is instructive to look at the Global Cure which accounts for both of these parameters (using the expanded definition for recurrence and the sponsor definition for clinical cure). The following table highlights those findings.

Table 23: Study P001: Global Cure (FAS)

	Placebo N= 395 n (%)	MK-6072 N= 386 n (%)	MK-3415A N= 383 n (%)	MK-3415 N= 232 n (%)	Diff. between Placebo and MK-6072
Global Cure (Expanded Recurrence*/Clinical	**164 (41.5%)**	**193 (50%)**	**182 (47.5%)**	**85 (36.6%)**	**-8.5%**

Cure)					

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

Table 24: Study P002 Global Cure (FAS)

	Placebo N= 378 n (%)	MK-6072 N= 395 n (%)	MK-3415A N= 390 n (%)	Diff. between Placebo and MK-6072
Global Cure (Expanded Recurrence*/Clinical Cure)	**172 (45.5%)**	**217 (54.9%)**	**185 (47.4%)**	**-9.4%**

*- Expanded includes those not tested for toxin upon diarrhea recurrence and those receiving fecal transplant, immunoglobulin, or active therapy after SOC

In both studies the rate of Global Cure favored MK-6072.

Data Quality and Integrity – Reviewers' Assessment

The overall data quality was acceptable from an efficacy perspective.

Efficacy Results – Secondary and other relevant endpoints

Many of the secondary and exploratory endpoints were included in the discussion of the primary endpoint because of their clinical relevance. However it is also instructive to look at rates of recurrence in those initially cured, regardless of toxin test results at the time of recurrence. The following table highlights those findings.

Table 25: Study P001: Recurrence Regardless of Toxin Results (FAS)

	Placebo N= 395 n (%)	MK-6072 N= 386 n (%)	MK-3415A N= 383 n (%)	MK-3415 N= 232 n (%)	Diff. between Placebo and MK-6072
Recurrence	**211 (53.4%)**	**147 (38.1%)**	**148 (38.6%)**	**128 (55.2%)**	**15.3%**

Table 26: Study P002: Recurrence Regardless of toxin Results (FAS)

	Placebo N= 378 (n)(%)	MK-6072 N= 395 (n)(%)	MK-3415A N= 390 (n)(%)	Diff. between Placebo and MK-6072
Recurrence	**166 (43.9%)**	**149 (37.7%)**	**147 (37.7%)**	**6.2%**

In both studies MK-6072 and MK-3415A performed considerably better than placebo using this more liberal definition of recurrence.

MAJOR AMENDMENT

To further assess the rick-benefit profile of bezlotoxumab, a Major Amendment was requested in order to evaluate 1) post hoc analyses the sponsor had conducted using expanded definitions of clinical cure and failure 2) the severity of recurrences and 3) clinical variables associated with clinical failure, particularly in subjects who might be reclassified from initial clinical failure to initial clinical cure in any post-hoc analyses. The materials provided in the amendment included additional datasets and CRFs.

To address concerns regarding differences in initial clinical cure, the sponsor conducted a post hoc analysis that altered the definition of initial clinical cure in two ways:

- There was no limit on the duration of SOC (previously clinical failure had been defined as use longer than 14 days [16 calendar days])
- Two consecutive days of < 3 stools/24 hours could occur at any point after finishing SOC in order to be considered a clinical cure (previously the two consecutive days had to occur immediately after the completion of SOC in order to be considered a clinical cure).

The expanded cure was assessed at various time points relative to start of infusion (2, 3, 4, 6, 8 and 12 weeks).

Clinical cure rates using the expanded definition are presented in the table below.

Table 27: Clinical Cure Rates Using Post Hoc Sponsor Definition of Clinical Cure, P001 and P002, FAS

	P001		P002	
	Bezlotoxumab	**Placebo**	**Bezlotoxumab**	**Placebo**
	N=386	N=386	N=395	N=378
Original Definition	299 (77.5)	327 (82.8)	326 (82.5)	294 (77.8)
Expanded Definition				
2 weeks	297 (76.9)	313 (79.2)	308 (78.0)	296 (78.3)
3 weeks	353 (91.5)	361 (91.4)	367 (92.9)	340 (89.9)
4 weeks	364 (94.3)	370 (93.7)	372 (94.2)	349 (92.3)
6 weeks	370 (95.9)	375 (94.9)	376 (95.2)	353 (93.4)
8 weeks	371 (96.1)	377 (95.4)	377 (95.4)	354 (93.7)
12 weeks	373 (96.6)	379 (95.9)	377 (95.4)	356 (94.2)

As can be noted, by 2 weeks, differences in clinical cure between the two arms start to reduce and are gone by 3 weeks.
To better understand whether differences in clinical cures under the original clinical cure definition were possibly related to strict definition criteria, which may not be clinically relevant and whether expanded definition of clinical cure was clinically acceptable, clinical characteristics of subjects reclassified as cures were analyzed, see the table below.

Reference ID: 3989587

Table 28: Characteristics of Initial CDI Episode in Subjects Reclassified from Clinical Failure to Clinical Cure Using Post Hoc Sponsor Definition of Clinical Cure, P001 and P002

	P001		P002	
	MK-6072	Placebo	MK-6072	Placebo
Number Reclassified	**74 Subjects**	**52 Subjects**	**51 Subjects**	**61 Subjects**
Median Duration of SOC (25%-75%)	**16 days (10.5-19)**	**16 days (12.25-21.75)**	**17 days (15-21)**	**16 days (13.75-20)**
Median Duration from End of SOC to Resolution of Diarrhea (25%-75%)	**3 days (2-5.25)**	**3 days (2-7.75)**	**3 days (2-5)**	**3.5 days (2-6)**
Number with Recurrences	**15 subjects**	**15 subjects**	**20 subjects**	**21 subjects**

Looking at the reclassified bezlotoxumab subjects, their clinical course did not look particularly severe relative to the original definition of clinical cure and failure. For example, a subject could have taken 16 calendar days of SOC under the original definition for clinical cure. Similarly, under the expanded definition in the bezlotoxumab arm the median duration of SOC treatment is 16-17 days. Similarly, under the original definition, a cured subject would be classified as a cure if he/she had no diarrhea for the two days immediately following the last day of SOC(i.e. - a duration of 2 days from the end of SOC to the defined resolution of diarrhea). The median duration for the reclassified bezlotxumab subjects was 3 days, which implied that relative to the original definition, subjects had an extra day of diarrhea following the last day of SOC. Thus, from this descriptive analysis there are signals that some of the original clinical failures may not have been particularly clinically severe.

Exploration of the CRF's of a subset of reclassified bezlotoxumab subjects in P001 appeared to support these analyses in that it was difficult to pinpoint in the majority of cases a direct relationship between receipt of bezlotoxumab and clinical failure. Cases were often confounded, such as in cases where a subject may have received short durations of SOC. If a link was plausible between drug and clinical failure it often caused only a modest prolonging of the initial illness.

With the use of the expanded definition of clinical cure, the number of clinical cures in all arms expanded, and thus the number of subjects that could be evaluated for recurrences also expanded. Using this definition of clinical cure and the original definition for recurrences, the number of

recurrences was as noted in the table below.

Table 29: Recurrences Using Post Hoc Sponsor Definition of Clinical Cure, FAS, Study P001

	MK-6072 N= 386	Placebo N=395	Difference (Placebo – MK6072)
Recurrences at 12 Weeks	**82 (21.2%)**	**124 (31.4%)**	**10.2%**

Table 30: Recurrences Using Post Hoc Sponsor Definition of Clinical Cure, FAS, Study P002

	MK-6072 N= 395	Placebo N=378	Difference (Placebo – MK6072)
Recurrences at 12 Weeks	**82 (20.8%)**	**118 (31.2%)**	**10.4%**

As can be seen, using this altered clinical cure definition, a reduction in recurrences can still be noted with bezlotoxumab in both studies

These recurrences were evaluated in terms of severity based on multiple parameters including need for/type of treatment, need/location of hospitalization, and clinical characteristics. Note the table below (note that this table focuses on 1st recurrences).

Table 31: Clinical Characteristics of 1st Recurrences Using Post Hoc Sponsor Definition of Clinical Cure, Study P001

Recurrences at 12 weeks	MK-6072 N=82	Placebo N= 124	Difference (Placebo-MK6072)
Treatment in health care facility	**28 (34.2%)**	**40 (32.3%)**	**-1.9%**
ICU Admission	**4 (4.9%)**	**3 (2.4%)**	**-2.5%**
Any SOC use	**46 (56.1%)**	**76 (61.3%)**	**5.2%**
Vancomycin	**30 (36.6%)**	**59 (47.6%)**	**11.0%**
Metronidazole	**13 (15.8%)**	**21 (16.9%)**	**1.1%**
Fidaxomicin	**4 (4.9%)**	**5 (4.0%)**	**-0.9%**
Mean Duration of SOC, Days (Range)	**11 (0-63)**	**12 (0-60)**	**1 day**
Zar Score ≥ 2	**4 (4.9%)**	**12 (9.7%)**	**4.8%**
WBC > 15,000	**1 (1.2%)**	**7 (5.6%)**	**4.4%**

cells/µL			
Temperature > 38.3°C	**1 (1.2%)**	**3 (2.4%)**	**1.2%**
Median Number of Loose Stools at Onset (25-75%)	**3 (3-4)**	**4 (3-6)**	**1 stool**
Median Time to Resolution of Recurrence, Days (25-75%)	**2 (1-4)**	**3 (1-5)**	**1 day**
Eventual Need for Fecal Transplant	**3 (3.7%)**	**6 (4.8%)**	**1.1%**

Table 32: Clinical Characteristics of 1st Recurrences Using Post Hoc Sponsor Definition of Clinical Cure, P002

Recurrences at 12 weeks	MK-6072 N=82	Placebo N= 118	Difference (Placebo-MK6072)
Treatment in health care facility	**30 (36.6%)**	**41 (34.8%)**	**-1.8%**
ICU Admission	**4 (4.9%)**	**3 (2.5%)**	**-2.4%**
Any SOC use	**45 (54.9%)**	**77 (65.3%)**	**10.4%**
Vancomycin	**30 (36.6%)**	**60 (50.8%)**	**14.2%**
Metronidazole	**21 (25.6%)**	**24 (20.3%)**	**-5.3%**
Fidaxomicin	**5 (6.1%)**	**5 (4.2%)**	**-1.9%**
Mean Duration of SOC, Days (Range)	**10 (0-52)**	**12 (0-73)**	**2 days**
Zar Score ≥ 2	**7 (8.5%)**	**13 (11%)**	**2.5%**
WBC > 15,000 cells/µL	**1 (1.2%)**	**12 (10.2%)**	**9.0%**
Temperature > 38.3°C	**1 (1.2%)**	**3 (2.5%)**	**1.3%**
Median Number of Loose Stools at Onset (25-75%)	**4 (3-5)**	**4 (3-6)**	**0 stools**
Median Time to Resolution of Recurrence, Days (25-75%)	**2 (1-5)**	**3 (1-4)**	**1 day**
Eventual Need for Fecal Transplant	**2 (2.4%)**	**16 (13.6%)**	**11.2%**

Overall, clinical implications of CDI recurrences were not trivial indicating clinical benefit of

preventing recurrences. More than a half of the patients required a repeat course of SOC and about a third of the patients required hospitalization including ICU admission in 2.5-5% of cases.

Though not particularly striking for all parameters, in P001 trends toward less severity were noted in the bezlotoxumab arm, particularly with regards to need for treatment. Similarly, in P002, trends toward less severity were noted in the bezlotoxumab arm in terms of need for treatment and fecal transplant.

Dose/Dose Response

Dose response was not studied in the Phase 2 or 3 trials. All subjects received the 10 mg/kg dose of study drug in these trials.

Durability of Response

As has already been discussed, both studies P001 and P002 assessed for recurrences over a period of 12 weeks. However, in P002, a subset of 300 subjects were followed for an additional nine months (the extension cohort). 293 of these subjects were a part of the FAS (112 subjects in the MK-3415A arm, 99 in the MK-6072 arm, and 82 in the placebo arm). When looking at global cures over the 12 month period in the extension cohort, the MK-6072 arm did considerably better than placebo (69.7% global cures vs. 40.2%, respectively). However, the extension cohort appeared to differ significantly from the full P002 FAS. In fact the rate of recurrence over the 12 week study period in the full FAS set in the placebo arm was 25.7% while it was 41.5% over the same period for the placebo arm of the extension cohort (using the sponsor definition for recurrence). Thus, it is difficult to draw any real conclusions regarding durability given the atypical characteristics of the extension cohort.

Persistence of Effect

See above discussion regarding durability of response.

7 Integrated Review of Effectiveness

7.1. Assessment of Efficacy Across Trials

7.1.1. Primary Endpoint

7.1.2. Secondary and Other Endpoints

Because of the disparate findings across the individual trials, particularly as regards clinical cure,

pooling for analysis of the primary and secondary endpoint is not considered appropriate. Moreover, studies P001 and P002 had some differences as regards racial makeup as well as location of investigative sites.

7.1.3. Subpopulations

Because of the disparate findings across the individual trials, the differences in study design as regards study sites and racial makeup, pooling results is not appropriate in this setting.

7.1.4. Dose and Dose-Response

Only a single dose (10mg/kg) was studied in the phase 2 and phase 3 trials, thus dose response in the clinical setting cannot be adequately evaluated. The sponsor chose this dose because the exposure it afforded was similar to the protective exposure noted in hamster models of CDI. The sponsor notes that higher doses were not studied because it was felt that they would provide no increased benefit. Lower doses were not studied because the 10mg/kg dose was sufficiently tolerated in phase 1 and 2 studies,

7.1.5. Onset, Duration, and Durability of Efficacy Effects

The primary endpoint covered a 12 week period. Please see earlier discussions regarding efficacy.

7.2. Additional Efficacy Considerations

7.2.1. Considerations on Benefit in the Postmarket Setting

[See discussion in Section 7.3]

7.2.2. Other Relevant Benefits

See discussion in Section 7.3

7.3. Integrated Assessment of Effectiveness

In assessing the efficacy of MK-6072 in preventing the recurrence of CDI from the data presented in this review, it is helpful to methodically move through the salient issues.

First, what is the clinical impact of any therapy designed to reduce recurrence of CDI? Assuming that the therapy had no significant deleterious effects on clinical cure of the initial episode and had a tolerable safety profile, the clinical impact would be significant. Currently there is no approved therapy to treat the recurrence of CDI. Instead, clinicians have applied a heterogeneous group of unapproved therapies to treat recurrences. These range from repeating the intimal treatment course to pulsed vancomycin tapers to immunoglobulin therapy to fecal

transplant. Reductions in recurrences would have a significant impact on patient morbidity and health care costs directly and indirectly. Thus, there is no question that the clinical impact of an approved therapy to prevent recurrence would be significant.

Secondly, it is important to consider how clinicians might use any therapy directed toward preventing recurrence. In an ideal scenario, any adjunct therapy to prevent recurrence would be given at the time of/right after initial clinical cure. This would spare the clinician and patient the risk and cost associated with administering the adjunct therapy early in the course of the initial CDAD episode. Such risk would include administering adjunct therapy to a subject that was never cured initially (in which case preventing recurrence becomes moot) or administering therapy that might somehow actually interfere with the SOC treatment of the initial episode.

Bearing the above considerations in mind, one can now evaluate the data submitted in this application.

In regards to recurrences, no matter what definitions of recurrence or clinical cure was used (the original applicant definitions of recurrence and cure, the reviewer expanded definition of recurrence, or the applicant expanded definition of clinical cure), bezlotoxumab appeared to either have a statistically significant effect or a trend toward lessening recurrences.

Using the reviewer expanded definition of recurrence that captures subjects not tested for toxin upon recurrence of diarrhea as well as those who used therapies that could essentially treat CDI, MK-6072 appears to perform well relative to placebo in P001 and marginally better in study P002. In P001, MK-6072 had an absolute reduction in recurrence rate of 13.6%. Moreover, MK-6072 appeared to do better than placebo in preventing recurrence in almost all subgroups tested. In study P002, MK-6072 had absolute reduction in recurrence rate of 5.3%. Though this reduction was modest, the MK-6072 arm had lower recurrence rates in almost every subgroup (except, importantly, in males).

It is interesting to note that in P001 administration of SOC for ≤ 3 days before study drug infusion was associated with "less" favorable outcomes, and in P002, administration of SOC for 5-6 days before study drug infusion was associated with "more" favorable outcomes. Whether this is reflective of an overall improved activity of MK-6072 on recurrence when given later in the initial course of SOC is unclear, particularly given that the opposite effect was seen in subjects who had received >6 days of SOC before receiving study drug infusion (though the sample size in this subgroup is quite small). It should be noted that the original recurrence definition did not distinguish between those requiring treatment and those who did not. However, there appeared to be a trend toward less need for treatment in the bezlotoxumab arm; this was in conjunction with other exploratory analyses that seemed to suggest some decreased severity of recurrence with bezlotoxumab. Overall, there appears to be a modest to moderate effect of MK-6072 on recurrence.

In regards to clinical cure, MK-6072 appeared to perform more poorly relative to placebo in study P001 with an absolute reduction in clinical cure of 5.3%. MK-6072 also performed worse in virtually every subgroup analyzed in this study. In study P002, MK-6072 had the opposite effect with an absolute increase in the rate of clinical cure of 4.7%. There were, however, several subgroups where placebo performed numerically better than MK-6072. In terms of the duration of SOC prior to receiving study drug infusion, no clear trend can be seen from the two trials (although in subjects >6 days of SOC there was trend toward better clinical cures in the MK-6072 arm; again, the sample size was quite small). Thus, MK-6072's effect on initial clinical cure is unclear. However, in exploratory analyses using the expanded definition of clinical cure, the rates of clinical cure in study arms are similar and it appears that many of clinical failures defined according to the original cure definition were only modestly clinically significant. A subset review of CRFs of reclassified cures from P001 did not provide any clear link between administration of bezlotoxumab and initial clinical failure. In totality, bezlotoxumab may have a negative effect on clinical cure, but it appears to be clinically minor and such risk may be tolerable in high benefit scenarios.

Overall, when looking at the effect of MK-6072 on the two most important clinical parameters, recurrence and initial clinical cure, a positive risk benefit calculus seems plausible. There is certainly a level of uncertainty when relying on stool diaries, etc. but hopefully in this blinded trial those uncertainties are minimized. . In general, there appears to be a moderate reduction in recurrences at the possible expense of small increases in mostly minor initial clinical failures. Still, this would place the significant risk-benefit calculus involved in deciding whether or not to prescribe the study drug to a patient with CDI at the feet of clinicians, when ideally this process should have been definitively made much earlier- i.e.-during this drug development and regulatory review process.

It is imperative that clinicians use the drug judiciously, hopefully reserving its use for those subjects with significant risk factors for recurrence. Given the likely cost of this medication, such decision-making is even more imperative. Ideally, proper guidance regarding the timing of study drug relative to SOC would have been delineated from the trial results but given the heterogeneous nature of drug receipt relative to SOC as well as the heterogeneous results, no clear recommendation can be made with regard to timing of bezlotoxumab.
The Division and Applicant may consider a PMR/PMC for a study evaluating the timing of study drug administration relative to SOC in regards to effects on both clinical cure and recurrence. Such a study should have a mutually agreed upon definition of clinical cure and recurrence and could stratify subjects based on timing of SOC.

8 Review of Safety

8.1. Safety Review Approach

This safety review focuses on data from the safety population enrolled in the two Phase 3 randomized, placebo-controlled trials, study P001 and P002, supporting this application. These two studies evaluated the efficacy, safety and tolerability of a single infusion of bezlotoxumab (bezlo) alone and in combination with actoxumab (acto) for prevention of *Clostridium difficile* infection (CDI) in patients receiving standard of care antibacterial therapy (oral vancomycin, fidaxomicin, oral or intravenous metronidazole). Both bezlo and acto are fully human monoclonal antibodies: bezlo is an anti- type B *Clostridium difficile* toxin antibody and acto is an anti-type A *Clostridium difficile* toxin antibody.

The two studies are similar in design with the following exceptions: the first study, study P001, followed an adaptive design. It also had four arms study arms: subjects that received acto infusion only (acto arm), subjects that received bezlo infusion alone (bezlo arm), subjects that received combination of acto and bezlo infusions (acto+bezlo arm), and subjects that received normal saline infusion (placebo arm). In addition, study P001 had a planned interim analysis at 40% enrollment to evaluate each study arm and to drop one of the single therapy arms if it did not demonstrate efficacy more than the combination arm. Based on the interim analysis, which showed worsening outcome in the acto arm as compared to all three arms, the acto arm was dropped for the post-interim analysis study period. Study P002 did not have planned interim analysis and was designed with only 3 arms: bezlo arm, acto+bezlo arm and placebo arm. In addition, study P002 had an extended safety cohort of 300 participants that were followed for additional 9 months after the 12-week study period. Given the similarity in study design, including adverse event (AE) reporting period, pooled data from study P001 and P002 are presented.

The safety population included subjects in Study P001 and P002 that received partial or full infusion of bezlo, acto+bezlo or placebo. Of note, subjects that received acto alone were excluded from the safety population. However, data from the acto arm is presented separately for completeness of the review since the safety population includes subjects that received the combined acto+bezlo infusion. Also, data from a phase 2 study that compared the combination of acto and bezlo with placebo, study P017, is not pooled with the phase 3 studies mentioned above due to difference in AE reporting time frame The safety data for this study is presented separately.

Since bezlo is a monoclonal antibody, special attention was given to adverse events related to immunogenicity, and infusion-related adverse reactions.

For the integrated safety analysis, MAED version 18.0, JMP version 12.0 and JReview 9.2 analytic software were utilized.

8.2. Review of the Safety Database

8.2.1. Overall Exposure

Overall, there were 1790 subjects exposed to bezlo either alone or in combination with acto; 126 were healthy subjects in Phase 1 studies (30 subjects received bezlo alone and 96 received acto + bezlo), and 1664 were subjects in Phase 2 and 3 (786 received bezlotoxumab alone and 878 received acto+bezlo). Of these, 1741 subjects were exposed to bezlo dose at 10mg/kg, while 30 and 19 were exposed to dose of <10mg/kg dose and 20mg/kg, respectively. With the exception of subjects in one phase 1 study (study P004) who received two doses of acto+bezlo (n=30), all subjects were exposed to a single dose of bezlo or acto+bezlo.

Majority of subjects in the two Phase 3 trials that make up the safety population received bezlo alone or in combination with acto at the intended bezlo dose of 10mg/kg, with intended dose defined as ≥9.5 to <10.5 mg/kg. Of the 786 that received bezlo alone, 97.1% (n=763) received the intended bezlo dose. Only 1.8% (n=9) subjects received less than the intended dose (defined as <9.5 mg/kg) and 1.1% (n=9) received more than the intended dose (defined as ≥10.5 mg/kg). No subjects received bezlo at a dose >20 mg/kg. Among the 777 that received acto+bezlo, 96.7% (n=751) received the intended dose of 10mg/kg, and 2.3% (n=18) and 1 %(n=8) received less than and more than the intended dose, respectively.

Since bezlo is dosed on a milligram per kilogram(mg/kg) basis, the range of doses was large given the range of subject weights (range of weights 29.8 to 194 kg; range of doses about 300 to 2000 mg). The median dose was 700 mg, consistent with the median weight of 70 kg among bezlo-treated subjects.

Table 33 summarizes the overall extent of exposure to bezlo throughout the drug development program.

Table 33: Bezlotoxumab-exposed Population during Development Program

Safety Database for Bezlotoxumab Individuals exposed to the study drug in this development program for the indication under review N=1790		
Clinical Trial Groups	**New Drug (n=1790)**	**Placebo (n= 892)**
Phase 1 Studies (Healthy Subjects)		

P020	**54**	**0**
bezlo		
<10mg/kg	18	
10mg/kg	6	
>10mg/kg	6	
acto+bezlo		
<10mg/kg	12	
10mg/kg	6	
>10mg/kg	6	
P005	**29**	**6**
acto+ bezlo(10mg/kg)	29	
P006	**13**	**6**
acto+bezlo(10mg/kg)	6	
acto+bezlo(20mg/kg)	7	
P004	**30**	**0**
acto+bezlo(10mg/kg)-two doses	30	
Controlled trials conducted for this indication		
Phase 2-Study P017	**101**	**99**
acto+bezlo (10mg/kg)	101	99
Phase 3-Study P001	**777**	**400**
bezlo (10mg/kg)	390	
acto+bezlo (10mg/kg)	387	
Phase 3-Study P002	**786**	**381**
bezlo (10mg/kg)	396	
acto+bezlo (10mg/kg)	390	

bezlo: bezlotoxumab; acto+bezlo: Combination of actoxumab and bezlotoxumab
Note: Subjects in the acto+bezlo arm received the stated dose for each immunoglobulin.

8.2.2. Relevant characteristics of the safety population:

The safety population is comprised of all subjects in the two phase 3 studies that received partial or full infusion of the study medication. The subjects that received the same study treatment in the two Phase 3 trials, i.e. acto+bezlo, bezlotoxumab, or placebo are pooled. For the safety analysis, subjects were grouped in to the treatment arm they received (as treated), as opposed to the group they were randomized to. There were total of 7 subjects that received treatments

different from their randomized arm. In study P001, two subjects randomized to bezlo received placebo, and one subject randomized to acto+bezlo received placebo. In study P002, one subject randomized to bezlo received placebo; a second subject randomized to placebo received acto+bezlo; a third subject randomized to acto+bezlo received bezlo alone, and a fourth subject randomized to acto+bezlo received acto alone.

Of note, the safety population did not include subjects that received acto alone in study P001. The data from the acto only arm are included in the tables for comparison only.

Tables 34, 35 and 36 summarize the relevant baseline demographic, clinical and laboratory characteristics of the pooled safety population, respectively.

Table 34: Demographic Characteristics of Safety Population

	Bezlo N, %	**Acto+ Bezlo** N, %	**Placebo** N, %	**Total** N, %	**Acto*** **N, %**
Subjects in population	786	777	781	2344	236
Gender					
Male	343 (43.6)	348 (44.8)	326 (41.7)	1017 (43.4)	103 (43.6)
Female	443 (56.4)	429 (55.2)	455 (58.3)	1327 (56.6)	133 (56.4)
Age (Years)					
Median	62	64.2	63.5	63.2	66
Range	18-100	18-95	18-98	18-100	21-99
<40	105 (13.4)	85 (11)	85 (10.8)	275 (11.8)	21 (8.9)
40-59	211 (26.9)	186 (23.9)	199 (25.4)	596 (25.5)	67 (28.4)
60-79	327 (41.6)	343 (44.2)	344 (44.1)	1014 (43.2)	103 (43.6)
≥80	143 (18.6)	183 (21)	153 (19.6)	459(19.6)	45 (19.1)
Race					
American Indian or Alaska Native	5 (0.6)	2 (0.3)	4 (0.5)	11 (0.5)	0
Asian	68 (8.7)	70 (9)	59 (7.6)	197 (8.4)	3 (1.3)
Black or African American	46 (5.9)	35 (4.5)	28 (3.6)	109 (4.7)	16 (6.8)
Multiple	15 (1.9)	12 (1.5)	8 (1)	35 (1.5)	3 (1.3)
White	652 (83)	658 (84.7)	682 (87.3)	1992 (85)	214 (90.4)
Ethnicity					
Hispanic or Latino	79 (10.1)	86 (11.1)	103 (13.2)	268 (11.4)	23 (9.7)
Non-Hispanic or Latino	683 (86.9)	668 (86)	655 (83.9)	2006 (85.6)	211 (89.4)
Not Reported/unknown	24 (3)	23 (2.9)	23 (2.9)	70 (3)	2 (0.8)
Region of Enrollment					
US	311 (39.6)	314 (40.4)	322 (41.2)	947(40.4	124 (52.5)

	Bezlo N, %	**Acto+ Bezlo** N, %	**Placebo** N, %	**Total** N, %	**Acto*** **N, %**
Subjects in population	786	777	781	2344	236
Ex-US	47 5(60.4)	463 (59.6)	459 (58.8)	1397(59.6)	112 (47.5)

Table 35: Key Baseline Clinical Characteristics of Safety Population

	Bezlo N, %	**Acto+ Bezlo** N, %	**Placebo** N, %	**Total**	**Acto Only** N, %
Subjects in population	786	777	781	2344	236
Weight(kg)					
Median	70	68.5	70	69.6	69.3
Range	29.8-194	32.7-187.3	28.9-168.3	28.9-194	35.3-86.5
CDI severity[1]					
Severe	122 (15.5)	143 (18.4)	126 (16.1)	391 (16.7)	31 (13.1)
Non-severe	634 (80.7)	595 (76.6)	614 (78.6)	2038 (83.3)	190 (80.5)
Unknown	30 (3.8)	39 (5)	36 (4.6)	105 (4)	15 (6.4)
Charlson Score[2] (N,%)					
<3	465 (59.2)	450 (58)	476 (60.9)	1391(59.3)	130 (55.1)
3-4	141 (17.9)	152 (19.6)	134 (17.2)	427 (18.3)	52 (22)
≥5	180 (22.9)	175 (22.5)	171 (21.9)	526 (22.4)	54 (22.9)
CDI in past 12 months, N (%)					
Yes	237(30.1)	224 (28.8)	239 (30.6)	700 (29.9)	78 (33.1)
No	539 (68.6)	538 (69.2)	532 (68.1)	1609 (68.6)	156 (66.1)
Unknown	10 (1.3)	15 (1.9)	10 (1.3)	35 (1.5)	2 (0.8)
SoC tx[3] , n(%)					
Oral vanco	380 (48.3)	374 (48.1)	374 (47.9)	1128 (48.1)	115 (48.7)
Metronidazole	377 (48)	379(48.8)	380 (48.7)	1136 (48.5)	114 (48.3)
Fidaxomicin	29 (3.7)	24 (3.1)	27(3.4)	80 (3.4)	7 (3)
Hospitalization status, n(%)					
Inpatient	531 (67.6)	523 (67.3)	523 (67)	1577 (67.3)	158 (66.9)
Outpatient	255 (32.4)	254 (32.7)	258 (33)	767 (32.7)	78 (33.1)
Immune compromised					
Yes	171 (21.8)	153 (19.7)	147 (18.8)	471 (20.1)	55 (23.3)
No	615 (78.2)	624 (80.3)	634 (81.2)	1873 (79.9)	181 (76.7)
Renal disease[4]					
Severe	52 (6.6)	39 (5)	52 (6.7)	143 (6.1)	11 (4.7)

	Bezlo N, %	**Acto+ Bezlo** N, %	**Placebo** N, %	**Total**	**Acto Only** N, %
Subjects in population	786	777	781	2344	236
Moderate	46 (5.9)	42 (5.4)	47 (6)	135(5.8)	12 (5.1)
No or mild	688 (87.5)	696 (89.6)	682 (87.3)	2066 (88.1)	213 (90.3)

[1]CDI was classified as severe if subjects score ≥2points based on the following criteria: Age >60 years (1 point), body temperature >38.3°C (1 point), serum albumin<2.5mg/dL (1 point), peripheral white blood cell count >15000 cells/mm^3 within 48 hours (1 point), endoscopic evidence of pseudomembranous colitis (2 points), and treatment in ICU (2 points).
[2]Charlson index (Charlson et al, J. Chron Dis 1987) was used to assess co-morbid conditions
[3]SoC tx refers to standard of care treatment subjects received in conjunction with study drug
[4] Renal disease is classified as severe if subjects had one of the three conditions: uremia, on dialysis or have had transplant. Mild and moderate are classified with serum creatinine 2-3mg/dL and >3mg/dL respectively.
NOTE: Subjects in acto arm are NOT included in the safety population; data from this arm is presented for comparison only.

Table 36: Key Baseline Laboratory Features of Safety Population

	Bezlo **N, %**	**Acto+ Bezlo** **N, %**	**Placebo** **N, %**	**Total** **N, %**	**Acto Only** **N, %**
Subjects in population	786	777	781	2344	236
NAP1/027 strain, N(%)					
Yes	90(11.5)	76(9.8)	100(12.8)	290(12.4)	24(10.2)
No	394(50.1)	401(51.6)	390(49.9)	1305(55.7)	120(50.8)
Unknown	302(38.4)	300(38.6)	291(37.3)	985(42)	92(39)
Epidemic strains*					
Yes	211	222(28.6)	233	723	57(24.1)
No	273	255(32.8)	257	872	87(36.9)
Unknown	302	300(38.6)	291	985	92(39)
Baseline WBC >15x10^3/µL					
Yes	57 (7.3)	68 (8.8)	60 (7.7)	185 (7.9)	14 (5.9)
No	689 (87.7)	667 (85.8)	669 (85.7	2025 (86.4)	204 (86.4)
Unknown	40 (5)	42 (5.4)	52 (6.6)	134 (5.7)	18 (7.6)

*Epidemic strains included NAP1/027, 001, 0078 and106 strains of *C. difficile*

8.2.3. Adequacy of the safety database:

Overall, the safety database provided adequate numbers of exposed subjects at the proposed dose and frequency (one time use) for bezlo. Most of the baseline demographic, prior history of CDI, comorbidity, disease severity and laboratory features appear comparable among the three arms. More than half of the subjects in each arm were older than 60 years of age, which reflects CDI's clinical burden in the elderly. The range of severity of disease in the safety population represents the severity range encountered in clinical practice. Of note, there is underrepresentation of minority populations as subjects of white racial background accounted for >80% of subjects in each arm. There is some literature evidence of higher incidence of CDI in white vs non-white individuals, but there is no major difference in manifestation of CDI across racial groups. Hence, the information obtained from these studies could be generalized to the greater US population.

8.3. Adequacy of Applicant's clinical Safety Assessment

8.3.1. Issues Regarding Data Integrity and Submission Quality

In collaboration with the Office of Computational Science (OCS), data fitness assessment was performed on January 14, 2016. The result showed some inconsistencies in the data. For example, the death information, particularly timing, between the various domains was inconsistent with some fatal adverse events not having a death record in the disposition domain. There were instances where subject death information was captured in the Comments (CO) or Clinical Events (CE) domains. In addition, a full comprehensive file of all the deaths that occurred during the pivotal studies was not provided initially and was obtained by information request.

Case report forms were reviewed to assess consistency of the data submitted. A coding dictionary was created to assess how AEs were transcribed to MedDRA dictionary LLT and preferred term. All of the reported terms for AE matched the MedDRA dictionaries used during the study. There were few instances of some splitting terms. For example, the term cardiac failure was reported as "cardiac failure", "cardiac failure acute", "cardiac failure chronic", and "cardiac failure congestive".

Of note, clinical inspection of 4 study sites were requested from the Office of Scientific Investigation (OSI), and the results of the inspections are pending at time of filing of this review. The results of the clinical inspection will be amended when results are available.

8.3.2. Categorization of Adverse Events

The sponsor defined AE as any unfavorable and unintended sign (including an abnormal

laboratory finding, for example), symptom, or disease temporally associated with the use of a medicinal product or protocol- specified procedure, whether or not considered related to the medicinal product or protocol- specified procedure. Any worsening (i.e., any clinically significant adverse change in rate and/or intensity) of a preexisting condition that is temporally associated with the use of a pharmaceutical product is also an AE. AEs that occurred after infusion of study drug were classified as treatment emergent AEs (TEAEs). AEs were reported by System Organ Class (SOC), as specified by the Medical Dictionary for Regulatory Activities (MedDRA) Version 17.1 for study P001 and MedDRA Version 18.0 for study P002. MedDRA Version 18.0 was used for the pooled safety analysis.

Clinical AEs were recorded in case report forms over the 12-week post infusion study period during visits on days 4, 11, 29, 57 and 85. Vital signs were recorded pre-infusion, 30-minutes after start of infusion, end of infusion and during the above mentioned study visits. Electrocardiogram study was done immediately prior to start of infusion and within 2-hrs after end of infusion. Laboratory assessments (hematology, chemistry, and urinalysis) were carried out during study visits on days 1, 4, 11, and 29. Blood samples for anti-drug antibody were collected during study visit days 1, 29, 57 and 85.

AEs were assessed by frequency, with each event per patient reported. AE severity was categorized into mild, moderate, and severe. In addition, a serious AE (SAE) was defined as any AE occurring at any dose that:

- Results in death
- Is life-threatening
- Results in persistent or significant disability/incapacity
- Results in or prolongs an existing inpatient hospitalization
- Is a congenital anomaly/birth defect
- Is a cancer
- Is associated with an overdose
- Is another important medical event

In the two Phase 3 trials (P001, P002), non-serious AEs were collected from the time of study medication infusion until Week 4 post-infusion. Serious adverse events (SAEs) were collected from the time of study medication infusion until the Week 12 post-infusion visit. The study investigator assessed the duration, intensity and relationship of the AE to the study drug. In the 9-month extension for P002, only deaths and SAEs deemed treatment-related by the investigator were reported.

Since bezlo is a monoclonal antibody, the sponsor carried out AE monitoring specifically for infusion-specific reactions by selectively looking at AEs that occurred within the 24-hour post infusion of the study drug. The sponsor looked at the following specific terms to assess clinical events that may be related to infusion reaction: infusion-site adverse experience, pyrexia, chills, rash, arthralgia, myalgia, joint swelling, obstructive airways disorder, bronchospasm, stridor, dysphonia, headache, fatigue, pruritus, urticaria, hypotension, hypertension, nasal congestion,

nausea, vomiting, flushing, angioedema, dyspnea, and dizziness/lightheadedness. The information was collected for inpatient subjects while outpatient subjects, who made up about a third of the study population, were instructed to call the study staff if AEs occurred within the first 24 hours post infusion.

8.3.3. Routine Clinical Tests

The schedule of events for study P002 is presented in table 37 as an example for the timing of safety lab collection in the safety population. Similar laboratory testing schedule was employed in study P001.

Table 37: Study P002 Schedule of Clinical and Safety Evaluation

Activity	**Time Point**						
Study Visit	**#1**	**#2**	**#3**	**#4**	**#5**	**#6**	**Unscheduled**
Relative Day/Week of Study	**Day 1**	**Day 4 (± 1 day)**	**Day 11 (± 2 days)**	**Week 4**	**Week 8 (Day 57 ± 7**	**Week 12 (Day 85 ± 5**	**Day 1 to Week**
CDI Diagnosis (stool test for toxigenic *C.*	X						X
Physical examination	X		X			X	X
Vital Signs (pre- & post-infusion)	X	X	X	X	X	X	X
Non-serious AE	<----------X ----------						X
Serious AE Assessment	<---------- X ----------						X
Collection of Prior / Concomitant Medication Use	<---------- X ----------> (All medications)				<---------- X ----------> (Medications to treat CDI, other antibacterials, anti-diarrheal, and excluded		X
12-Lead ECG (pre- & post-infusion)	X						
Safety Lab Sample	X	X	X	X			X
Blood sample for anti-drug antibody (ADA)	X			X	X	X	

Loose Stool Count	<----------Daily loose stool count by the patient with stool count log----------

AE – adverse event,
Source: Adapted from Table 9-4, study P002 clinical study report

8.4. Safety Results

8.4.1. Deaths

Phase 1 and 2 Studies
There were no deaths reported in the four Phase 1 studies (P020, P004, P005, P006) involving administration of bezlo alone or in combination with acto.

In study P017, the placebo-controlled Phase 2 trial where subjects received combination of acto+bezlo, there were a total of 14 deaths, with 7 deaths (6.9%) in 101 acto+bezlo-treated subjects, and 7 (7.1%) in 99 placebo-treated subjects in the 12-weeks post infusion of study drug. Two of the deaths were due to serious AEs that were deemed possibly drug-related by the investigator; these subjects are marked with asterisk in the table 38. There was one additional death in the placebo group about 7 months post infusion and this subject was not included in the analysis.

List of deaths that occurred in the acto+bezlo arm in P017 are summarized in the table below.

Table 38: Summary of Deaths in Actoxumab+Bezlotoxumab arm of Study P017

Subject ID	Age (yrs), sex (F/M)	Cause of Death	Study Day of AE onset	Study Day of Death	Relationship to Study Drug		
					Investigator-defined	Applicant-defined	Reviewer-defined
02030401*	87,F	Urosepsis, Pneumonia with para-pneumonic effusions	4	15	Possibly related	Unlikely related	Unlikely related
02091001	83, M	Respiratory failure	24	33	Unrelated	Unrelated	Unlikely related
02530204	74, M	Respiratory distress, VRE sepsis, severe COPD	21	35	Unlikely related	Unlikely related	Unlikely related
02620908*	79, M	Sepsis, kidney infarction,	9	12	Possibly related	Unlikely related	Unlikely related

		embolism, atherosclerosis					
02620910	83, M	Sudden death	45	45	Unlikely related	Unlikely related	Unlikely related
02921701	39, F	Cardio-pulmonary arrest	22	24	Unlikely related	Unlikely related	Unlikely related
		P. jirovecii pneumonia	4				
02921709	101, M	COPD Cardiac failure congestive	8	8	Unlikely related	Unlikely related	Unlikely related

*denotes the two subjects with serious and drug-related AEs as assessed by the investigator.

Source: Sponsor's "Narratives of Deaths" submitted for study P017 (Section14.3.3.1)

The death narratives for the two subjects with serious AE resulted in death that were possibly related to study drug and as assessed by the investigator are presented below:

Subject 203-0401 was an 87-year-old female with significant cardiac disease (s/p coronary bypass grafting, congestive heart failure, atrial fibrillation, s/p pace maker placement, mitral regurgitation), deep venous thrombosis, hypertension, chronic renal failure, and recurrent urinary tract infection. She was hospitalized and diagnosed with *C. difficile* colitis associated with acute chronic renal failure secondary to volume depletion. After discharge to a skilled nursing facility, the subject was re-hospitalized on study day 4 with dyspnea and respiratory distress and found to have a left pleural effusion, significant leukocytosis, acute renal insufficiency and pyuria. The subject was diagnosed with urosepsis and pneumonia with a parapneumonic effusion. The subject expired on study day 15 after life support measures were withdrawn. The event was assessed by the Investigator as possibly related due to the temporal relationship of the event to the study infusion. The Sponsor's medical reviewer assessed the event as unlikely related to the study infusion.

Subject 262-0908 was a 79-year-old male with COPD, hypertension, artherosclerosis, s/p recent small bowel obstruction hospitalized for possible ischemic colitis and later diagnosed with *C. difficile* colitis. During the hospitalization the subject experienced worsening leukocytosis and on study day 9 was diagnosed with left renal infarct vs. pyelonephritis and new onset of dysarthritic speech. The subject was placed in end of life/comfort care and expired on study day 12. The immediate cause of death was sepsis; underlying causes were kidney infarction, embolism and atherosclerosis. The Investigator assessed the event as possibly related to study infusion and unexpected. The Sponsor's medical reviewer assessed the event as unlikely related to the study infusion

Reviewer Comment: Though relationship to the study drug cannot be ruled out in the above two deaths, both subjects had underlying co-morbidities that could explain the AEs. In all of the 7 deaths, there were co-morbidities that could contribute to and explain the fatal outcome; hence, there were no deaths that appeared related to the study drug.

Phase 3 Studies

Overall, there were 212 deaths during the Phase 3 trials (Figure 1). Of these, 166 deaths (77 in study P001 and 89 in study P002) occurred due to AE reported in the 12-weeks post-infusion follow-up period. The remaining 46 deaths occurred in the following groups: deaths in subjects excluded by sponsor for not meeting reporting criteria (n=9); death during the extension phase of study P002 (n=9), deaths among subjects that received acto alone (n=28). Table 39 and Figure 2 summarize the deaths that occurred in the two Phase 3 trials.

Table 39: Mortality Rate in Each Arm of the Safety Population over the 12-week Study Period

	Bezlo	Acto+Bezlo	Placebo	Any Bezlo exposure	Acto[a] N=236
Over 12 week study period					
P001	31/390(8%)	20/387(5.2%)	26/400(6.5%)	51/777(6.6%)	27/235(11.5%)
P002	25/396(6.3%)	31/390(8%)	33/381(8.7%)	56/786(7.1%)	1/1(100%) [b]
Total	56/786 (7.1%)	51/777(6.6%)	59/781(7.5%)	107/1563(6.8%)	28/236 (11.9%)
During Extension phase					
Study P002	5/100(5%)	2/112(1.8%)	2/83(2.4%)	3/212(3.3%)	

[a] The acto arm is not included in the safety population and data for acto arm is presented for comparison.
[b] This subject in study P002 was inadvertently given actoxumab only

The extension phase was comprised of 295 subjects with 112 subjects in the acto+bezlo arm, 100 subjects in the bezlo arm and 83 subjects in the placebo group; in general, subjects in the extension cohort were younger and had fewer bad prognostic factors. The mortality rate by arm for this group was as follows: 2 (1.8%) in the acto+bezlo arm, 5 in the bezlo arm (5%), and 2 in the placebo arm (2.4%).

Figure 2: Overall Deaths during Phase 3 Trials

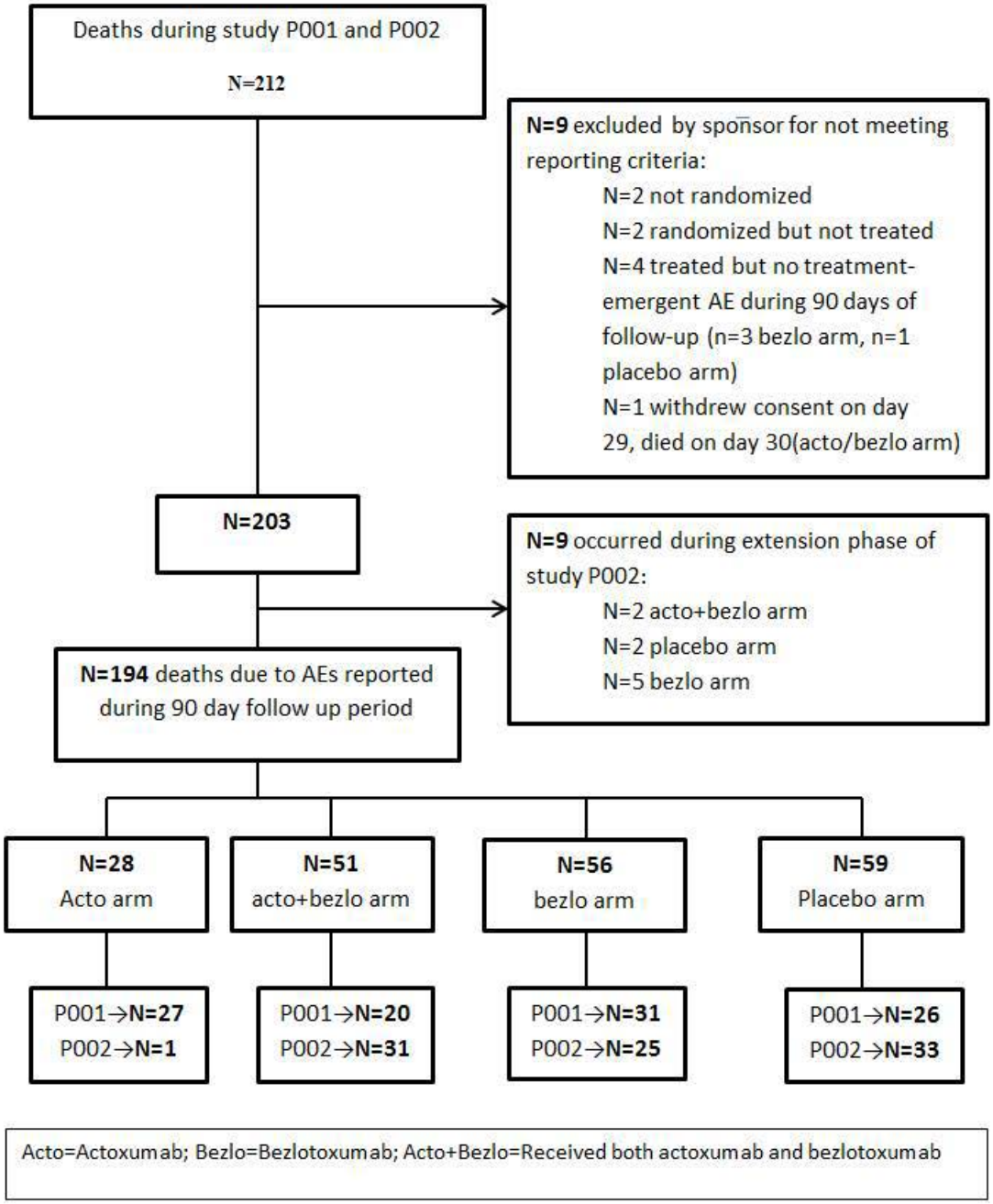


Table 40, 41, and 42 describe baseline demographic, clinical and laboratory features of subjects who died during the 12-week follow-up period. It should be noted that data for the acto only arm is provided for comparison and is not part of the safety population.

Table 40: Key Demographic Features of Subjects who Died due to AEs Reported during 12-week Follow-up

	Bezlo N=56	Acto+Bezlo N=51	Placebo N=59	Acto N=28
Gender-Female (n, %)	25(44.6)	23(45.1)	35(59.3)	10(35.7)
Race				
White	48(85.7)	41(80.4)	51(86.4)	26(92.9)
Non-white	8(14.3)	10(19.6)	8(13.6)	2(7.1)
Age (in yrs)				
Median(range)	73.5(24-97)	76(37-90)	72(33-96)	77(54-93)
25th-75th quartile	63.3-82.8	68-82	62-83	65.5-86.8
Weight (in kg)				
Median(range)	65(43.9-98)	60(34.5-96)	61(31-105.6)	67.9(43-129.3)
25th-75th quartile	54-76.9	49-73.8	53-70	55-81.5
Median BMI(range)	23.5(15.9-38.8)	22.7(14.5-37.2)	22.3(13-40.2)	22.1(16.6-59.6)
25th-75th quartile	18.7-26.5	19.2-25.7	19.6-25.8	20.2-29.2
Study site				
US	12(21.4)	7(13.7)	15(25.4)	7(25)
Non-US	44(78.6)	44(86.3)	44(74.6)	21(75)

Table 41: Key Clinical Features of Subjects who Died due to AEs Reported during 12-week Follow-up

	Bezlo N=56	Acto+Bezlo N=51	Placebo N=59	Acto N=28
CDI severity[1]				
Severe	22(39.3)	26(51)	24(40.7)	8(28.6)
Non-severe	32(57.1)	19(37.2)	31(52.5)	18(64.3)
Unknown	2(3.6)	6(11.8)	4(6.8)	2(7.1)
Charlson Score[2] (n,%)				
<3	13(23.2)	9(17.6)	17(28.8)	7(25)
3-4	20(35.7)	17(33.3)	13(22)	6(21.4)
≥5	23(41.1)	25(49)	29(49.2)	15(53.6)
CDI in past 12 months, n(%)				
Yes	12(21.4)	14(27.5)	14(23.7)	11(39.3)
No	41(73.2)	33(64.7)	44(74.6)	16(57.1)
Unknown	3(5.3)	4(7.8)	1(1.7)	1(3.6)
SoC tx[3] , n(%)				
Oral vancomycin	25(44.6)	24(47.1)	24(40.7)	12(42.9)
Metronidazole	29(51.8)	27(52.9)	32(54.2)	16(57.1)
Fidaxomicin	2(3.6)	0(0)	3(5.1)	0(0)
Hospitalization status, n(%)				
Inpatient	56(100)	49(96.1)	55(93.2)	24(85.7)
Outpatient	0	2(3.9)	4(6.8)	4(14.3)
Immune compromised				
Yes	16(28.6)	18(35.3)	12(20.3)	9(32.1)
No	40(71.4)	33(64.7)	47(79.7)	19(67.9)
Renal disease[4]				
Severe	9(16)	4(7.8)	7(11.9)	3(10.7)
Moderate	2(3.6)	6(11.8)	8(13.5)	2(7.1)
No or mild	45(80.4)	41(80.4)	44(74.6)	23(82.1)

[1]CDI was classified as severe if subjects score ≥2points based on the following criteria: Age >60 years (1 point), body temperature >38.3^{o}C (1 point), serum albumin<2.5mg/dL (1 point), peripheral white blood cell count >15000 cells/mm^3 within 48 hours (1 point), endoscopic evidence of pseudomembranous colitis(2 points), and treatment in ICU (2 points)
[2]Charlson index (Charlson et al, J. Chron Dis 1987) was used to assess co-morbid conditions

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[3]SoC tx refers to standard of care treatment subjects received in conjunction with study drug
[4] Renal disease is classified as severe if subjects had one of the three conditions: uremia, on dialysis or have had transplant. Mild and moderate are classified with serum creatinine 2-3mg/dL and >3mg/dL respectively
NOTE: Subjects in the acto arm are NOT included in the safety population; data from this arm is presented for comparison only.

Table 42: Key Laboratory Features of Subjects who Died due to AEs Reported during 12-week Follow-up

	Bezlo N=56	Acto+Bezlo N=51	Placebo N=59	Acto N=28
NAP1/027 strain, n(%)				
Yes	11(19.6)	9(17.6)	12(20.3)	3(10.7)
No	26(46.4)	29(56.9)	29(49.2)	13(46.4)
Unknown	19(33.9)	13(25.5)	18(30.5)	12(42.9)
Epidemic strains*				
Yes	19(33.9)	16(31.4)	21(35.6)	4(14.3)
No	18(32.1)	22(43.1)	20(33.9)	12(42.9)
Unknown	19(33.9)	13(25.5)	18(30.5)	12(42.9)
Median WBC x10^3/µL				
n (range)	10.2(1.8-41.7)	9.3(1.4-32.4)	10.2(0.5-36.5)	9.4(4.1-18.4)
25th-75th quartile	7.4-14	6.1-14	6.3-15.2	6.6-10.8
Median WBC x10^3/µL				
n (range)	9.3(1.4-32.4)	10.2(1.8-41.7)	10.2(0.5-36.5)	9.4(4.1-18.4)
25th-75th quartile	6.1-14	7.4-14	6.3-15.2	6.6-10.8
Pre-infusion Creatinine				
Median(range)	1.1(0.2-5.5)	0.9(0.2-8.3)	0.9(0.2-6.7)	1.05(0.4-4.7)
25th-75th quartile	0.7-1.8	0.6-1.5	0.6-1.3	0.75-1.45

*Epidemic strains included NAP1/027, 001, 0078 and106 strains of *C. difficile*

Reviewer Comment: Compared to the overall safety population, those who died were older (median age 77 years vs 63 years), had severe CDI disease (more than 40% in those who died vs. 15-17% in overall population), and high/very high Charlson co-morbidity score (70-80% for those who died vs. only 40-43% in overall population). *There were higher percentage of immunocompromised subjects at baseline (30%) compared to the overall safety population (20%). There was also slightly higher percentage of subjects with NAP1 strain (around 20%) as compared to the overall safety population (10-12%). About 80% of the deaths occurred in sites outside the US while the subjects outside the US accounted for 60% of the overall safety population. Of note, 19% of subjects enrolled outside the US had severe CDI while only 12% of the subjects enrolled the US were reported to have severe CDI. Hence, this may partly explain the observed mortality difference between US vs non-US group. In addition, the observed mortality difference may also partly be a reflection on variation in supportive care in the US vs. non-US sites.*

Looking at the three arms, the acto+bezlo arm had higher number of subjects with severe CDI at baseline(51% compared to 40% in the placebo and bezlo arm only) and higher percentage of immunocompromised individuals at baseline(35% vs. 28% and 20% in the bezlo and placebo arm, respectively). Otherwise, the baseline characteristics were comparable among those who died in each of the three arms.

Table 43: Broad Classifications of Reported Causes of Deaths

	Bezlo N=56	Acto + Bezlo N=51	Placebo N=59	Acto N=28
Sepsis/septic shock	7	7	14	9
Respiratory				
Bronchitis/pneumonia	5	5	7	4
Acute Respiratory failure	3	0	5	0
Chronic respiratory failure	3	5	0	0
Malignancy	1	1	1	1
Pulmonary edema	0	2	0	0
Cardiovascular				
Cardiac arrest	4	1	6	2
Myocardial infarction	1	1	2	0
Arrhythmia	1	0	0	0
Heart failure	8	6	4	1
Gastro-intestinal				
Malignancy	3	4	1	2
Liver dysfunction	1	1	0	0
Pancreatitis	0	0	0	1

	Bezlo N=56	Acto + Bezlo N=51	Placebo N=59	Acto N=28
Intestinal dilation/obstruction/hemorrhage	1	3	0	0
C. difficile infection	0	1	1	0
Central Nervous System				
CVA/Infarct/Hemorrhage	3	3	0	1
Malignancy	1	0	2	0
Infection	0	0	1	0
Senile dementia	0	1	0	0
Status epilepticus	0	1	0	0
Renal				
Acute renal failure	2	0	0	2
Chronic renal failure	2	1	1	0
Malignancy	1	0	1	0
Reproductive & Genito-urinary				
Breast cancer	0	1	1	0
Prostate cancer	1	0	0	0
Bladder cancer	1	1	0	0
Hematologic				
Malignancy	0	1	0	1
Febrile neutropenia	0	1	1	0
Other				
Cutaneous malignancy	1	0	2	0
Wound infection	1	0	0	0
Bone cancer	0	1	0	0
GVHD	0	0	0	1
Asthenia	0	0	1	0
Dehydration	0	0	1	0
Failure to thrive	0	0	0	1
General health deterioration	1	0	0	0
Multi-organ failure	3	1	2	1
Unknown cause of death	1	2	5	1
GVHD – graft versus host disease				

Death Narratives:
All death narratives provided by the sponsor were reviewed. The following death narratives are for the subjects in the safety population whose death was deemed to have a possible relationship

to the study drug. The first 3 were reported by the sponsor as possibly related and in the fourth one, the reviewer thought that the potential of bezlo contributing to the death cannot be excluded due to the temporal proximity of the AE that lead to death.

1. **Study P001-Subject ID 107888:** This 44-year-old African American woman with HIV and active tuberculosis was hospitalized for sepsis due to perianal fistulae and pneumonia. She subsequently developed CDI and was treated with oral metronidazole for 2 days and IV metronidazole for additional 4 days as standard of care CDI therapy**. Acto+bezlo** was infused on the 4th day of standard of care therapy. On Day 2 post infusion, the subject's condition deteriorated and she experienced confusion, hypotension (87/42 mmHg), hypoglycemia, and respiratory distress. She was treated with antibacterial drugs and IV fluids. The subject died on Day 2 due to worsening sepsis. The investigator felt that the death **may have been possibly related to study medication** due to temporal relationship of drug administration, but listed the following contributing factors: disseminated tuberculosis including possible bone marrow involvement, HIV and the fact that the subject had stopped antiretroviral and tuberculosis treatment. No autopsy was performed.

2. **Study P001-Subject ID 107040**: This 75-year-old white woman with myelodysplastic syndrome, alcohol-induced cirrhosis, cardiomyopathy, and esophagitis/gastritis. She developed CDI while hospitalized, and was treated with PO metronidazole as standard of care CDI therapy. She received **bezlo** two days after initiating standard of care therapy. On Day 8, while still receiving metronidazole, the subject developed sepsis and acute spontaneous bacterial peritonitis with symptoms of chills, diaphoresis, and abdominal distension, which was treated initially with ceftriaxone. Blood cultures grew methicillin-resistant *Staphylococcus aureus* (MRSA) and therapy was switched to IV vancomycin. Gentamicin was added 4 days later. The subject initially improved; however, on Day 19 (while still on intravenous antibacterials), the subject was found unresponsive to stimuli, with hypotonus and aniscoria (dilated right pupil). Brain computed tomography revealed left parenchymal hematoma in lenticulo-capsular and anterior frontal lobes with midline shift, reported as cerebral hemorrhage. Bleeding was attributed to disseminated intravascular coagulation caused by MRSA sepsis. Despite continued medical interventions, the subject died on Day 19. CDI has resolved while on metronidazole with no recurrence of diarrhea before the time of death. No autopsy was performed. The investigator **considered the event, sepsis due to MRSA, as related to study medication** with rationale that the study medication is an immunomodulator.

3. **Study P002-Subject ID116310**: A 90-year-old white woman with global cardiac failure/decompensation, osteoporosis, and prior episodes of pneumonia developed CDI as an inpatient. She received **acto+bezlo** 6 days after standard of care CDI treatment was initiated with PO vancomycin. Diarrhea resolved and the subject did not have a recurrence prior to death. On Day 3, the subject developed an

acute episode of vomiting, and imaging suggested evidence of an acute small bowel obstruction. By Day 5, the subject had progressed into septic shock, which was treated with antibacterial drugs and vasopressors (including ticarcillin, gentamicin, cefoxitin, and norepinephrine). On Day 7, a laparotomy revealed evidence of the small bowel obstruction and a small bowel resection was performed. The subject's condition failed to improve, and she died later that day. No autopsy was performed. The primary cause of death was small intestine obstruction, including septic shock due to occlusive syndrome of the small bowel. The event was assessed by the investigator as **severe and related to study therapy**.

4. **Study P002-Subject ID 116952**: This was an 89-year-old white female with hypertension, atrial fibrillation, heart failure, diabetes mellitus, chronic renal failure, dementia, and repeated episodes of urinary tract and respiratory infections. While hospitalized for presumed respiratory infection, the subject developed a CDI and was treated with PO metronidazole. She received **acto+bezlo** 2 days after the start of metronidazole therapy. On day 2, the subject experienced involuntary clonic movements in all four limbs, with mandibular trismus, increased stiffness, and increased disorientation. The subject was treated with levetiracetam, clonazepam, and diazepam, with little effect. The condition progressed, and on Day 17, the subject was treated with morphine as part of palliative care. The subject died on Day 19. Death was attributed worsening of senile dementia. No autopsy was performed. The Investigator considered death **not related to study therapy**.

Reviewer Comment: I have reviewed all the death narratives provided by the sponsor. Table 43 summarizes the reported causes of death by broad primary causes of death categories reported in the death narratives and patient disposition data. Since bezlo is a novel substance and the mechanism of action is not fully elucidated, it is difficult to anticipate all of the potential off-target effects and hence, potential contribution to the deaths. However, there were no safety signals in the animal models and there was no excess mortality noted comparing the bezlo and the acto+bezlo arms to placebo; the number and causes of deaths that occurred in the Phase 3 trials are comparable among the study arms. Overall, the majority of the deaths appear to be related to subjects' co-morbid conditions.

8.4.2. Serious Adverse Events

Phase 1 and 2 Studies
There were no serious AEs, including deaths, reported in the four Phase 1 trials involving administration of bezlo alone or in combination with acto (P020, P004, P005, P006). There were

no AEs that resulted in study drug interruption or discontinuation. There were two subjects with tachycardia (n=1 bezlo-exposed, n=1 acto+bezlo exposed).

Table 44 below summarizes the AEs that occurred during study P017. There were no discontinuations of the active study drug due to an AE. The overall AE rate and the mortality rate in the acto+bezlo arm were comparable to the placebo arm.

Table 44: Summary of Adverse Events for Study P017

	Acto+Bezlo arm N	(%)	Placebo arm N	(%)	Total N	(%)
Subjects in population	101		99		200	
With serious AEs	18	17.8	28	28.3	46	23
With serious drug-related AEs*	3	3	4	4	7	3.5

* determined by the investigator to be related to the drug
Data source: Modified from sponsor's table 2.7.4:15

Phase 3 Studies
Table 45 summarizes the serious adverse by MedDRA System Organ Classification (SOC).

Table 45: Number of Subjects with Serious Adverse Events by System Organ Class (SOC)

System Organ Class (SOC)	Bezlo N=786 n (%)	Acto+ Bezlo N=777 n (%)	Placebo N=781 n (%)	Acto* N=236 n (%)
Subject with at least one SAE	231 (29.4)	212 (27.3)	255 (32.7)	105 (44.5)
Infection and infestation	104 (13.2)	93 (12)	138 (17.7)	50 (21.2)
Gastrointestinal disorders	49 (6.2)	37 (4.8)	42 (5.4)	20 (8.5)
Cardiac disorders	36 (4.6)	24 (3.1)	27 (3.5)	14 (5.9)
Respiratory, thoracic& mediastinal disorders	27(3.4)	26(3.4)	24 (3.1)	9 (3.8)
General disorders & administration site conditions	19 (2.4)	18 (2.3)	19 (2.4)	8 (3.4)
Neoplasms benign, malignant and unspecified	18 (2.3)	16 (2.1)	16 (2.1)	6 (2.5)

Reference ID: 3989587

System Organ Class (SOC)	Bezlo N=786 n (%)	Acto+ Bezlo N=777 n (%)	Placebo N=781 n (%)	Acto* N=236 n (%)
Subject with at least one SAE	231 (29.4)	212 (27.3)	255 (32.7)	105 (44.5)
Renal and urinary disorders	16 (2)	6 (0.8)	18 (2.3)	5 (2.1)
Nervous system disorders	13 (1.7)	24 (3.1)	8 (1)	13 (5.5)
Blood & lymphatic disorders	13 (1.7)	10 (1.3)	8 (1)	3 (1.3)
Metabolism & nutritional disorders	11 (1.4)	10 (1.3)	16 (2.1)	6 (2.5)
Injury, poisoning & procedural complications	10 (1.3)	7 (0.9)	9 (1.2)	2 (0.9)
Hepatobiliary disorders	7 (0.9)	8 (1)	2 (0.3)	2 (0.9)
Psychiatric disorders	7 (0.9)	4 (0.5)	5 (0.6)	2 (0.9)
Vascular disorders	5 (0.6)	10 (1.3)	13 (1.7)	8 (3.4)
Investigations	3 (0.4)	3 (0.4)	4 (0.5)	0
Musculoskeletal & connective tissue disorders	3 (0.4)	3 (0.4)	4 (0.5)	0
Skin & subcutaneous tissue disorders	1 (0.1)	4 (0.51)	3 (0.38)	2 (0.9)
Immune system disorders	1 (0.1)	1 (0.1)	2 (0.3)	2 (0.9)
Endocrine disorders	1 (0.1)	1(0.1)	0 (0)	0 (0)
Eye disorders	1 (0.1)	0 (0)	0 (0)	0 (0)
Ear & labyrinth disorders	0(0)	0 (0)	1 (0.1)	0 (0)
Reproductive systems & breast disorders	0(0)	1 (0.1)	0(0)	0 (0)

Table 46 and Table 47 summarize serious AEs (SAEs) that occurred in more than >4 subjects in the active arm. The SAEs are presented in a descending order for the active arm.

Table 48 summarizes SAEs that occurred in >2 subjects in the acto only arm.

Table 46: Serious Adverse Events that Occurred in the Bezlotoxumab Arm

MedDRA Preferred Term	Bezlo N=786 n (%)	Placebo N=781 n (%)
Cardiac failure*	**17 (2.2)**	**7 (0.9)**
Urinary tract infection	**15 (1.9)**	**9 (1.2)**

MedDRA Preferred Term	Bezlo N=786 n (%)	Placebo N=781 n (%)
Diarrhea	**16 (2)**	**12 (1.53)**
Abdominal pain	**7 (0.89)**	**4 (0.51)**
Anemia	**5 (0.64)**	**3 (0.38)**
Chronic kidney disease	5 (0.64)	5 (0.64)
Respiratory failure	5 (0.64)	6 (0.77)
Acute kidney injury	6 (0.76)	10 (1.28)
Pneumonia and pneumonia aspiration	17 (2.2)	22(2.8)
Sepsis and urosepsis	14(1.78)	27(3.46)
Clostridium difficile infection	24 (3.1)	54 (6.9)

*Cardiac failure terms "cardiac failure", "cardiac failure acute", "cardiac failure congestive" and "cardiac failure chronic" were combined.
MedDRA=Medical Dictionary for Regulatory Activities

Table 47: Serious Adverse Events that Occurred in the Actoxumab+Bezlotoxumab Arm

MedDRA Preferred Term	Acto+Bezlo N=777 n (%)	Placebo N=781 n (%)
Cardiac failure*	**17 (2.19)**	**7 (0.9)**
Bacteremia	**6 (0.77)**	**3 (0.38)**
Urinary tract infections	**11 (1.42)**	**9 (1.15)**
Abdominal pain	**6 (0.77)**	**4 (0.51)**
Chronic obstructive pulmonary disease	**5 (0.64)**	**4(0.51)**
Respiratory failure	5 (0.64)	6 (0.77)
Diarrhea	10 (1.28)	12 (1.54)
Septic shock	7 (0.9)	10 (1.28)
Pneumonia and pneumonia aspiration	17(2.19)	22 (2.81)
Sepsis and urosepsis	11 (1.42)	27 (3.46)
Clostridium difficile infection	31 (4)	54 (6.9)

*Cardiac failure terms "cardiac failure", "cardiac failure acute", "cardiac failure congestive" and "cardiac failure chronic" were combined.
MedDRA=Medical Dictionary for Regulatory Activities

Table 48: Serious Adverse Events that Occurred in >2 Subjects in the Actoxumab Arm

MedDRA Preferred Term	Acto N=236 n (%)	Placebo N=781 n (%)
Clostridium difficile infection	**26 (11.02)**	**54 (6.91)**
Cardiac failure*	**7 (2.97)**	**7 (0.9)**
Urinary tract infection	**6 (2.54)**	**9 (1.15)**
Cerebrovascular accident	**3 (1.27)**	**0**
Constipation	**3 (1.27)**	**0**
Abdominal pain	**4 (1.69)**	**4 (0.51)**
Nausea	**3 (1.27)**	**2 (0.26)**
Diarrhea	**6 (2.54)**	**12 (1.54)**
Acute kidney injury	**4 (1.69)**	**10 (1.28)**
Septic shock	3 (1.27)	10 (1.28)
Pneumonia and pneumonia aspiration	8	22
Sepsis and urosepsis	10	27

*Cardiac failure terms "cardiac failure", "cardiac failure acute", "cardiac failure congestive" and "cardiac failure chronic" were combined.
MedDRA=Medical Dictionary for Regulatory Activities

Reviewer Comment: Overall, there is no major imbalance in the frequency of serious adverse events among the three arms: 29.4%, 27.3% o and 32.7% of subjects in the bezlo, acto+bezlo and placebo arm, respectively, experienced at least one SAE in the 12-week study period. In all three arms.The infection and infestation and gastrointestinal MedDRA System Organ Classes accounted for the majority of the SAEs. A detailed look of the SAEs revealed cardiac failure, urinary tract infection, diarrhea, abdominal pain to have occurred more frequently in the bezlo arm as compared to the placebo arm. From these, cardiac failure showed the largest risk difference, with 17 subjects (2.2%) in the bezlo arm as compared to placebo (n=7, 0.009%). Same trend of more subjects with cardiac failure was seen in the acto+bezlo arm (n=17, 2.2%) as compared to placebo. Of note, preferred terms that describe cardiac failure (cardiac failure, cardiac failure acute, cardiac failure congestive, cardiac failure chronic) were combined for the analysis. Please refer to section 8.5.3 for detailed subgroup analysis of subjects by presence or absence of congestive heart failure (CHF) at baseline.

Narratives of SAEs were reviewed, including the four acto+bezlo subjects who had SAEs deemed related to the study drug by the investigator. Synopses of these four subjects are presented below:

- 91-year-old Asian woman with multiple co-morbidities including hypertension, and dyslipidemia received acto+bezlo infusion on day 3 of standard of care therapy with metronidazole. On day 8, she displayed signs and symptoms of cerebral infarction. She concurrently had atrial fibrillation. She was treated for cerebral infarction and recovered with sequelae. The atrial fibrillation was reported ongoing and was assessed by the investigator as being moderate and related to the study therapy.
- 73 year-old white woman with cutaneous lupus erythematous, hypercholesterolemia, severe hypoproteinemia received acto+bezlo on day 7 of SoC therapy with metronidazole. She subsequently developed colitis and anasarca due to hypoalbuminemeia on day 21. She was also diagnosed with alveolar hemorrhage on day 22. The alveolar hemorrhage resolved and she was discharged on day 35. The alveolar hemorrhage was assessed by the investigator as moderate and related to the study therapy.
- 84-year-old Asian man with stomach cancer and adhesive ileus was diagnosed with CDI while hospitalized. He received acto+bezlo infusion 5 days after initiation SoC with PO metronidazole. He was discharged on Day 3. On day 34, the subject experienced 2 loose stools (diarrhea) which tested positive for CDI. He was hospitalized again on day 34 and treated with PO metronidazole. The subject recovered and was discharged on day 36. The investigator stated that the relationship of the CDI recurrence to study medication was unknown due to the temporal sequence of AE to administration of study medication.

- 85 year old white man with prostate adenocarcinoma, asthma, atrial fibrillation and hepatitis received acto+bezlo infusion 3 days after initiation of standard of care PO metronidazole treatment. While in the extension phase of the study, on day 249, he fell at home, and was hospitalized for fractured right femur. He underwent hemiarthroplasty and was discharged from the hospital on Day 258. The investigator noted the event (femur fracture) could be related to study medication.

Reviewer Comment: Although the relationship between the SAEs and the study drugs cannot be completely ruled out, all four subjects had other concurrent illnesses or demographics that could have contributed to the SAEs. No clear relationship between the SAEs and the study drug could be concluded.

8.4.3. Dropouts and/or Discontinuations Due to Adverse Effects

Bezlo, acto+bezlo and placebo (0.9% sodium chloride) were given as one-time infusion in their respective arms; hence, drug discontinuation in the traditional term is not applicable. The infusion volume ranged from 200-250ml, and an in-line filter was used for the infusions. There were 47 infusion interruptions, with 9, 34 and 4 interruptions in the bezlo, acto+bezlo and acto arm, respectively. Majority of these were due to technical difficulties. Sponsor's explanation for the higher technical difficulties in the acto+bezlo arm was propensity of acto to form visible particles at low concentrations. This is supported by the higher treatment interruptions seen in

acto only arm in study P001 (27/236) and formation of visible particles at low concentration.

There was only one infusion interruption that led to permanent discontinuation. This occurred in a 32 year-old- white male with AIDS who developed ventricular arrhythmia, chills, and dizziness with an onset approximately 36 minutes after the start of bezlo infusion. His pre-infusion electrocardiogram (ECG) showed atrial and ventricular rate of 103 beats per minute (bpm). The bezlo infusion was discontinued, and the subject was treated with fenistil (an antihistamine/anticholinergic drug), prednisolone, and ranitidine intravenously. The chills s and ventricular tachyarrhythmia resolved within 2-5minutes, and dizziness resolved within 90 minutes. ECG performed about 30 minutes after discontinuing the infusion showed atrial and ventricular rates of 99, and signs of pre-excitation syndrome. The investigator and the sponsor classified the event as drug-related.

Reviewer Comment: There is a strong temporal relationship between the infusion of bezlo and occurrence of the serious adverse events of ventricular arrhythmia, chills, and dizziness in the subject described above. The subject experienced significant infusion reaction and the reviewer agrees with the investigator and sponsor's classification of the event as drug-related.

Table 49 summarizes the overall subject disposition in the safety population.

Table 49: Subject Disposition in the Safety Population

	Bezlo N=786 n, (%)	Acto+Bezlo N=777 n, (%)	Placebo N=781 n, (%)	Total N=2344 n, (%)	Acto N=236 n, (%)
Completed	675 (85.9)	664(85.5)	654 (83.7)	1993 (85)	201 (85.2)
Death [a]	52 (6.6)	47 (6)	57 (7.3)	156 (6.7)	27 (11.4)
Death including subjects reported in other categories [b]	64 (8.1)	54 (6.9)	62 (7.9)	180 (7.7)	28 (11.9)
Withdrawal by subject	34 (4.3)	30 (3.9)	39 (5)	103 (4.4)	3 (1.3)
Lost to follow-up	21 (2.7)	26 (3.3)	22 (2.8)	69 (2.9)	2 (0.85)
Physician decision	3 (0.38)	5 (0.64)	3 (0.38)	11 (0.47)	1 (0.42)
Adverse event	1 (0.13)	1 (0.13)	2 (0.26)	4 (0.17)	1 (0.42)
Protocol violation	0	3 (0.39)	2 (0.26)	5 (0.21)	0
Progressive disease	0	1 (0.13)	2 (0.26)	3 (0.13)	0
Lack of efficacy	0	0	0	0	1 (0.42)

[a] As reported by the sponsor
[b] Includes all deaths among subjects who received study treatment including those who died outside of a 90-day follow-up period, died during extension Phase of study P002, withdrew consent, or were reported under the completed or discontinued due to AE categories. The numbers in this row should not be included when counting the total.

Of note, in addition to the subjects classified in the death category in Table 49 , there were subjects who died but reported under other disposition categories. Thus, in the bezlo arm, four additional subjects had fatal outcomes (n=2 from the completed category, n=1 from discontinued due to AE category, and n=1 from withdrawal by subject category).

Among the three subjects who were removed from the study by physician decision, 2 were reportedly non-compliant with study visits/ procedures; there was no reported reason for the third subject.

Reviewer Comment: Overall, about 85% of subjects in each arm completed the study. There was no significant imbalance in subject withdrawn from study in the active arms, including the bezlo arm, when compared to placebo.

8.4.4. Significant Adverse Events

Please refer to the laboratory section for analysis of hematological AEs by study arm. In general, there did not appear to be an imbalance in degree of anemia, thrombocytopenia, or neutropenia comparing the bezlo arm to the placebo arm.

8.4.5. Treatment Emergent Adverse Events and Adverse Reactions

Among the 2344 subjects that make up the safety population, 1622 reported one or more treatment emergent adverse events over the 12-weekstudy period, with 5394 events reported. Of these, 1418 subjects reported TEAE during the first 4 weeks of the study, with total of 3819 events (1209 in the acto+bezlo, 1319 in bezlo and 1291 in placebo arms).

Table 50 summarizes the number of AE events and subjects that experienced at least one AE over the 4-week period.

Table 50: Treatment Emergent Adverse Events (TEAE) Over the 4-week and 12-week Study Period

	4-week period	12-week period
Number of reported AEs	3819	5349

Acto*	159	716
Acto+Bezlo	1209	1696
Bezlo	1319	1803
Placebo	1291	1850
Number of individuals with AE	1418	1622
Acto*	483	184
Acto+Bezlo	455	529
Bezlo	485	548
Placebo	478	545

*Acto arm is not included in the safety population

Reviewer Comment: It should be noted that the sponsor collected all TEAEs in the first 4-week of the study; after that, only serious adverse events were reported. Hence, the data for the 12-week period includes all the TEAEs reported in the 4-week period. In general, there does not seem to be a significant imbalance in the frequency of adverse events for the bezlo and acto+bezlo arm as compared to the placebo arm.

Table 51 summarizes the reported TEAEs by MedDRA SOC classification.

Table 51: Treatment Emergent Adverse Events by System Organ Class

SOC	Bezlo N=786 n, (%)	Acto+Bezlo N=777 n, (%)	Placebo N=781 n, (%)	Acto N=236 n, %)
Gastrointestinal	360(45.8)	332(42.7)	319(40.8)	150(63.6)
Infection & infestation	304(38.7)	268(34.5)	340(43.5)	120(50.8)
General disorders & administration site conditions	178(22.6)	164(21.1)	149(19.1)	69(29.2)
Nervous system disorders	104(13.2)	123(15.8)	112(14.3)	46(19.5)
Respiratory, thoracic & mediastinal disorders	104(13.2)	85(10.9)	99(12.7)	38(16.1)
Investigations	96(12.2)	80(10.3)	86(11)	27(11.4)
Skin & subcutaneous tissue disorders	68(8.65)	56(7.210	88(11.3)	24(10.2)
Metabolism & nutrition disorder	66(8.4)	80*10.3)	93(11.9)	34(14.4)
Musculoskeletal and connective tissue disorders	64(8.14)	78(10)	72(9.22)	23(9.75)
Cardiac disorders	54(6.87)	55(7.08)	56(7.17)	21(8.9)
Renal and urinary disorders	54(6.87)	40(5.15)	59(7.55)	12(5.08)

Blood and lymphatic system disorders	47(5.98)	40(5.15)	41(5.25)	16(6.78)
Injury, poisoning & procedural complication	45(5.73)	56(7.21)	53(6.79)	22(9.32)
Psychiatric disorders	38(4.83)	39(5.02)	37(4.74)	14(5.93)
Vascular disorders	38(4.83)	44(5.66)	48(6.15)	22(9.32)
Neoplasms(benign, malignant and unspecified)	19(2.42)	18(2.32)	17(2.18)	7(2.97)
Eye disorders	13(1.65)	16(2.06)	12(1.54)	6(2.54)
Hepatobiliary disorders	12(1.53)	13(1.67)	9(1.15)	4(1.69)
Reproductive system & breast disorders	6(0.76)	13(1.67)	6(0.77)	2(0.85)
Immune system disorders	5(0.64)	4(0.51)	6(0.77)	2(0.85)
Ear and labyrinth disorders	4(0.51)	4(0.51)	7(0.9)	2(0.85)
Endocrine disorders	2(0.25)	3(0.39)	2(0.26)	0
Social circumstances	0	1	0	0

Table 52 and Table 53 summarize the common TEAEs that occurred in >2% of subjects and were more frequent in the active arms (risk difference>0). Table 54 provides summary of TEAEs that occurred in >3% of subjects in the acto arm. As indicated throughout the review, the data for the acto arm is not included in the safety population, and is presented for completeness.

Table 52: Common Treatment Emergent Adverse Events that Occurred in >2% of Bezlotoxumab-Treated Subjects

MedDRA Preferred Term	Bezlo N=786 n (%)	Placebo N=781 n (%)
Nausea	52(6.62)	39(4.99)
Diarrhea	47(5.98)	45(5.76)
Pyrexia	36(4.58)	27(3.46)
Headache	35(4.45)	24(3.07)
Vomiting	31(3.94)	21(2.69)
Peripheral edema	19(2.42)	14(1.79)
Fatigue	18(2.29)	12(1.54)
Cough	17(2.16)	8(1.02)
Dyspnea	17(2.16)	13(1.66)

Table 53: Common Treatment Emergent Adverse Events that Occurred in >2% of Actoxumab+Bezlotoxumab-Treated Subjects

MedDRA Preferred Term	Acto+Bezlo N=777 n (%)	Placebo N=781 n (%)
Nausea	47(6.05)	39(4.99)
Diarrhea	46(5.92)	45(5.76)
Headache	33(4.25)	24(3.07)
Pyrexia	31(3.99)	27(3.46)
Dizziness	26(3.35)	23(2.94)
Vomiting	24(3.09)	21(2.69)
Fatigue	21(2.7)	12(2.69)
Constipation	16(2.06)	10(1.28)

Table 54: Common Treatment Emergent Adverse Events that Occurred in >3% of Actoxumab-Treated Subjects

MedDRA Preferred Term	Acto N=236 n, (%)	Placebo N=781 n, (%)
Nausea	28(11.9)	39(4.99)
Clostridium difficile infection	21(8.9)	48(6.15)
Abdominal pain	15(6.36)	34(4.35)
Headache	14(5.93)	24(3.07)
Urinary tract infection	13(5.51)	35(4.48)
Pyrexia	11(4.66)	27(3.46)
Fatigue	11(4.66)	12(1.54)
Vomiting	10(4.24)	21(2.69)
Peripheral edema	9(3.81)	14(1.79)

8.4.6. Laboratory Findings

The ADLB datasets for study P001 and P002 as well as the integrated summary of safety (ISS) ADLB datasets were reviewed. Safety laboratory investigations included blood tests for complete blood counts with white blood cell differential and platelets, liver function panel, serum electrolytes and urine for urinalysis with microscopic analysis. Blood and urine samples were taken for initial testing within 24 hours of the infusion on Day 1 (Visit 1). Thereafter, blood and urine were collected at the scheduled post-infusion study visits at Day 4 (± 1 day), Day 11 (± 2 days), and Week 4 (Day 29 ± 3 days). Blood samples were also taken to assess the occurrence of anti-drug antibody (ADA) after treatment with the study drugs. Details of ADA analysis are presented separately in section 8.4.10

Assessment of Liver Tests

Elevations of alanine transaminase (ALT), and aspartate transaminase (AST) by 3-times, 5-times and 10-times the upper limit of normal, elevation of total bilirubin 2-times the upper limit of normal, and ≥50% increase in alkaline phosphates for each arm by study visit are summarized in Table 55.

Table 55: Liver investigations by Study Visit

	Bezlo n (%)	Acto+ Bezlo n (%)	Placebo n (%)	Acto n (%)
Day 4 visit	N=756	N=740	N=749	N=226
ALT 3X ULN	10(1.3)	12 (1.6)	18 (2.4)	3 (1.3)
ALT 5X ULN	2 (0.26)	4 (0.54)	4 (0.53)	1 (0.4)
ALT 10X ULN	0	0	2 (0.27)	0
AST 3X ULN	10 (1.3)	10 (1.4)	13 (1.7)	0
AST 5X ULN	3 (0.4)	4 (0.54)	4 (0.53)	0
AST 10X ULN	1 (0.13)	2 (0.27)	2 (0.27)	0
Total bilirubin 2X ULN	7 (0.93)	9 (1.2)	8 (1.1)	4(1.8)
Alkaline phosphatase 2XULN	29 (3.8)	23 (3.1)	24 (3.2)	7 (3.1)
Day 11 visit	N=733	N=713	N=728	N=219
ALT 3X ULN	7 (0.95)	11 (1.5)	9 (1.23)	1 (0.46)
ALT 5X ULN	0	4 (0.56)	6 (0.82)	1 (0.46)
ALT 10X ULN	0	0	5 (0.69)	0
AST 3X ULN	5 (0.68)	9 (1.26)	9 (1.2)	2 (0.91)
AST 5X ULN	2 (0.27)	2 (0.28)	6 (0.82)	1 (0.46)
AST 10X ULN	1 (0.14)	0	3 (0.41)	0
Total bilirubin 2X ULN	5 (0.68)	13 (1.8)	6 (0.82)	6 (2.7)
Alkaline phosphatase 2XULN	25 (3.4)	26 (3.6)	23 (3.2)	6 (2.7)
Week 4	N=677	N=671	N=666	N=210
ALT 3X ULN	6 (0.89)	10(1.5)	4(0.6)	3 (1.4)
ALT 5X ULN	0	4(0.6)	2(0.3)	1(0.48)
ALT 10X ULN	0	1(0.15)	0	0
AST 3X ULN	4(0.59)	9(1.3)	3(0.45)	2(0.95)
AST 5X ULN	1(0.15)	5(0.75)	0	0
AST 10X ULN	0	1(0.15)	0	0
Total bilirubin 2X ULN	9(1.3)	101.5)	0	4(1.9)
Alkaline phosphatase >2XULN	24(3.5)	26(3.9)	24(3.6)	8(3.8)

ALT: Alanine transaminase; AST: Aspartate transaminase; N indicates number of subjects with the available lab for the visit in the specified arm; n indicates the number of subjects with the lab abnormality in that arm.

Reviewer Comment: Overall, there is no significant difference in the liver labs. There were slightly more subjects in the placebo arm with elevated transaminases and total bilirubin compared to the bezlo or acto+bezlo arm for each study visit. Since bezlo is a monoclonal antibody, it is catabolized in the body, and not metabolized in the liver. Hence, we did not anticipate significant impact on the liver, and the results support the hypothesis.

Additional analyses were performed to assess for potential drug-induced liver injury (DILI), and liver abnormalities that fulfill the criteria for Hy's law are presented in Table 56.

Table 56: Subjects that Fulfilled Hy's Law by study Visit and study arm

Subjects with ALT or AST 3X ULN AND total bilirubin 2XULN	Visit	Bezlo	Acto+ Bezlo	Placebo	Acto
	Day 4	1	2	3	0
	Day 11	0	2	1	1
	Week 4	1	3	0	1
Subjects with ALT or AST 3X ULN AND total bilirubin 2XULN AND ALP within the normal range	Visit	Bezlo	Acto+ Bezlo	Placebo	Acto
	Day 4	0	0	1	0
	Day 11	0	0	1	0
	Week 4	0	2	0	0

ULN=Upper limit of normal; ALP=alkaline phosphatase; ALT=alanine transaminase; AST=Aspartate transaminase

For day 4 visit, one subject from the bezlo arm, a 78 year-old female with multiple co-morbidities including dyslipidemia and congestive heart failure, had a 3 times the upper limit of normal value for aminotransferases and 2 times the normal value for bilirubin. This subject also had four-fold increase in alkaline phosphatase from baseline (>3xULN). The subject's ALT normalized during day 11 visit, with continued elevation in ALP and bilirubin; the subject withdrew from the study between week 3 and 4.

Reviewer Comment: In addition to elevated transaminases, the subject also had elevated alkaline phosphatase; it is possible that the subject had cholestasis and this may have contributed to elevated transaminases.

For week 4 visit, one subject from the bezlo arm, a 32 year-old female with a history of cirrhosis, thalassemia and iron overload, had a 4 times the ULN value for ALT(130 IU/L) and a 3 times the ULN for AST(92 IU/L), with bilirubin 2 times the ULN(2.3mg/DL) and slightly elevated ALP of 126 IU/L(ULN defined as 105IU/L). The hepatic enzyme elevations were attributed to the subject's underlying conditions. The subject was reported to have recovered.

Reviewer Comment: The reviewer agrees with the sponsor's assessment of subject's underlying conditions (cirrhosis) contributing to the observed hepatic enzyme elevations.

Assessment of Kidney Function

Changes from baseline in kidney function were evaluated for each study visit. The results are summarized in Table 57 and Table 58.

Table 57: Clinically Significant Changes in Creatinine from Baseline

	Bezlo Proportion, (%)	Acto+Bezlo Proportion, (%)	Placebo Proportion, (%)	Acto Proportion, (%)
Increase by≥0.3mg/dL				
Day 4	59/760(7.8)	55/741(7.4)	49/750(6.5)	23/226 (10.2)
Day11	91/739(12.3)	67/718(9.3)	78/731(10.7)	38/220 (17.3)
Week 4	81/680(11.9)	83/676(12.3)	84/671(12.5)	42/210 (20)
Post baseline creatinine>2mg/dL				
Day 4	70/760(9.2)	41/741(5.5)	57/750(7.6)	21/226 (9.3)
Day 11	68/739(9.2)	42/718(5.8)	65/731(8.9)	18/220 (8.2)
Week 4	61/680(9)	40/676(5.9)	50/671(7.5)	19/210 (9)

Table 58: Changes in Blood Urea Nitrogen (BUN) from Baseline

	Bezlo Proportion, (%)	Acto+Bezlo Proportion, (%)	Placebo Proportion, (%)	Acto Proportion, (%)
BUN increase by≥50%				
Day 4	35/760 (4.6)	24/741 (3.2)	32/750 (4.3)	13/226 (5.8)
Day11	74/739 (10)	63/718 (8.8)	65/731 (8.9)	35/220 (15.9)
Week 4	87/680 (12.8)	80/676 (11.8)	88/671 (13.1)	37/210 (17.6)

Post baseline BUN >31mg/dL				
Day 4	92/760 (12.1)	55/741 (7.4)	83/750 (11.1)	26/226 (11.5)
Day 11	85/739 (11.5)	62/718 (8.6)	78/731 (10.7)	32/220 (14.4)
Week 4	81/680 (11.9)	66/676 (9.8)	91/671 (13.6)	26/210 (12.4)

Reviewer Comment: There is no significant difference in creatinine and BUN changes from baseline when comparing the bezlo arm to the placebo at each study visit. The acto+bezlo arm had overall slightly less subjects with increase in creatinine and BUN from baseline.

Assessment of Hematologic Labs

Table 59 summarizes significant changes in hemoglobin, platelet count, and neutrophil counts from baseline for subjects in each study arm.

Table 59: Changes in Hemoglobin, Platelet, and Neutrophil Count from Baseline

	Bezlo n (%)	Acto+ Bezlo n (%)	Placebo n (%)	Acto n (%)
Hemoglobin decrease by ≥3gm/dL from baseline				
Day 4	2/738 (0.3)	3/722 (0.4)	5/737 (0.7)	1/217(0.5)
Day11	9/718 (1.3)	10/690(1.4)	10/710 (1.4)	4/213(1.9)
Week 4	7/661 (1.1)	8/660 (1.2)	13/648(2)	6/207(2.9)
Post baseline platelet count <100x10^3/µL				
Day 4	31/716 (4.3)	30/694 (4.3)	36/720 (5)	6/210(2.9)
Day 11	35/703 (5)	25/671 (3.7)	28/695 (4)	8/203 (3.9)
Week 4	26/641 (4.1)	21/650 (3.2)	27/638 (4.2)	9/200 (4.5)
Post baseline neutrophil count <1000/µL				
Day 4	12/728 (1.6)	10/717 (1.4)	7/722 (0.9)	2/216 (0.9)
Day 11	11/708 (1.6)	14/685 (2)	14/702 (2)	3/210 (1.4)
Week 4	5/649 (0.8)	9/656 (1.4)	11/642(1.7)	4/203 (2)

Reviewer Comment: There was no imbalance in the number of subjects with drop in hemoglobin, thrombocytopenia or neutropenia when comparing the bezlo arm to placebo.

8.4.7. Vital Signs

Vital signs were recorded at baseline, 30-minutes after start of infusion, end of infusion, and at day 4 and day11 visits. A summary of clinically significant changes from baseline in vital signs are presented in Table 60 and

Table 61 for 30-minute post infusion and end of infusion, respectively.

Table 60: Changes from Baseline in Vital Signs 30-minutes after Start of Infusion

	Bezlo N=786, n (%)	Acto+ Bezlo N=777 n (%)	Placebo N=781 n (%)	Acto N=236 n (%)
Drop in SBP by ≥30% from baseline	1 (0.13)	5 (0.64)	1 (0.13)	0
SBP<90mmHg	9 (1.1)	11 (1.4)	7 (0.9)	3 (1.3)
Drop from BL≥5mmHg	3 (0.38)	5 (0.64)	3 (0.38)	1 (0.4)
Drop in DBP by ≥20mmHg	17 (2.2)	19 (2.4)	24 (3.1)	3 (1.3)
DBP<60mmHg	89 (11.3)	95 (12.2)	95 (12.2)	31 (13.1)
Drop from BL by ≥5mmHg	40 (5.1)	49 (6.3)	48 (6.1)	8 (3.4)
HR>120bpm	3 (0.38)	2 (0.25)	6 (0.77)	3 (1.3)
Increase from BL by ≥10bpm	2 (0.25)	1 (0.13)	2 (0.25)	1 (0.42)
HR<60bpm	68 (8.7)	74 (9.5)	64 (8.2)	20 (8.5)
Drop from BL by ≥10bpm	6 (0.76)	12 (1.5)	12 (1.5)	7 (3)
Temp >37.8°C	25 (3.2)	15 (1.9)	16 (2)	3 (1.3)
Increase from BL by ≥1°C	3 (0.38)	3 (0.39)	0	1 (0.42)
Temp<35 °C	2 (0.25)	0	3 (0.38)	0
Drop from BL by≥1°C	0	0	2 (0.26)	0
Temp increase by ≥2 °C from BL	1 (0.13)	1(0.13)	0	0
Temp decrease by ≥2 °C from BL	0	1 (0.13)	1 (0.13)	0
RR>24 per minute	29 (3.7)	22 (2.8)	25 (3.2)	8 (3.4)
Increase from BL by ≥3 per minute	14 (1.8)	12 (1.5)	10 (1.3)	4 (1.7)
RR<12 per minute	12 (1.5)	7 (0.9)	14 (1.8)	0
Decrease from BL by ≥3 per minute	1 (0.13)	2 (0.26)	1 (0.13)	0
BL – baseline				

Table 61: Changes from Baseline in Vital Signs at End of Infusion

	Bezlo N=786, n (%)	Acto+ Bezlo N=777 n (%)	Placebo N=781 n (%)	Acto N=236 n (%)
Drop in SBP by ≥30% from baseline	2 (0.25)	5 (0.64)	2 (0.26)	0
SBP<90mmHg	11 (1.4)	9 (1.16)	5 (0.64)	4 (1.7)
Drop from BL by≥ 5mmHg	3 (0.38)	5 (0.6)	2 (0.26)	1 (0.42)
Drop in DBP by ≥20mmHg	19 (2.4)	21 (2.7)	15 (1.9)	11 (4.7)
DBP<60mmHg	86 (10.9)	83 (10.7)	82 (10.5)	34 (14.4)
Drop from BL≥5mmHg	43 (5.5)	34 (4.34)	41 (5.2)	15 (6.4)
HR>120	4 (0.5)	4 (0.51)	5 (0.64)	2 (0.84)
Increase from BL by ≥10bpm	4 (0.5)	2 (0.26)	2 (0.26)	1 (0.42)
HR<60	68 (8.7)	65 (8.4)	68 (8.7)	16 (6.8)
Drop from BL by ≥10bpm	9 (1.1)	6 (0.77)	14 (1.8)	3 (1.3)
Temp ≥37.8 ^{o}C	21 (2.7)	18 (2.3)	19 (2.4)	4 (1.7)
Increase from BL by ≥1°C	8 (1)	3 (0.39)	1 (0.13)	2 (0.84)
Temp<35 ^{o}C	1 (0.13)	1 (0.13)	3 (0.38)	0
Drop from BL by ≥1°C	0	0	1 (0.13)	0
Increase by≥2°C from BL	2 (0.25)	0	0	0
Decrease by≥2 ^{o}C from BL	0	1	1 (0.13)	0
RR>24	25 (3.2)	17 (2.2)	27 (3.5)	6 (2.5)
Increase from BL≥3 per minute	9 (1.1)	7 (0.9)	11 (1.4)	2 (0.84)
RR<12	8 (1)	7 (0.9)	10 1.3)	2 (0.84)
Decrease from BL≥3 per minute	0	2 (0.26)	2 (0.26)	1 (0.42)
BL - baseline				

Looking at the bezlo arm at 30 minutes after start of infusion, there were 10 subjects that fulfilled Sampson's criteria of decrease in systolic BP with either drop in SBP by 30 % (n=1) or SBP< 90mmHg (n=9). Among those with SBP<90mmHg, 3 had drop from baseline by >5mmHg. At the end-of-infusion, there were 13 subjects that met the criteria of a drop by 30% in SBP (n=2) or SBP<90mmHg (n=11). Among those with SBP<90mmHg, three had drop from baseline by >5mmHg

One subject in the bezlo arm and 4 subjects in the acto+bezlo arm had drop in SBP by 30% at both 30-minute after start of infusion and end of infusion. Among subjects with SBP less than 90mmHg, 16 (n=5 in the bezlo arm, 6 in acto+bezlo arm, 3 in placebo arm and 2 in acto arm) had SBP<90mmHg at both 30-minutes after start of infusion and end of infusion. There was only one subject in the bezlo arm that had a drop of>5mmHg from baseline at both times, and synopsis of the patient is presented below.

The subject in the bezlo arm is a 91 year-old female with multiple comorbidities including hypertension, atrial fibrillation, and hypercholesterolemia with the following baseline (pre-infusion) vital signs: BP 150/70 mmHg, HR 88bpm, RR 16 and temperature of 37.2. At 30 minutes after start of infusion, her blood pressure decreased to 104/60, and heart rate and respiratory rate increased to 160 and 23, respectively. She remained afebrile. At the end of infusion, she had further drop in blood pressure to 90/64 mmHg, heart rate 166 bpm, RR 20 and temperature 37 C. Her concomitant medications on day of infusion included metronidazole, furosemide, terbutaline sulfate, ipratropium bromide, ceftriaxone, and spironolactone.

Reviewer Comment: Overall, there was no significant difference between the treatment arms and placebo in regards to changes from baseline of heart rate, systolic blood pressure, diastolic blood pressure, body temperature, and respiratory rate. Majority of subjects with SBP below 90mmHg had baseline SBP below 90mmHg, which likely is due acute illness (infection) rather than infusion reaction. The one bezlo subject presented in the synopsis above appears to have had an infusion reaction with significant changes in SBP and tachycardia; however, there may also be contribution of her vasoactive concomitant medication to the changes in vital signs. In addition, the one bezlo subject described in section 8.4.3 that developed ventricular tachycardia dizziness and chills during bezlo infusion is considered to have had infusion reaction.

8.4.8. Electrocardiograms (ECGs)

There was no dedicated QT study during the drug development program. In the Phase 1 studies P004, P005 and P006, healthy subjects had ECG monitoring and time-matched pharmacokinetic sampling after administration of 10mg/kg to 20mg/kg of acto+bezlo. The sponsor use a QT interval adjusted by population-derived correction (QTcP), and noted a trend of decreasing change from baseline QTc with increasing bezlo serum concentration that was statistically significant but was felt to not be clinically meaningful.

In the two Phase 3 studies, electrocardiogram (ECG) measurements were performed pre-infusion and within 120 minutes post infusion during the Phase 3 studies.

8.4.9. QT

ECG was performed just prior to start of infusion (pre-infusion) and within 2 hours after the end of infusion; summary of the QT findings corrected for heart rate using Frederica correction is presented below in Table 62.

Table 62: Summary of QTc Measurements

	Bezlo n,(%)	Acto+ Bezlo n,(%)	Placebo n,(%)	Acto n, (%)
Baseline ECG	N=771	N=761	N=768	N=228
QTc 450-480 ms	82 (10.6)	86 (11.3)	94 (12.2)	24 (10.5)
QTc 481-500 ms	29 (3.8)	29 (3.8)	18 (2.3)	4 (1.8)
QTc >501 ms	21 (2.7)	13 (1.7)	12 (1.6)	10 (4.4)
Post-infusion ECG	N=765	N=755	N=758	N=224
QTc 450-480 ms	96 (12.5)	96 (12.7)	108 (14.2)	33 (14.3)
QTc 481-500 ms	26 (3.4)	25 (3.3)	13 (1.7)	3 (1.3)
QTc >501 ms	17 (2.2)	16 (2.1)	15 (2)	13 (5.8)
QTc change from BL≤30 ms	713 (93.2)	692 (91.7)	711 (93.8)	203 (90.6)
QTc change from BL >30-≤60msec	35 (4.6)	43 (5.7)	32 (4.2)	15 (6.7)
QTc change from BL>60ms	10 (1.3)	15 (2)	15(2)	4 (1.8)

ms=milliseconds; BL=baseline; QTc= Frederica-corrected QT interval

Reviewer Comment: There were numerically more subjects in the bezlo and acto arm with baseline and post-baseline QTc 481-500ms; otherwise, there were no significant differences in the number of subjects with QT changes from baseline when comparing the active arms (bezlo and acto+bezlo) vs. the placebo arm.

8.4.10. Immunogenicity

Development of anti-drug antibody (ADA), anti-bezlotoxumab antibody, were assessed at baseline (Day1), visits 4 (Day29), 5 (Day57), and 6 (Day 85). In addition, in the extension subset of 300 subjects in study P002, presence of ADA was assessed on Day168. Summary of the results of these tests are summarized in Table 63.

Table 63: Presence of Anti-drug Antibody (ADA) in study P001 and P002

Trial	Study arm	Subjects dosed bezlo	Total evaluable* N	Negative#,1 N (%)	Inconclusive#,2 N (%)	Non-TE Positive#,3 N (%)	TE-Positive#,4 N (%)

P001	Bezlo	390	351	239(68.1)	112(31.9)	0(0)	0(0)
	Acto+ Bezlo	387	356	237(66.6)	114(32)	5(1.4)	0(0)
P002	Bezlo	359	359	271(75.5)	88(24.5)	0(0)	0(0)
	Acto+ Bezlo	348	348	266(76.4)	78(22.4)	4(1.1)	0(0)
P001+ P002		1563	1414	1013(71.6)	392(27.7)	9(0.6)	0(0)

*:Includes patients with at least one ADA sample available after treatment with bezlo or acto+bezlo
#: Denominator was total number of evaluable subjects; TE=Treatment Emergent
[1]: Negative result is defined as all pre- and post-treatment samples were negative in the ADA confirmatory assays and the drug concentration in the last post-dose sample was below the respective drug tolerance level (DTL) for the ADA assay
[2]: Inconclusive is defined as all pre-and post-treatment samples were negative in the ADA confirmatory assay and the drug concentration in the last post-dose sample was equal or above the respective DTL for the ADA assay
[3]: TE- positive result is defined as pre-treatment sample was negative and at least one post-treatment sample was positive in the ADA confirmatory assay (Treatment-induced positive) OR pre-and post-treatment treatment samples were both positive in the ADA confirmatory assay AND the post-treatment titer increased by ≥2-fold (treatment boosted positive)
[4]: Non-TE positive result is defined as pre-treatment sample was positive and post-dose samples were negative in the ADA confirmatory assay OR pre-and post-treatment samples were positive in the ADA confirmatory assay with a <2-fold increase in titer in the post-treatment sample

Data source: Modified from sponsor's tables 2.7.2:8 and 2.7.2:10 in Summary of Clinical Pharmacology Studies section

Reviewer Comment: Among the 710 bezlo subjects that had at least one post-infusion ADA testing, none were reported to have developed ADA, although 28% had inconclusive test results. Among the 704 subjects that received acto+bezlo, there were no subjects reported to have post treatment ADA; however, 9(1.3%) had non-treatment emergent ADA.

In this review, several factors were considered regarding potential immunogenicity of bezlo, including patient-specific factors, route of administration, and origin of the study product. The intravenous administration of bezlo, its fully human origin, and its target being a non-endogenous makes it less likely candidate for eliciting immunogenicity. In addition, pre-clinical studies of repeated dose administration and tissue cross-reactivity studies did not show significant immunogenicity. Please refer to Dr. Terry Miller's, pharmacology-toxicology review, for details of pre-clinical studies. The results of the ADA testing are consistent with the expectation of low immunogenicity; however, some drug reactions are not related to pharmacological activity of the drug and are not predictable. Thus, continued

pharmacovigilance is needed in the post marketing setting. In addition, the present data only supports low immunogenicity of bezlotoxumab when given as one-time dose. Only 30 subjects in one Phase 1 study received two doses of bezlotoxumab; hence, the potential immunogenicity of bezlotoxumab after multiple doses cannot be assessed with the available data.

8.5. Analysis of Submission-Specific Safety Issues

Infusion-related adverse events were of particular interest since bezlo is a monoclonal antibody. The sponsor defined post-infusion reaction as any of the following symptoms occurring within 24 hours post-infusion: infusion-site adverse experiences, pyrexia, chills, rash, arthralgia, myalgia, joint swelling, obstructive airways disorder, bronchospasm, stridor, dysphonia, headache, fatigue, pruritus, urticaria, hypotension, hypertension, nasal congestion, nausea, vomiting, flushing, angioedema, dyspnea, and dizziness/lightheadedness. The sponsor used the Division of Microbiology and Infectious Diseases (DMID) Adult Toxicity Table (November 2007, draft) grading for the reporting of these adverse events.

This reviewer looked at clinical signs and symptoms of infusion –related reaction that occurred within the 24 hours following infusion of study drugs. These include:

- Cutaneous -infusion-site reaction, rash, pruritus, urticaria, flushing, angioedema
- Respiratory -dyspnea, bronchospasm, stridor, dysphonia, sudden nasal congestion, wheeze, sensation of throat closure or choking
- Cardiovascular-faintness, tachycardia, bradycardia, chest pain, hypotension, hypertension, sense of impending doom
- Gastrointestinal-nausea, vomiting, diarrhea
- Musculoskeletal -arthralgia, myalgia, joint swelling
- Nervous system-headache, loss of consciousness, dizziness/lightheadedness
- General conditions-Body temperature alteration(pyrexia, diaphoresis, chills), fatigue, and asthenia

Sampson's criteria of a combination of signs/symptoms from the different organ systems were applied to focus on subjects who potentially experienced infusion-related reactions.

8.5.1. Infusion-related Reactions

For this analysis, adverse events that started on day of infusion (day 1) and the following day (day 2) were evaluated. Comparing the occurrence of infusion-related reactions as defined above, 261 subjects (n=106 bezlo, n=77 acto+bezlo, n=78 placebo) had at least one AEs that appeared to be infusion-related. Summary of incidence of TEAE and infusion-related AEs are presented in

Table 64.

Table 64: Summary of TEAEs and Infusion -related AEs during Days 1 and 2 After Infusion

	Bezlo N=786, n (%)	Acto+ Bezlo N=777 n (%)	Placebo N=781 n (%)	Total N=2344 N, (%)
Subjects with any TEAE on day 1 or day 2	148 (18.8)	120 (15.4)	114 (14.6)	382 (16.3)
Subjects with serious TEAEs	7 (0.9)	8 (1.03)	9 (1.2)	24 (1)
Subjects with infusion related AE*	106 (13.5)	77 (9.9)	78(10)	261 (11.1)
Subjects with serious AE	2(0.2)	4 (0.5)	3 (0.4)	9 (0.4)

*Reviewer's definition of infusion related AEs as defined in section 8.5 were used.

In the bezlo arm, there was one subject that had respiratory and gastrointestinal symptoms: this was a 72-year-old man with significant medical history for coronary and peripheral artery disease, hypertension, hyperlipidemia who was admitted for abdominal pain and diarrhea, and was found to have *C. difficile* infection and pancolitis. He has a small brownish emesis prior to infusion of bezlo that was not known to the study staff prior to infusion bezlo. Following the infusion, he had two small brown-colored emesis. His vital signs and respiratory status were stable during infusion. However, he had an acute respiratory decompensation that necessitated intubation less than 24 hours after infusion of bezlo. He was taken to the OR emergently for colectomy. The subject did well initially but had complications that required additional surgical intervention, and subsequently died 25 days after infusion of bezlo.

No other bezlo subject fulfilled the combination of cutaneous, respiratory, cardiac, and gastrointestinal symptoms outlined in Sampson's criteria for signs and symptoms of anaphylaxis. Four subjects in the placebo arm and one subject in the acto+bezlo arm had cutaneous and gastrointestinal symptoms. The acto+bezlo subject was an 83-year-old Japanese male who had reported nausea and pruritus about an hour prior to infusion.

Reviewer Comment: The symptoms of the bezlo subject described in the above synopsis appear to have been related to his C. difficile colitis and less likely due to infusion reaction. There were 4 subjects with GI and skin manifestations, mostly pruritus. The cutaneous symptoms may be due to other concomitant medication the subjects were receiving and the GI symptoms may partly be due to CDI's clinical manifestation.

Reference ID: 3989587

Please refer to section 8.4.7 for details of subjects that fulfilled Sampson's criteria of decrease blood pressure.

In the adverse event dataset, there were seven (n=2 bezlo, n=1 acto+bezlo, n=4 placebo) subjects who were diagnosed with hypotension on the day of or the day after infusion (days 1 and 2). A synopsis for the two subjects in the bezlo arm is presented below:
The first subject was a 76-year-old male with diabetes with baseline BP of 118/51 mmHg, HR 89 bpm, RR 14, temperature of 37.3C; at 30-min post start of infusion, his vital signs were BP 133/65 mmHg, HR 100, RR 13, temp 37.6, and end of infusion BP of 103/57 mmHg, HR 100 bpm, RR 12, temp 37.6C. The reported low blood pressure lasted for one day.

The second subject was a 76-year-old Native American female with diabetes, atrial fibrillation hypothyroidism, and primary squamous cell. Her pre-infusion vital signs were BP 122/54 mmHg, HR 75 bpm, temp of 36.8C; 30 minutes after start of infusion, her blood pressure dropped to 92/49 mmHg; HR was 77 bpm. The low blood pressure lasted for 10 minutes and she recovered with the following vitals at end of infusion: BP of 135/48 mmHg, HR of 84 bpm, and temperature of 36.9 C.

Neither of these two subjects' decrease in blood pressure fulfilled the Sampson's criteria of anaphylaxis.

Reviewer Comment: In addition to the two subjects reported above, there was one additional subjects described in section 8.4.7 that had decrease in blood pressure. However, parsing out the contribution of ongoing infection to the decrease in blood pressure is difficult. Also, the one subject described in section 8.4.3 who developed ventricular arrhythmia, chills and dizziness is considered to have infusion reaction. In total, there were 4 potential infusion-related reactions.

8.5.2. Safety Analysis by Body Weight

Since bezlo is dosed by body weight, potential effects of higher body weight, and overall higher dose, was assessed. Table 65 summarizes the results of this analysis.

Table 65: Safety Analysis by Body Weight

	Body weight ≤70kg N=1233 n, (%)	Body weight >70kg N=1111 n, (%)	Total N=2344 n, (%)
All TEAE	870 (70.6)	752 (67.7)	1622 (69.2)
Bezlo	283 (23)	265 (23.9)	548 (23.4)
Acto+Bezlo	295 (23.7)	234 (21.1)	529 (22.6)

Placebo	292 (23.7)	253 (22.8)	545 (23.3)
Serious AE	387 (31.4)	311 (28)	698 (29.8)
Bezlo	122 (9.9)	109 (9.8)	231 (9.9)
Acto+Bezlo	128 (10.4)	84 (7.6)	212 (9)
Placebo	137 (11.1)	118 (10.6)	255 (10.9)
Death	121 (9.8)	45 (4.1)	166 (7.1)
Bezlo	34 (2.8)	22 (2)	56 (2.4)
Acto+Bezlo	38 (3.1)	13 (1.2)	51 (2.2)
Placebo	49 (4)	10 (0.9)	59 (2.5)

TEAE: All subjects with at least one TEAE during the 12-week study period
SAE: All subjects with at least one serious AE during the 12-week study period
Death: All subjects who died due to an AE that occurred during the 12-week study period
Bezlo: Bezlotoxumab; Acto+Bezlo: Actoxumab and Bezlotoxumab

Reviewer's Comments: There were numerically higher number of TEAE, SAE, and death in subjects weighing ≤70kgs as compared to subjects weighing >70kg. The extreme low weight subjects potentially being malnourished may partly explain this. As reported in Table 35, there were subjects with body weight as low as 29.8kg.

8.5.3. Safety analysis by baseline Congestive heart failure

There were numerically more subjects in the bezlo arm who were reported to have congestive heart failure as SAE, and subgroup analysis of subjects with baseline CHF for each study arm is presented below in Table 66 and Table 67.

Table 66: Subgroup Analysis by Baseline Congestive Heart Failure (CHF)

	Baseline CHF N=325 n,(%)	No baseline CHF N=2019 n, (%)	Total N=2344 n, (%)
TEAE	250 (66.9)	1372 (68)	1622 (69.2)
Bezlo	99 (30.5)	449 (22.2)	548 (23.4)
Acto+Bezlo	78 (24)	451 (22.3)	529 (22.6)
Placebo	73 (22.5)	472 (23.4)	545 (23.3)
SAE	159 (48.9)	539 (26.7)	698(30)

Bezlo	63 (19.4)	168 (8.3)	231 (9.9)
Acto+Bezlo	46 (14.2)	166 (8.2)	212 (9)
Placebo	50 (15.4)	205 (10.2)	255 (10.9)
Death	54 (16.6)	112 (5.5)	166 (7.1)
Bezlo	23 (7.1)	33 (1.6)	56 (2.4)
Acto+Bezlo	18 (5.5)	33 (1.6)	51 (2.2)
Placebo	13 (4)	46 (2.3)	59 (2.5)

Baseline CHF by arm: Bezlo, n=118; acto+bezlo, n=103; placebo, n=104
TEAE: Treatment emergent adverse event; SAE: serious adverse event

Table 67: Reported Adverse Events among Subjects with Baseline Congestive Heart Failure (CHF) status

	Bezlo N=118 n, %	Acto+bezlo N=103 n,%	Placebo N=104 n,(%)
TEAE	99 (83.9)	78 (75.7)	73 (70.2)
SAE	63 (53.4)	46 (44.7)	50 (48)
Death	23 (19.5)	18 (17.5)	13(12.5)

TEAE: All subjects with at least one TEAE during the 12-week study period;
SAE: All subjects with at least one serious AE during the 12-week study period
Death: All subjects who died due to an AE that occurred during the 12-week study period
Bezlo: Bezlotoxumab; Acto+Bezlo: Actoxumab and Bezlotoxumab

Reviewer's Comment: It appears that patients with baseline CHF had more AEs, SAEs and death, in general. Furthermore, there was a distinct numerical difference in the occurrence of AEs, SAEs, and death among the bezlo-treated subjects with baseline CHF.

8.6. Safety Analyses by Demographic Subgroups

Subgroup analysis of adverse events by sex, race, and age are presented in Table 68, Table 69, and Table 70, respectively. Of note, in Table 69, due to the small numbers of subjects in each non-white racial group, the data for the non-white racial groups were pulled and presented together.

Table 68: Safety Analysis by Sex

	Female	Male	Total

	N=1327	N=1017	N=2344
TEAE	943 (71)	679 (66.8)	1622 (69.2)
Bezlo	318 (24)	230 (22.6)	548 (23.4)
Acto+Bezlo	302 (22.8)	227 (22.3)	529 (22.6)
Placebo	323 (24.3)	222 (21.8)	545 (23.3)
SAE	369 (27.8)	329 (32.4)	698 (29.8)
Bezlo	122 (9.2)	109 (10.7)	231 (9.9)
Acto+Bezlo	108 (8.1)	104 (10.2)	212 (9)
Placebo	139 (10.5)	116 (11.4)	255 (10.9)
Death	83 (6.3)	83 (8.2)	166 (7.1)
Bezlo	25 (1.9)	31 (3)	56 (2.4)
Acto+Bezlo	23 (1.7)	28 (2.8)	51 (2.2)
Placebo	35 (2.6)	24 (2.4)	59 (2.5)

TEAE: All subjects with at least one TEAE during the 12-week study period
SAE: All subjects with at least one serious AE during the 12-week study period
Death: All subjects who died due to an AE that occurred during the 12-week study period
Bezlo: Bezlotoxumab; Acto+Bezlo: Actoxumab and Bezlotoxumab.

Table 69: Safety Analysis by Race

	Non-white* N=352 n, (%)	White N=1992 n, (%)	Total N=2344 n, (%)
TEAE	250 (71)	1372 (84.6)	1622 (69.2)
Bezlo	96 (27.3)	452 (27.9)	548 (23.4)
Acto+Bezlo	81 (23)	448 (27.6)	529 (22.6)
Placebo	73 (20.7)	472 (29.1)	545 (23.3)
Serious AE	127 (36.1)	571 (35.2)	698 (29.8)

Bezlo	42 (11.9)	189 (11.7)	231 (9.9)
Acto+Bezlo	46 (13.1)	166 (10.2)	212 (9)
Placebo	39 (11.1)	216 (13.3)	255 (10.9)
Death	26 (7.4)	140 (8.6)	166 (7.1)
Bezlo	8 (2.3)	48 (3)	56 (2.4)
Acto+Bezlo	10 (2.8)	41 (2.5)	51 (2.2)
Placebo	8 (2.3)	51 (3.1)	59 (2.5)

*Non-whit races include African American or black, American Indian or Alaskan Native, Asian and multi-racial subjects
TEAE: All subjects with at least one TEAE during the 12-week study period
SAE: All subjects with at least one serious AE during the 12-week study period
Death: All subjects who died due to an AE that occurred during the 12-week study period
Bezlo: Bezlotoxumab; Acto+Bezlo: Actoxumab and Bezlotoxumab

Reviewer Comment: There is no significant difference in the proportion of TEAEs, SAEs and deaths when comparing non-white subjects to white subject across each arm, or comparing men and women.

Table 70: Safety Analysis by Age

	Age<65 years N=1100 n, (%)	Age≥65 years N=1244 n, (%)	Total N=2344 n, (%)
All TEAE	739 (67.2)	883 (71)	1622(69.2)
Bezlo	270 (24.5)	278 (22.3)	548 (23.4)
Acto+Bezlo	221 (20.1)	308 (24.8)	529 (22.6)
Placebo	248 (22.5)	297 (23.9)	545 (23.3)
SAE	260 (23.6)	438 (35.2)	698 (29.8)
Bezlo	98 (8.9)	133 (10.7)	231 (9.9)
Acto+Bezlo	69 (6.3)	143 (11.5)	212 (9)
Placebo	93 (8.5)	162 (13)	255 (10.9)
Death	40 (3.6)	126 (10.1)	166 (7.1)
Bezlo	15 (1.4)	41 (3.3)	56 (2.4)
Acto+Bezlo	10 (0.9)	41 (3.3)	51 (2.2)
Placebo	15 (1.4)	44 (3.3)	59 (2.5)

TEAE: All subjects with at least one TEAE during the 12-week study period
SAE: All subjects with at least one serious AE during the 12-week study period
Death: All subjects who died due to an AE that occurred during the 12-week study period
Bezlo: Bezlotoxumab; Acto+Bezlo: Actoxumab and Bezlotoxumab

Reviewer Comment: There were more SAEs and deaths in subjects ≥65 years old as compared to <65 years in all treatment arms including placebo. Older age is one of the prognostic factors for severe CDI. Hence, it is likely that the results are reflecting that. However, there were no significant differences in adverse outcomes between the bezlo arm as compared to the placebo arm in each age cohort.

8.7. Specific Safety Studies/Clinical Trials

There was no specific study or clinical trial that was conducted to evaluate a specific safety concern.

8.8. Additional Safety Explorations

8.8.1. Human Carcinogenicity or Tumor Development

This subsection is not applicable.

8.8.2. Human Reproduction and Pregnancy

There was one pregnancy reported in study P002, and the narrative is included below. It occurred in a 23 year old, multi-racial (African American and white) subject with a negative result on a serum pregnancy test prior to receiving infusion with acto+bezlo. On day 10, she presented to an emergency room with dizziness, abdominal cramping, nausea, and occasional retching. She was discharged from the hospital with a diagnosis of moderate, persistent CDI. She subsequently had positive urine and a confirmatory serum pregnancy test. On day 258, the subject had a spontaneous, live, vaginal birth of a healthy, female baby with normal anthropometric measurements.

8.8.3. Pediatrics and Assessment of Effects on Growth

The pediatric population was excluded from the Phase 3 trials. The sponsor submitted an initial Pediatric Study Plan (iPSP) on January 28, 2014 for acto+bezlo. The submission included a deferral of pediatric assessments until after approval of the Biologics Licensing Application (BLA) in adults. Subsequently, after consulting with the FDA, the sponsor submitted an Agreed Amended iPSP for bezlo (referred to as MK-6072) on November 6, 2015 as bezlo became the selected product for marketing authorization (eCTD sequence number 0291).

8.8.4. Overdose, Drug Abuse Potential, Withdrawal, and Rebound

Overall, there were 1790 subjects exposed to bezlo either alone or in combination with acto; 126 were healthy subjects in Phase 1 (30 subjects received bezlo alone and 96 received acto + bezlo), and 1664 were subjects in Phase 2 and 3 (786 received bezlotoxumab alone and 878 received acto+bezlo). Of these, 1741 subjects were exposed to bezlo dose at 10mg/kg, while 19 were exposed to 20mg/kg. With the exception of subjects in study P004 who received two doses of acto+bezlo (n=30), all subjects were exposed to a single dose of bezlo or acto+bezlo.

Since bezlo is a human monoclonal antibody, subsection on potential for abuse, withdrawal and rebound are not applicable.

8.9. Safety in the Postmarket Setting

8.9.1. Safety Concerns Identified Through Postmarket Experience

The study drug is a novel biologic, and thus, there is no postmarket experience.

8.9.2. Expectations on Safety in the Postmarket Setting

Since the mechanism of action for this monoclonal antibody is not fully elucidated, it is difficult to anticipate all the potential off-target effects. Hence, postmarketing pharmacovigilance is recommended.

In addition, there are only 29 subjects that received more than one dose of bezlo in a Phase 1 study of healthy volunteers. Hence, there is minimal data as to the safety of repeated doses of bezlo. In the post-marketing setting, it is possible that patients may receive more than one dose of bezlo, especially patients who attend multiple health care facilities.

As mentioned earlier, there is paucity of minorities in the Phase 3 trials. Although the subgroup analysis based on ethnicity did not show major differences in AE rates or mortality, the small numbers may not allow for detection of some safety signals in subpopulation of minority groups.

8.10. Additional Safety Issues From Other Disciplines

There were no safety concerns raised by product quality, microbiology, and clinical pharmacology and pharmacology-toxicology disciplines.

8.11. Integrated Assessment of Safety

Summary

Bezlotoxumab, given as a single IV infusion, had an overall favorable safety profile with the rates of treatment emergent adverse events (TEAE), serious adverse events (SAE) and deaths similar to placebo. No increase in adverse events (AEs) related to potential immune-mediated reactions was observed in the bezlotoxumab- treated subjects as compared to placebo- treated subjects. A safety finding identified during the review of data from the two Phase 3 trials is the numerically higher numbers of serious adverse events and deaths in bezlotoxumab-treated subjects with baseline congestive heart failure as compared to placebo.

Discussion

There was adequate number of subjects exposed to bezlo at the proposed dose during the drug development program. There were 1790 subjects exposed to bezlo either alone or in combination with acto; 126 were healthy subjects in Phase 1 studies (30 subjects received bezlo alone and 96 received acto + bezlo), and 1664 were subjects in Phase 2 and 3 (786 received bezlotoxumab alone and 878 received acto+bezlo). Of these, 1741 subjects were exposed to bezlo dose at 10mg/kg. With the exception of subjects in study P004 who received two doses of acto+bezlo (n=30), all subjects were exposed to a single dose of bezlo or acto+bezlo.

The safety population is comprised of all subjects in the two Phase 3 studies, studies P001 and P002), that received partial or full infusion of the study medication. For the safety analysis, subjects were grouped in the treatment arm they received (as treated), as opposed to the group they were randomized to. Throughout the two pivotal studies, P001 and P002, there were 212 deaths. Among these, 166 occurred in the safety population because of AEs reported during the 12-week study period. There was no significant difference in the mortality rate of the bezlo (7.1%) and acto+bezlo (6.6%) arm as compared to placebo (7.1%).

There is no major imbalance in the frequency of serious adverse events among the three arms: 29.4%, 27.3% o and 32.7% of subjects in the bezlo, acto+bezlo and placebo arm, respectively, experienced at least one SAE in the 12-week study period. In all three arms, the infection, infestation, and gastrointestinal MedDRA System Organ Classes accounted for the majority of the SAEs. A detailed look of the SAEs revealed cardiac failure, urinary tract infection, diarrhea, abdominal pain to have occurred more frequently in the bezlo arm as compared to the placebo arm. From these, cardiac failure showed the largest risk difference, with 17 subjects (2.2%) in the bezlo arm as compared to placebo (n=7, 0.009%). Among subjects with baseline diagnosis of congestive heart failure, more subjects in the bezlo died (23/119 vs. 13/104).

There were 47 (n=9 bezlo, n=34 acto+bezlo, n=4 placebo) infusion interruptions. The majority of these were due to technical difficulties. Only one subject in the bezlo arm discontinued study infusion due to adverse reactions of ventricular tachycardia, chills, and dizziness.

Analysis to evaluate occurrence infusion-related reactions did not reveal a significant difference in adverse events that occurred on days 1 and 2 after infusion among the study arms. Looking at changes in systolic pressure over the infusion period, numerically more subjects had either decrease in SBP by 30% from baseline or SBP<90mmHg when comparing the bezlo arm at 30 minutes after start of infusion (10 vs. 8) and end of infusion (13 vs. 7). In total, there were 4 subjects with potential infusion-related reactions.

Among the 710 bezlo subjects that had at least one post-infusion anti-drug antibody (ADA) testing, none were reported to have developed ADA, although 28% had inconclusive test results. The subgroup analysis of adverse events by age, race, sex and body weight did not reveal significant differences by the abovementioned variables.

Overall, one time dose of bezlo appears safe; however, safety, including, immunogenicity, of multiple doses of bezlo cannot be assessed based on the data provided.

9 Advisory Committee Meeting and Other External Consultation

An Advisory Committee (AC) meeting was held on June 9th, 2016. The AC consisted of infectious disease specialists, pharmacologists, statisticians, a gastroenterologist, and patient representatives, as well as key division and office personnel. The following key findings were noted:

- On the single voting question of "Has the applicant provided substantial evidence of the safety and effectiveness of bezlotoxumab for the prevention of *C. difficile* infection recurrence in patients aged 18 years and older?," there were 10 votes for Yes, 5 votes for No, and 1 abstention.
- Many members felt that benefit with regards to recurrence had been demonstrated though some members had concerns as to any potential negative impact on clinical cure.
- Some committee members were concerned about particular trial design characteristics, such as a primary endpoint based on recurrence and the receipt of study drug while also receiving initial SOC treatment.
- Several committee members recommended a warning related to use in subjects with underlying CHF.
- Some committee members felt that benefit had not been demonstrated and that an additional trial to resolve the regulatory and clinical issues could easily be conducted

- Committee members recommended evaluating further the impact of the drug on the severity of recurrences as well as evaluating the clinical significance of subjects noted to be initial clinical failures.

10 Labeling Recommendations

A warning on the possibility of cardiac failure in subjects with a history of congestive heart failure is being considered to be included in the bezlotoxumab label. Overall, the discussions of the content of the bezlotoxumab label continue at the time this review is written.

11 Risk Evaluation and Mitigation Strategies (REMS)

This reviewer does not recommend REMS for this product

12 Postmarketing Requirements and Commitments

This reviewer believes a properly designed and conducted PMR study could provide clarity regarding the timing of study drug relative to SOC and clarify whether study drug (and its timing) have any effects on clinical cure and recurrences. Given the prevalence of CDI, such a trial should not be difficult to perform.

12.1. References

Mandell, Douglas, and Bennett's principles and practice of infectious diseases/Gerald L. Mandell, John E. Bennett, Raphael Dolin. 7th ed. 2010

12.2. Financial Disclosure

None.

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/s/

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09/22/2016

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